

Chapter 1

1.1 Introduction

Tara works as a data capturer at a prominent technology (tech) company. Her job is to record client details, file reports, search for important information in company documents and check financial transaction records. This is very time-consuming work, and because the company has so many customers, she is often behind on her deadlines. One day, Tara goes to work and is told by her employer that they are implementing a software program that employs artificial intelligence (AI) to help her perform her tasks more easily, by automating many repetitive and boring functions. Tara is anxious about the software at first, but after some training, she finds that the AI is highly efficient and that she can do the same amount of work in half the time she used to. After a few months and some software updates, the AI has become even more efficient and does 70 per cent of the work for her. As a result, her employer decides to put Tara onto shift work, because she is no longer required full-time. While Tara is glad that AI made her job easier, she feels resentful because her status in the company is now lower than it was, and she makes less money than she used to. In order to live comfortably, Tara gets a second part-time job as a waitress.

The world today faces many important challenges: trillions of dollars in global debt, millions of displaced people, disappearing polar ice caps, a rapidly growing population, and the rising cost of living are just a few of the more pressing concerns that confront modern societies. Many people believe that new technologies can solve some of these problems, making the world a safer, happier and more prosperous place. Humans have made some incredible technological advances over the past few decades, and the future promises many more. What is clear is that the way we do work is changing, as many repetitive and data-heavy tasks are already being performed by computers and AI. In the coming years, we will see further automation, and an overall change in the way we as humans see ourselves and the world around us.

People who are aware of these developments are more likely to benefit from them than people who are not. As technology changes, we will see changes to the way we approach work, as new technology increasingly automates many aspects of different jobs. At the same time, new skills and abilities might become more important in the workplace. A high degree of digital literacy will be required. This may sound intimidating, especially because we are still uncertain about the effects that technology will have on society and the economy. Many people believe that this shift toward a robotic workforce will have a major impact on unskilled labour (and even some types of skilled labour), and will probably cause significant changes and disruptions. For this reason, governments and businesses will need to be very careful in their approach to the fourth

industrial revolution (4IR). While the potential benefits of mechanisation are indeed significant, there are risks and challenges that might lead to strong resistance to these changes. While it is almost impossible to resist the progress of technology, provisions need to be made to lessen the harm caused in the process. This is no small task and will require cooperation and commitment from a number of different key role players across the world.

After studying this unit, you should be able to:

- define the fourth industrial revolution;
- identify and discuss global leadership challenges posed by the fourth industrial revolution;
- comment on how artificial intelligence drives the future world of work, from a global perspective;
- discuss how the fourth industrial revolution will reinvent the future of jobs, globally; and
- identify workplace skills necessary to flourish in the fourth industrial revolution, in a global context.

1.2 Key terms and definitions

The following key terms and definitions are applicable to this unit:

- *Algorithm*: A repetitive procedure used to solve mathematical equations in a set number of steps.
- *Artificial intelligence (AI)*: Relating to a machine or software program that can ‘think’ and act like humans.
- *Cognitive*: Any activity relating to thinking, reasoning, problem-solving and/or remembering.
- *Cryptocurrency*: A kind of digital currency that makes use of a decentralised network for exchanges.
- *Deep learning*: A process in which computer data is fed into a model that is based on the human brain.
- *Industrial Revolution*: A period of rapid mechanisation and technological development.
- *Machine learning*: A branch of artificial intelligence concerned with the study of algorithms.

- *Neurobiology*: Relating to the biology of the brain and the central nervous system.
- *Technocentrism*: A philosophy that believes technology can control and protect the environment.
- *VUCA*: An acronym used to describe the world today, which stands for volatility, uncertainty, complexity and ambiguity.

• 1.3 Defining the fourth industrial revolution

Developments in industry and technology have had major effects on the lives of people all around the world, as well as on the environment. For example, in 2015, the number of vehicles driving on roads worldwide reached over 1.25 billion (Statista 1, 2018). Try to imagine well over a billion cars moving from place to place, consuming petrol and contributing to air pollution in cities and towns all around the globe. The massive global infrastructure that provides us with what we want and need is driven by technology that uses oil and petrol. Consider the expense of a global rise in petrol prices, and of what might happen if we run out of petrol one day. Petrol and diesel are finite resources derived from crude oil, a fossil fuel that is becoming scarcer with each passing day. As the number of cars continues to grow, the demand for petrol increases, pushing up petrol prices, and therefore also the price of our food and goods requiring transport. As our fuel reserves decline and pollution continues, the question of how transport will work in the future has become very important.

What does all of this have to do with the fourth industrial revolution? An industrial revolution is driven by resources and technology, which we apply in an attempt to address problems. The previously mentioned example focused on petrol, a key resource that revolutionised industry in the past. The following table lists the three previous industrial revolutions in order to place the current one in context.

Year	Industrial Revolution	Description
1765	First	Water and steam power, creation of machines, railways and the cottage industry
1870	Second	Mass production, large factories, automobiles, electricity, telecommunication and aeroplanes

1969	Third	Digitisation, computerisation, machine learning, networking, semiconductors and automation
2016	Fourth	Artificial intelligence, autonomy, self-driving vehicles and decentralisation

(Source: *EDGE Learning Media (Pty) Ltd, 2019; adapted from SENTRYO, 2017; and Xu et al., 2018: 91*)

Table 1.1: The four industrial revolutions

Before we consider this progression more closely, it is worth remembering that there were periods of transition between each of these industrial revolutions (Xu et al., 2018: 91). During these transition periods, technologies and key resources were phased in and out of use. Different parts of the world experienced these changes differently. It is also important to note that different people have categorised these revolutions in different ways, especially in terms of when they started and ended.

The first industrial revolution began in 1765 and saw the emergence of some of the world's first modern industries, which were very unlike the massive factories and foundries that we see all over the world today. Instead, these were small-scale mines and factories, in addition to many small workshops in people's homes (White, 2009). Certain machines and inventions helped to improve the outputs of these relatively small and simple factories and workshops. At the same time, new mining and metalworking techniques were developed for the first time in 18th-century Britain (White, 2009). Before this, production required long-established technologies, like waterwheels and windmills. In 1712, the first steam-driven piston engines were used to pump water from mines. This new technology improved at a rapid rate, and was soon applied throughout the industrial sector (White, 2009). These improved mining techniques made it easier to produce iron by supplying the necessary coal. Coal also drove the other steam-powered machines that were used in factories. In many respects, however, the first industrial revolution came about because of new agricultural techniques. A rise in agricultural production, with larger yields, meant that more food was available for people, and the human population began to grow. Finally, the major mode of transportation developed during this period was the railway, as steam-powered trains began transporting people and resources (Xu et al., 2018: 91).

The second industrial revolution overlapped in time with the first, occurring in America and Europe from the middle of the 19th century to the middle of the 20th century (*Encyclopedia Britannica*, n.d.). This industrial revolution was characterised by a change in the materials being used in the production process, such as lighter metal alloys

(mixtures of metals with improved properties), and later, plastic. Improvements in machines and the eventual appearance of early computers allowed for goods to be produced more quickly in streamlined and semi-automated factories. This is called mass production (*Encyclopedia Britannica*, n.d.). In this system, unskilled labourers worked on a mechanised production line. They each controlled their own particular station on the line, performing repetitive tasks for long periods of time. This system led to a reduction in the cost of consumer goods.

Another key development during this period was the increasing use of electricity, both in industries and in the home, starting in the late 19th century. During the second industrial revolution, railway networks expanded massively in response to growing industries. Along with this came the rise of new sources of energy, especially petrol and diesel, which started being used in the 20th century to power the internal combustion engine. This allowed cars and trucks to become major forms of transportation. The 20th century also demonstrated the destructive powers of these new technologies in two deadly conflicts, World War I (1914–1918) and World War II (1938–1945). New technologies, such as armoured vehicles and aeroplanes, were deployed in these conflicts, and mass production techniques were used to supply soldiers with the vast amount of ammunition that their new rapid-firing weapons required. The end of World War II was marked by the deployment of atomic bombs on Hiroshima and Nagasaki. The ability to unleash nuclear energy was a brand new and potentially devastating technology, which the subsequent industrial revolution exploited to generate electricity.

The third industrial revolution, spanning a period sometimes called the Information Age, began in 1969 with the creation of the Advanced Research Projects Agency Network (ARPANET). ARPANET laid the early groundwork for the Internet, because it enabled networked computers to send and receive ‘packets’ of data (Roberts, 2015: 2). In addition to the birth of the Internet, another key development in this period was the harnessing of new energy sources, particularly nuclear energy (*Sentryo*, 2017). Much like the industrial revolutions before it, the third industrial revolution was driven by significant technical advancements in manufacturing, product development, transportation and power generation. However, unlike the previous two revolutions, the third industrial revolution is a global phenomenon that has been driven largely by the growing number of connections that people have with one another as a result of the World Wide Web (Roberts, 2015: 2). The invention of mobile devices and personal computers were enormously important for the third industrial revolution: word processing, video conferencing, emails, instant messaging, social networking, access to information and online businesses are just a few examples of the uses of these new technologies. However, along with these developments, automation in factories and other industries has reduced the need for unskilled workers. As more and more tasks are handled by computers, unemployment becomes a problem. At the same time,

certain skilled fields, such as programming and robot automation, are creating new jobs.

What exactly is the fourth industrial revolution then? The truth is that this revolution has only just begun, which means that it is still being defined. At this early stage, we can say that the fourth industrial revolution builds on the achievements and infrastructure provided by the third industrial revolution. One of the main features of the fourth industrial revolution is the attempt to harness new forms of renewable energy that have little or no impact on the environment (Sentryo, 2017). The past three industrial revolutions used coal, petrol and nuclear power as energy sources. This had significant negative impact on the world's ecosystem, in the form of pollution, deforestation and so on. Fossil fuels are efficient, but ultimately destructive forms of energy. Our reliance on fossil fuels has increased the levels of greenhouse gases in the atmosphere, and led to climate change and global warming. Unpredictable and dangerous weather, ecosystem collapse and a loss of biodiversity (the variety of plants and animals in the world) are all threats to life on earth. The fourth industrial revolution hopes to undo the damages caused by previous generations, and bring stability to the planet's ecosystem within the next few decades.

In order to understand the fourth industrial revolution, we must consider some of the driving forces and developments behind this new wave of advanced industrialisation. The following list describes some of the changes that are currently taking place on a global scale (Schwab, 2016):

- *Mobile supercomputing:* The power and usefulness of smartphones and tablets have already made them an important part of daily life for many people around the world. Given the rate at which mobile technology continues to develop, people will have access to increasingly powerful computers. These can be used to carry out increasingly advanced functions, and will change the way we work, as smaller devices become better at handling vast amounts of data and complex programs. It is possible that we will see more decentralised work in the future, as mobile supercomputers and high-speed Internet allow workers to perform tasks remotely.
- *Intelligent robots:* Robots have been used in manufacturing and production for many years, but they are still quite simplistic and have a rudimentary artificial intelligence that allows them to carry out one or more basic tasks. However, robots that can self-diagnose (identify their own problems), make predictions and optimise their work patterns, and even self-maintain, are becoming more and more of a reality as artificial intelligence becomes more developed. What was once science fiction is coming true, and intelligent, autonomous robots could form an integral part of the workforce in the near future.

- *3D printing*: The ability to use a machine that can use a 3D computer model to build real objects made of plastic or other materials is a key component of the 4IR. In 3D printing, a significantly smaller amount of materials can be used to create specifically-designed objects and tools. From customised prosthesis, to medical implants, to specific machine components, the uses of 3D printing are varied. Furthermore, it is increasingly becoming affordable enough for individuals to build or buy private 3D printers, bringing the technology into the home.
- *Self-driving cars*: Would you trust a driverless car to transport you from one location to another? Humans constantly have to make judgements while they are driving, and errors happen regularly, with disastrous consequences. Self-driving cars could make travel safer, more efficient and cost-effective for commuters and pedestrians alike. Self-driving cars are equipped with an artificial intelligence that can sense the surrounding environment, interface with other driverless vehicles and plan travel routes according to traffic flow patterns and other inputs.
- *Neurotechnological brain implants*: Our understanding of the human brain and nervous system has improved dramatically over the past few years, and now technology is approaching a point where implants (technology connected directly to the body) could supplement human thinking and abilities. For example, people who are unable to walk or are paralysed as a result of a birth defect or an acquired injury, may be able to overcome their disabilities thanks to neurobiological implants that could act as a bridge between different parts of the brain.
- *Genetic editing*: Our DNA forms the very basis of what and who we are. However, genetic editing offers the chance to tweak or change parts of an organism's DNA, such as altering the size or resilience of plants, or the eye or hair colour of humans, or reduce the risk of genetically inherited disorders. This technology is already being used to alter the genes of our food, allowing us to potentially feed more people at a lower cost. Instead of leaving things to chance when it comes to birth, foetuses can be imbued with certain genetic traits so that when children are born, they stand a better chance of living a happier, healthier and more satisfying life. Of course, the idea of removing birth defects and engineering people is a highly controversial issue – as are neurotechnological implants. We may face important ethical and legal dilemmas during the fourth industrial revolution, as people's belief systems come into conflict with what is technologically possible.

These are just some of the developments driving the fourth industrial revolution. It is clear, then, that the fourth industrial revolution has certain characteristics that are quite

unique. We can see that the fourth industrial revolution is taking place right now, and that it will fundamentally change the ways in which we do business, interact with one another and the environment, and that it will change our lives in profound ways (Schwab, 2016). Klaus Schwab, founder and Executive Chairman of the World Economic Forum (WEF), believes that the technologies at the heart of the fourth industrial revolution fuse the domains of the physical, digital and biological, and will impact on just about every aspect of human experience. While he is mostly optimistic about the future, he does share some concerns. The following list briefly outlines some of his key reservations (Schwab, 2016):

- Many organisations may lack the capacity to adapt to and keep up with the rapid advancements occurring in the world today. What this means is that many businesses may be forced to close or at least retrench staff members in order to stay afloat, meaning that unemployment will most likely increase.
- Governments may struggle to regulate and control new technologies, or may even apply them in less than desirable ways. If political leaders cannot see the benefits and potential dangers attached to new technologies, there could be far-reaching negative consequences.
- It is likely that most undeveloped nations will find it difficult to adopt and implement new technologies, because they already lack the infrastructure and wealth necessary to utilise existing technology to the benefit of their citizens. This threatens to continue the poverty trap that many developing countries find themselves in.
- The fourth industrial revolution will probably cause shifts in power, which will in turn create brand new security concerns. These will have to be addressed by governments and businesses.
- A higher degree of digitalisation means that more resources will have to be spent on keeping virtual spaces safe.
- It is entirely possible that the people who control technology will also control wealth, and so inequality (the gap between the rich and the poor) may increase.
- Technology has already fragmented societies and nations in many ways. This fragmentation could become even worse as new technologies replace old ones, and identities are placed under pressure.

Whatever the reality turns out to be, there are some significant challenges that need to be overcome. Some people are optimistic about the future. However, people who understand how complex human nature and history can be, are more sceptical about what is currently happening. The world needs decisive leadership and a careful strategy in order to successfully overcome the problems associated with rapid technological

development. This requires leaders who understand the possible consequences of new technologies. Determining how these technologies should be implemented and regulated will require collaboration. It is important to remember that such widespread change is highly disruptive, and will upset many people who made their living and enjoyed success thanks to the technology that is now being displaced.

For example, just a few decades ago, people still used to buy analogue cameras and film, and took their photos for development by professionals at a specific outlet or kiosk. Today, practically every smartphone has a camera, and most people share and store their pictures digitally. This obviously had a major impact on the photography industry, and a great number of the companies that used to develop photos have gone out of business or been forced to radically change their business models. This is the nature of change: once a better or more efficient way of doing things is invented, previous technologies become obsolete. This is why we do not see steam engines or pagers being used on a mass scale – they have been replaced by more efficient and newer technologies, namely the internal combustion engine and the mobile phone. The fourth industrial revolution will ultimately change the way we see ourselves and do business forever.

1.4 Global leadership challenges posed by the fourth industrial revolution

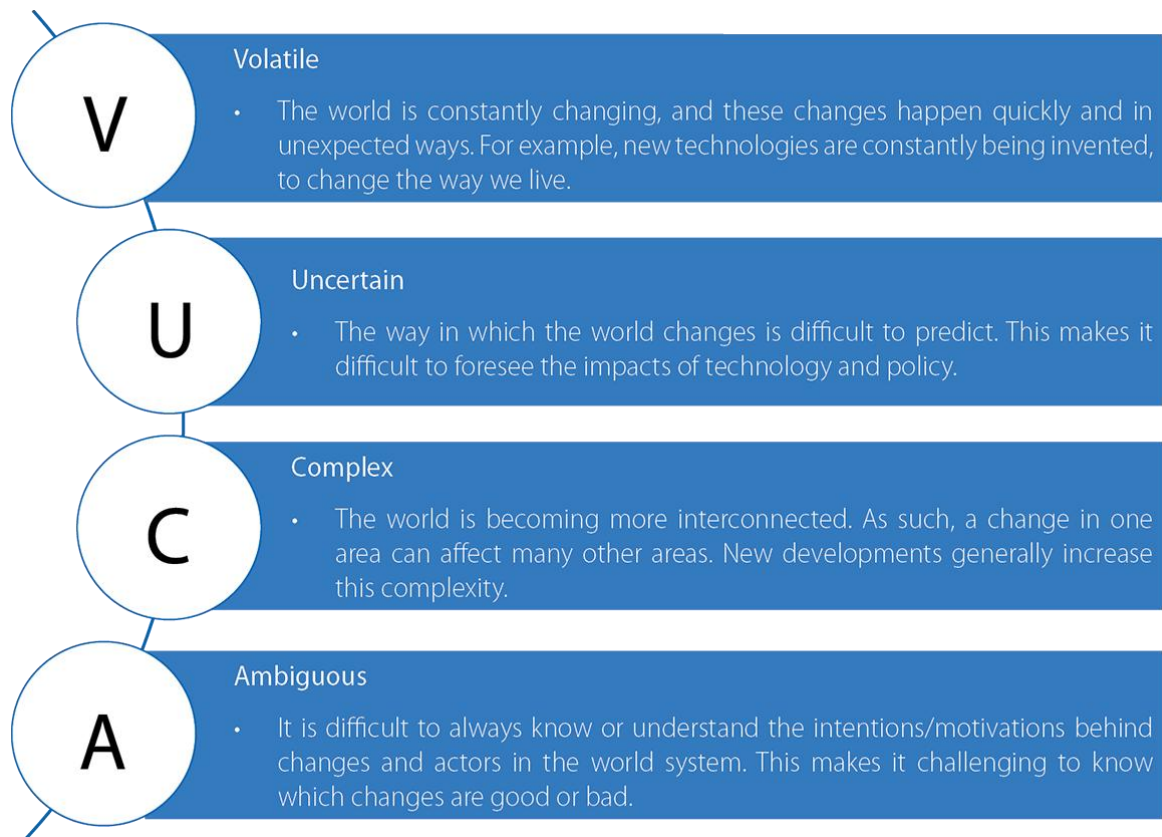
The success or failure of the fourth industrial revolution depends on the ability of governments, businesses and other entities to harness new technology in a responsible way, and create long-lasting benefits for all people across the planet. Unfortunately, in the face of such uncertainty and the increasing rate of change, it is likely that many nations and organisations will struggle to keep up, and may very well face some significant social and political problems as a result. One thing in particular distinguishes the 4IR from previous industrial revolutions: with the advent of neural implants, synthetic prostheses, artificial intelligence and genetic engineering, the fourth industrial revolution will alter humans at their base level, and probably create a new kind of human being that is the ‘next step’ in our evolution, which is now driven by our technologies. As Von Reiche (n.d.) states, the fourth industrial revolution ‘will see a deeper integration of technology into societies and the human body’. While the third industrial revolution brought some incredible changes to global society, the rate of change in the 4IR is likely to be exponentially greater than in the case of its predecessor. This means that instituting and regulating new developments will become increasingly more challenging.

The fourth industrial revolution will require that businesses that want to remain afloat and achieve some sort of success in the future will need to innovate relentlessly to stay ahead of their competitors. Of course, this could make an already highly competitive

marketplace even worse, and so it may become even more challenging for governments and political leaders to exercise control over this space. While many technologies may well be used to make life better for people and improve the efficiency of industrial production, they could also be abused and used in questionable or criminal ways. Since many highly industrialised and developed countries hold just about all of the power and wealth, it seems that the gap between developed and developing nations will increase in the years to come. This means that already disenfranchised and excluded populations will be at an even greater disadvantage, because they will simply be incapable of competing on any meaningful level. Perhaps we should be asking whether access to technology should be a human right, or if it should remain something that is reserved only for the very select elite few.

Leadership in a volatile, uncertain, complex and ambiguous (VUCA) world

In the past decade or so, life has become far more complicated, as human needs and patterns of interaction have fundamentally shifted. Our ability to access vast stores of information via the Internet has made us question our own beliefs and opened our eyes to the many wonders and horrors that the world has to offer. For this reason, things that once seemed certain and absolute to many people have become significantly more open-ended: nations and states, citizenship, abortion, sexuality, gender, the value of money, immigration and the very concept of truth have come under scrutiny, thanks to the mass sharing of ideas via the World Wide Web and other technological changes. VUCA stands for 'volatile, uncertain, complex and ambiguous' and was coined by the US Army in the early 1990s to describe the world after the Cold War came to an end (Forsythe et al., 2018). Since then, the acronym has been widely embraced by leaders and politicians all over the world.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Forsythe et al., 2018)Figure 1.1: A brief explanation of the VUCA acronym

VUCA therefore refers to the various developments that make the world of the 4IR more difficult to navigate successfully. The world can change rapidly and in unexpected ways, so leaders of all kinds will need to ensure that they and their governments, organisations or companies are capable of acting and reacting quickly in a VUCA world. As populations and the demands on the environment increase, leaders must be able to take considered and decisive action. Many of the issues surrounding the 4IR are ethical, so companies and leaders will need to consider the following questions:

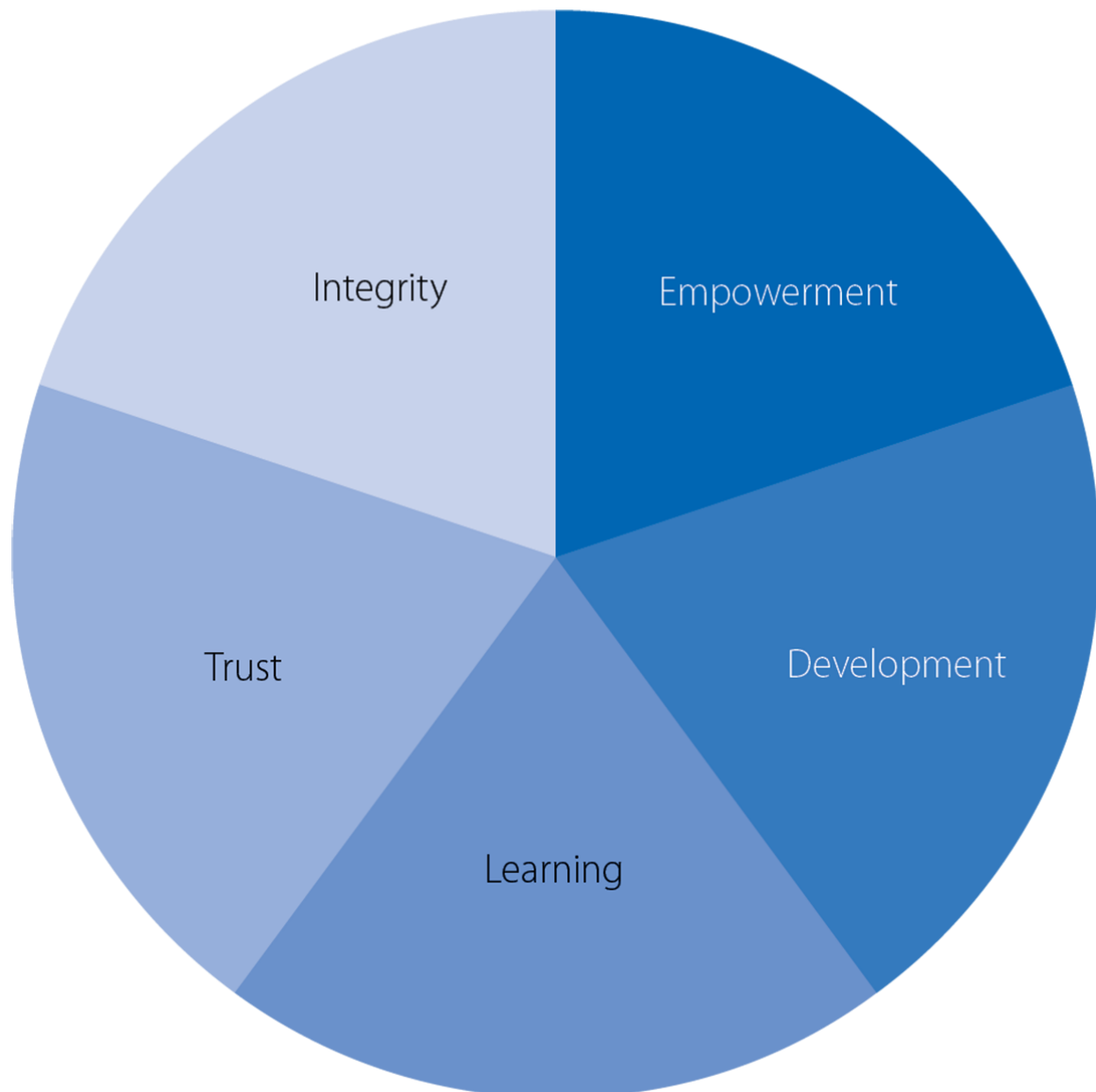
- What possible benefits and/or challenges will a particular form of technology bring along with it?
- Which individuals or groups will use the technology and for what purpose?
- How rapidly is a form of technology likely to develop, given the current rate of change in the world?
- Is it possible that a certain technology may be banned in one or more parts of the world?
- Will an innovation effectively increase inequality or will it help to address longstanding social issues?

- Is there the possibility that control over a technology may be lost if it is made freely available?
- How does a new technology or set of technologies fundamentally alter or augment the human experience?
- How profitable is a new technology and is its existence financially and socially justified?
- What countries have access to technology and/or control the rate of technological development?
- What role do businesses and governments play in regulating a rapidly developing digital space?

These questions are just a small sample of the questions that leaders from all parts of the world and in all spheres will need to ask, and answering them is no simple task. In fact, it may be nearly impossible to answer many of them, simply because the world is so volatile and unpredictable. How, then, can businesses and governments cope with the pressure to perform and deliver on their promises? According to Forsythe et al. (2018), the answer is for organisations to ‘build a responsive culture’. In other words, it is not always possible to prepare for every eventuality – in fact, every possible outcome cannot be anticipated, so preparing for these kinds of changes is a monumental task. For this reason, companies operating and participating in the 4IR will need to be highly responsive, and capable of adapting to new developments at the drop of a hat. This entails constantly surveying the local and global environment and always being highly vigilant.

Executives’ need for a paradigm shift in thinking

High levels of uncertainty and volatility demand that executives and business leaders should change their thinking in fundamental ways, so that their employees and stakeholders will be kept satisfied in the long run. It is important for organisations in the 4IR to be responsive to stakeholders’ needs (Forsythe et al., 2018). For this reason, the following aspects need to enjoy a strong focus in a VUCA world if modern organisations are to achieve optimal levels of responsiveness.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Forsythe et al., 2018) Figure 1.2: Areas of focus in order to achieve a responsive organisation in a VUCA world

The role of executives is to manage and facilitate the shaping of organisational culture by focusing on *integrity*, *trust*, *empowerment*, *development* and *learning* (Forsythe et al., 2018). While these aspects are essential in any organisation, they will arguably become even more critical as the rate of change accelerates over time. One of the key challenges of a VUCA world is that it is nearly impossible to know everything concretely. This is why the interrelated dimensions of trust and integrity are vital to the survival of organisations in the 4IR.

Perhaps the best example of a betrayal of trust and integrity is the ‘Steinhoff saga’, which is ‘one of the biggest cases of corporate fraud in South African business history’ (CNBCAfrica.com, 2018). Steinhoff’s empire came crashing down on 5 December 2017,

when the company's CEO, Markus Jooste, suddenly resigned from his position, while the company's board announced that it had become aware of 'accounting irregularities' (CNBCAfrica.com, 2018). These announcements caused the company's share prices to fall by 85 per cent – and by 11 May 2018, Steinhoff shares were valued at only R 1.60. As a result, Steinhoff has faced various charges, including two class action lawsuits in both Germany and the Netherlands (CNBCAfrica.com, 2018).

It is no wonder, then, that many South Africans are wary of large corporations, and see them as entities that only exist to serve themselves. Given the uncertainty around the coming 4IR, and fears that it may serve only to benefit the wealthy, the interrelated metrics of integrity and trust are key factors that can ensure a fairer future for more people. Empowerment and development are also connected to these areas. Employees who trust their organisations are more inclined to participate in development programmes and feel empowered as a direct result of the positive culture that permeates their organisation. Similarly, customers who feel they can trust an organisation feel empowered because they believe that by supporting such organisations, they are not adding to the hardships of others or causing harm in any major way. Employees need to be empowered so that they can grow and develop as workers and innovate in a world that will become ever more reliant on brilliant ideas. Learning is absolutely necessary for all stakeholders in a business: from the CEO to interns, every individual in an organisation needs to learn about the newest technologies and find ways to use them in novel and effective ways.

The 4IR and challenges for government

Lye (2017) points out that the 4IR presents governments with both good and bad news. The following table points out some of the positive and negative aspects that governments will need to take into consideration.

Positive aspects	Negative aspects
Fosters an open and dynamic economy	Governments may be overridden by exponential corporations
Industry becomes skill- and knowledge-based	Central banks and institutions may lose power
Promotes trade outside of traditional blocs	Renewables may decrease reliance on national grid
Improves effectiveness of health and social care	Systems may develop independently of government

'First mover' opportunity for defence and security	Greater unemployment may lead to social unrest
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(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Lye, 2017)

Table 1.2: Potential benefits and challenges posed by the 4IR for governments

The 4IR may grant governments the ability to run their nations more efficiently and intelligently, but there is also the possibility that businesses, rather than governments, will decide how human societies progress and are developed. The 4IR is likely to see the rise of the 'exponential organisation' – one that does not adhere rigidly to the existing structural and operational frameworks of current organisations (Exponential Organizations, n.d.). Exponential organisations will come into existence because businesses will begin to leverage and develop exponential technologies that will form the foundation and backbone of the 4IR.

However, because many new companies are moving toward decentralisation and are less 'physical' in nature, they will be able to exist in a number of different spaces and will not be restricted to any one location. For this reason, we can already see organisations that are able to bypass regulations and laws imposed by certain nations, because they are present on a global level and have access to a wealth of different resources, and do not 'belong' to any one person in particular. This means that bringing charges against companies that contravene laws may be an impossible task. Businesses and individuals may find even more ways to 'cut governments out', by dealing directly with one another via encrypted networks and using digital currencies such as Bitcoin. Many people have effectively lost faith in the global economy, and see the 4IR as an opportunity to reshape society and economics in a digital sphere, where the concept of value can be totally redefined.

The decentralisation of power and trade means that governments may have the opportunity to trade in commodities and assets with new players in completely new environments (Lye, 2017). What this means is that countries that were once incapable of participating in trade and other related economic activities may be able to do so, as a result of more relaxed global policies and digital networks. Although the 4IR economy may continue to reproduce global inequalities, it has also improved the ability of previously disenfranchised countries to trade more efficiently through global networks. This may provide an opportunity for these nations to become more powerful economic players, if the 4IR is prepared for effectively and implemented justly.

The emphasis in the 4IR economy will be on sustainability and socioeconomic development, and so there will be concerted efforts to try to harness cheap and clean renewable energy in order to run the world of the (near) future. This means that a great deal of government infrastructure may be rendered null and void, since fossil fuels will

become less important, and the generation of electrical energy will increasingly become something that cities, towns or even individuals can participate in. In this sense, governments should be attempting to migrate toward renewable energy themselves, so that they do not become irrelevant when private companies begin supplanting antiquated infrastructure. There is certainly the possibility that health and social care will become more affordable as new technologies allow us to live longer, happier and healthier lives (Lye, 2017).

However, depending on how governments institute and regulate such technology, it may only be available to those who can afford it, rendering any universal benefits non-existent. Perhaps the greatest concerns governments have is that new systems and technologies may develop without government involvement or intervention. In addition, new technologies may replace many unskilled human workers. High levels of unemployment do not bode well for governments, who may need to redefine social welfare programmes, or take preventative steps to prevent social upheaval. While high unemployment is not guaranteed, governments need to consider that this terrible reality may come to pass in the near future. One of the other possible benefits for governments that choose to adopt new technologies sooner rather than later, is that they may be able to secure themselves both militarily and virtually (Lye, 2017). New technologies may be deployed to make militaries and weapons more powerful, escalating the danger posed by conflicts as the 4IR continues. Perhaps even more pressing are the risks posed by cybercrime and cyberterrorism, which misuse or attack the Internet and interconnected devices and technologies we depend on in our modern economy and daily life.

With all of this in mind, what can governments do to deal with the numerous challenges posed by the 4IR? Lye (2017) proposes the following four scenarios. These are explained in further detail under the following bullet points.

- *Manage the market:* In this scenario, the general aim is to create a stable environment and ensure that new technologies are instituted fairly and reasonably. This is already taking place in the European Union (EU), which hopes to define a 'Digital Single Market' that creates a framework within which new technologies can be created and developed.
- *Take control:* Certain countries that are more authoritarian in nature may do their utmost to take control of new technologies for their own benefit. According to Lye, this is already happening to some degree in places like China, where control and obedience are encouraged by the government. A risk with this approach is that countries with fewer regulations and controls may develop more quickly, as free innovation is encouraged. More heavily regulated nations, on the other hand, may therefore be left behind. In China's case, their successful technology sector is something of a 'double-edged sword' (something with positives and negatives)

when it comes to this approach (Bey, 2017). While the continued competitiveness of China's tech companies is vital to the country's economic prosperity, the more the tech sector grows, the more the Chinese Communist Party's power will be undermined (Bey, 2017). For now, China's interests are in line with the tech sector, but this may change in the future.

- *Open for business*: This is the opposite of the 'take control' paradigm, and would effectively see governments doing whatever they can to attract foreign investment by encouraging technological innovation with little or no restrictions whatsoever. This may be the best route for smaller countries that have been disenfranchised and excluded in the past, as it may well bring newfound wealth and prosperity.
- *Hands-off*: This is essentially hypothetical at present, but involves a government willingly relinquishing some or even most of its power so that organisations and individuals can negotiate power and economics in new virtual spaces. Governments may still fulfil roles such as defence and security and managing foreign trade, while all other concerns are left in the hands of exponential organisations.

Regardless of which approach governments choose to adopt, they will have to strategise and prepare to deal with the impacts that new technologies will bring (Lye, 2017). How these approaches are applied will determine if new jobs are created as others disappear (due to automation, outsourcing or industries changing). Inevitably, there will be issues that governments and businesses will need to manage. In a best-case scenario, new jobs and development will provide opportunities people who are affected by industrial changes. In a worst-case scenario, high levels of unemployment and/or underemployment will lead to a loss in tax revenues and, render governments much less capable of dealing with issues effectively. The reality will probably be somewhere in-between these two paradigms, and the way the 4IR affects a country or region may differ from country or region. The future is uncertain, and that there is no way to fully predict exactly what will come to pass. The previous four scenarios are just some of the general approaches that governments may choose to adopt in order to facilitate the social and economic changes that the 4IR will bring. What is certain is that leadership is changing, in large part because of developments in technology. As Lye (2017) states, the last decade or so has seen trust in existing politics almost completely dissolved. Parties that have maintained largely moderate or centrist policies, as well as left-leaning ones (such as the Democratic Party in the USA), have suffered the greatest losses. This has led to the rise of nationalist right-wing parties, as well as more extreme left-wing ones (Lye, 2017).

People are anxious about the future for several reasons, so they seek certainty and

reassurance from politicians that their needs will be adequately addressed. This makes it important for political leaders to research and forecast the consequences of new technology, and to ensure that it delivers on its promises, while guarding against negative consequences. For example, President Cyril Ramaphosa appointed a commission specifically for the 4IR during his State of the Nation Address (SONA) in February 2019 (*ITWeb Africa*, 2019). In doing so, he showed his commitment to preparing South Africa for the 4IR, and to understanding its potential consequences.

Approaches to the role of technology

Over time, humanity has become more dependent on technology to survive and thrive (Razak, 2017). The 4IR will see a further migration toward and reliance on technology for economic activities. There are numerous terms describing the roles that technology could play in humanity's future. While these terms and the ideas they represent are complex, the following list shows just three approaches for thinking about what technology could achieve in the future:

1. *Technocentrism*: This approach puts almost total faith in the powers of technology and industry, thinking of them as central to handling the challenges of the world going forward, from human systems to the environment (Razak, 2017). In this view, technological developments are considered in a very positive light, as they increase the number of tools we have at our disposal to respond to problems and opportunities.
2. *Anthropocentrism*: 'Anthro' refers to humans, so anthropocentrism places supreme importance on humanity. For this reason, this approach views technology and our faith in it as short-sighted (Razak 2017). Ultimately, anthropocentrism views humans as the primary species on the planet, with all other resources, plants and animals at our disposal (Boslaugh, 2013). It also sees humans as superior to their technology, with abilities that technology cannot and should not replicate.
3. *Ecocentrism*: In this view, the environment and non-human life is just as important as humanity, because we depend on the global ecosystem for survival (Sustainable Environment, 2018). Technology has led to severe environmental problems, including pollution and climate change, and technology should not be seen as a cure-all solution for the future. This view is therefore critical of both technocentrism and anthropocentrism.

What this very brief summary demonstrates is that there are many different and competing ways of approaching the development of new technology. Technocentrists embrace technology and its potentials. Anthropocentrists question a reliance on technology which could affect human lives in negative ways. Ecocentrists question both

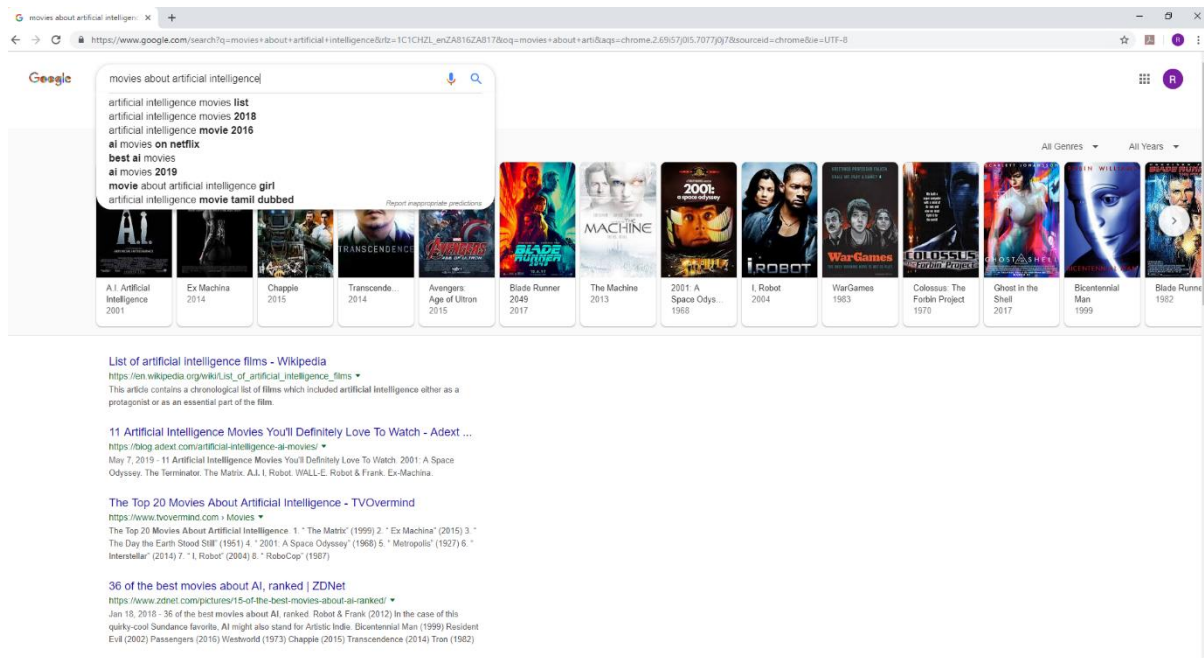
approaches, emphasising the global ecosystem and the role played by humans and technology in both harming and rehabilitating our shared biological environment.

1.5 Artificial intelligence's directive on how we work and build new industries

Artificial intelligence is by no means a new invention. AI has been employed in manufacturing, global positioning systems, video games, robotics and computing for decades. Work on AI began during World War II, when Alan Turing (a British computer scientist) developed a machine that could intelligently decipher encoded Nazi messages (Ray, 2018). In 1956, following further developments in computer intelligence, such as a computer mastering the game of checkers, the term 'artificial intelligence' first came into use. In 1972, the first 'intelligent' humanoid robot, WABOT-1, was created in Japan with rudimentary AI, which was a breakthrough at the time. Only in the late 1990s did interest in AI begin to grow in earnest, especially after IBM's computer, Deep Blue, became the first machine to defeat a world champion in chess. In recent years, big corporations such as Google, Amazon and Baidu have invested heavily in AI and machine learning (Ray, 2018).

Understanding AI

Machine learning (which is a subtype of AI) is something that has been used for a number of years on many websites, such as Amazon and Google. When you conduct a Google search, or browse for items on Amazon, the sites often make recommendations based on what you have previously searched for and looked at. This is an example of machine learning. Where Amazon is concerned, a simple AI aggregates all the items that users from all over the world view, along with the product in question, and then chooses the most commonly viewed items to recommend them to other site visitors. Similarly, Google's search engine can extract useful, personalised information from websites based on what you and other users have searched for before. It can also recommend other topics you might want to search for – as shown in the following figure.



Source: ©2018 Google LLC, used with permission. Google and the Google logo are registered trademarks of Google LLC. Figure 1.3: Google's search engine suggesting related topics and search terms

Machine learning has been around for a while, but in recent years, a new kind of AI has been developed in the form of deep learning (Campos, 2018). Simply speaking, deep learning is a subtype of machine learning, but instead of programming rules into a system so that it mimics human behaviour, deep learning feeds data into a model that is based on the human brain itself. This is called a neural network, and uses a number of layers that are connected to behave in the same way as a real human brain does when it processes and connects information between neurons (the cells in the brain) (Dormehl, 2019). What is important here is that there are 'hidden' layers that can test the results of the network, searching backward to identify where the AI makes mistakes, and then learning from them so that the AI does not make the same mistake again (*Techopedia*, 2019). In other words, an artificial or virtual brain teaches a computer or AI how to think or react to a certain situation, depending on the various inputs it might receive. Deep learning has the potential to break the limitations of machine learning and open up brand new territories of knowledge, learning and understanding for AI (Campos, 2018). Whereas AI in machine learning involves a computer that is programmed to think and carry out actions using algorithms, deep learning is a computer that models actual thinking. This may sound inconceivable, but if we consider that we process information in the form of electrical impulses between the various synapses in our brains, then it is not impossible for a machine to engage in the same processes. The only difference is that deep learning is possibly more effective, since a machine does not need to be made to concentrate and can basically avoid any distractions that we as humans often encounter.

In this sense, modern AI will be able to process information and learn at an exponential rate. This means that a relatively primitive and simple AI could become tens or hundreds of times more intelligent over the course of a few months. The smarter an AI gets, the more effectively it can learn, and the smarter it becomes. We could even go as far as to say that deep learning is 'true' artificial intelligence, because it basically refers to machines that learn on their own, with little human input – if anything, the number of inputs from humans required by a machine with deep learning capability should decrease over time. However, we cannot be sure how soon this kind of deep learning will become fully realised. The world of AI is constantly evolving. In any case, AI is here to stay, and it will form the backbone of the 4IR economy.

Opportunities and challenges for AI and the 4IR

In an interview between futurists and technologists that was published in *Emerj* (2018), it becomes clear that AI is seen by many people as the main early driver of the fourth industrial revolution. In certain respects, AI might be as transformative and revolutionary as steam and electricity were for the first two industrial revolutions. There are several benefits to employing AI in industry. The following list outlines some of these advantages and explains exactly how AI will define and shape the 4IR (*Emerj*, 2018):

- Advances in AI have already been commercially implemented, due to the rapid development of semiconductors (materials used in circuitry), which have improved the speed of processing data, and have led to a year-on-year doubling of the amount of data we are able to store. With ongoing improvements such as these, AI will become more capable of managing technical tasks.
- People in skilled roles that require them to examine and sort through vast amounts of quantitative data may find their work increasingly performed by thinking machines instead. Given the pattern-recognition skills of neural networks and AI, identifying patterns in data is something that is already easily accomplished. AI will be faster than humans at sorting through the vast amounts of data required by the legal, medical and financial sectors. This may not necessarily mean that these data-sorting jobs will disappear altogether, but jobs such as those performed by paralegals and financial data analysts may find their role changing significantly because of AI.
- AI could be used to support humans in complex qualitative roles. In other words, AI can analyse data gathered alongside human employees to run simulations and make predictions. For example, in psychotherapy, an AI may assist psychologists in recording and interpreting various kinds of data to profile their patients.

We can therefore see that AI is especially useful in white-collar work scenarios, where AI can help to analyse and process vast amounts of qualitative and quantitative data,

making work more efficient. Another industry where AI is likely to be deployed, is in education (ROSS, n.d.), where it could soon assist teachers and lecturers with repetitive tasks like marking, record-keeping and even tutoring. AIs may be able to generate tests and exams within seconds or minutes in the near future. This is highly significant because these tasks consume a great amount of educators' time – with fully and semi-automated systems in place, teachers will be able to spend more time on teaching and responding to learners' more pressing needs. Recent research suggests that the use of AI in education will grow by about 47.5 per cent by 2021 (EdTech in ROSS, n.d.). Multinational companies like IBM, Microsoft and Google are at the forefront of AI in education. While some educators may be wary of AIs, those who embrace it will probably find that their jobs will be made much easier and that some assessment tasks can be performed in a fraction of the time they require now, thanks to AI. Teachers and educators who have more freedom may turn out to be more effective than those who are bogged down with marking and setting tests. If evaluations can be automated and standardised, education should theoretically improve dramatically.

In an ideal scenario, AI and humans will complement each other, leading to the development of a highly educated and more skilled workforce that is faster and more effective than that of previous generations. AI could empower people to devote more time and energy to complex problems, such as mental and physical health issues. However, there are a number of concerns about artificial intelligence and its future development.

One of the main debates surrounding current AI technology is the question of bias in algorithms. As artificial intelligences have been trained on data fed into algorithms, they can only respond to issues in the data being given to them. While algorithms and AI are often seen as free of human biases, it is instead becoming clear that in many cases, these algorithms tend to reflect human biases, especially biases around race, gender and ideology (IBM, 2018). This is a major concern raised by Google's head of AI, John Giannandrea, because AI systems are far too complex for the people who will need to use them to aid in decision-making (in fields like law and medicine) to understand (Knight, 2017). There are already concerns about systems such as, for instance, an AI that is used by some judges in the USA to determine whether prison inmates should be paroled, because the system may mirror a human racial bias against minority groups (Knight, 2017). As we start to use AI to consult in key areas, it will be important for the way these intelligences work to be transparently presented (made clear and understandable to the people using them). One solution suggested by IBM is to use AI's advantage, which is its ability to recognise patterns and learn from them, to understand and work against the human biases in the data they have learned from (IBM, 2018). If this could be accomplished, there is even a possibility that AI with this ability could be used to help humans recognise their own biases and counteract prejudice in society.

Another concern for the future of AI is the fact that artificial intelligence lacks certain components of human intelligence, most notably emotions and empathy. Human beings have a very complex psychology, and there are concerns that AI may not be able to understand emotions in the same way as humans. However, some futurists believe that artificially intelligent devices may soon be more knowledgeable about a person's emotional state than their family. There are already studies suggesting that some AIs are able to detect signs of human emotions better than humans (Kleber, 2018). This still does not mean that the AI can actually feel, but it shows that the pattern recognition of AI can be used to detect and analyse data like facial expressions, tone of voice and body language to understand and respond to different emotional states in people. Ideally, these kinds of systems could be used to help people, and assist organisations in better responding to the emotional needs of their customers. However, there are concerns that this technology could easily be used to manipulate our behaviour or understand human emotions in order to strategically target customers – an ethical dilemma that will need to be addressed in the near future. Before considering using emotionally capable AI, Kleber (2018) suggests that company leaders should consider the following questions:

- Is what your company is offering something that requires emotional input from end-users and clients?
- Can you honestly justify the inclusion of emotional elements for users to have a better experience?
- What are the emotional intentions and motivations of your clients when interacting with your brand?
- What is the nature of interactions and discussions that your customers have about your business?
- Do you ask for permission from your clients before accessing and analysing their emotional data?
- Are users able to stay in control of their information, and can they deny access if they so choose?
- Is your system advanced enough to accurately record and react to your users' emotions?
- Are there any possible dangers or problems if your system should fail, both in terms of your users and/or your brand?

Considering some of these challenges currently faced by AI, we can see why Campos (2018) states that the current ability of AI is at the same level as a young child, which humans must responsibly guide into maturity and equip to the benefit of society. Others, including Tesla Inc. CEO Elon Musk, are much more sceptical of the benefits of

AI. In fact, Musk suggests that intelligent computing poses a greater threat to humanity than nuclear weapons (Campos, 2018). There are risks that AI could be used for criminal purposes, terrorist activity and for new forms of warfare.

What AI means for the future

In the previous section, we saw that AI will be a key technology underlying the 4IR, altering jobs in fields as diverse as law and medicine. The skilled professionals of the future may well be using intelligent machines daily. As a result, education will need to be adjusted to prepare the professional of the future for a highly technological workplace (Butler, 2018). From school level, technology, programming and artificial intelligences will need to form a central part of the curriculum. Furthermore, we need to prepare ourselves to work alongside AI, as intelligent machines increasingly begin to perform skilled work that, until now, has been solely done by people. Butler argues that over half the workforce will need to build and develop new skills, as AI begins to handle more of the workload (Butler, 2018). This is especially true in the age of deep learning, because AI will begin to understand and improve itself, becoming more skilled at more tasks than ever before. Technocentrists argue that AI might soon be able to build and improve on itself more effectively than humans can.

This brings us to a very important topic related to AI, namely ‘technological singularity’ (Shanahan, 2015: vxi). This is a hypothetical scenario (a theory or potential scenario) where AI reaches a level of self-awareness and is able to self-replicate (i.e. intelligent machines become able to create new and better versions of themselves). These would be called ‘artificial superintelligences’ (ASIs), and would be able to understand and process the world in a way humans and current machines cannot. Such machines would be able to rapidly and continuously improve their capabilities. This may lead to a future where humans share the world with intelligent machines, along with other humans who have augmented themselves through technology and genetic editing. What such a future may involve is purely speculative at this point, but with the rapid development of AI, this future may start becoming a reality by the middle of this century (Shanahan, 2015: xix).

Closer to the present, AI research is also driving and being driven by the push to create highly complex quantum computers (ROSS, n.d.). Understanding how these computers use particles smaller than atoms is very complex, but computer scientists are using the way matter and energy behave at a quantum (i.e. very small) level to reshape what can be achieved with computers (*ScienceDaily*, n.d.). In this way, the development of better AI encourages other technologies which might be instrumental in developing other areas of the 4IR. Quantum computers that are capable of more advanced AI would also be able to store and quickly process the vast amounts of data that businesses in the 4IR will be required to manage.

AI has the power to completely change the world. Whether for better or for worse, it is already changing the way we do business, learn, socialise and run our industries. AI could very well lead humanity toward a kind of utopia, or it could ultimately place our well-being and safety at risk. What is clear is that businesses and governments will need to be aware of the opportunities and risks involved in AI's development, because it is such an important aspect of the 4IR.

1.6 Future jobs shaped by the fourth industrial revolution

The previous industrial revolutions brought about massive shifts in the way that societies and economies functioned. The 4IR is no exception to this rule. With the development of new technologies, new industries and jobs are bound to be created as well. Assuming that technological singularity is at least a few decades away, humans will remain relevant in the fourth industrial revolution and will be required to build, maintain and develop new technologies so that they can continue to make a living and earn their positions in society. Before proceeding further, it is worth considering some observations made by Erik Brynjolfsson, the director of the MIT Initiative on the Digital Economy. First, Brynjolfsson points out that people working in the economy of the future will need to be capable of doing things that machines cannot do well – in his word, ‘You want to be working with them, not competing with them’ (in *World Economic Forum* 1, n.d.). This is critical because it is virtually guaranteed that machines will soon be able to outperform humans in many areas, and with time, the number of tasks machines will handle will probably increase. For this reason, Brynjolfsson makes the following observations, all of which are relevant to young people (in *World Economic Forum* 1, n.d.):

- The creativity of young people needs to be actively encouraged, especially because Brynjolfsson believes that there are various parties that are doing their very best to stamp out creativity in all its forms.
- The development of strong interpersonal and leadership skills along with a teamwork is of crucial importance because, as Brynjolfsson states, ‘People may not have degrees, but they can have high emotional intelligence’. Humans will still play an important role in the continued development of other human beings.
- People’s passions are critically important in the 4IR. People who love what they do and excel in their chosen field are almost certain to succeed, regardless of the presence of AI and other technologies. It may seem clichéd, but following your dreams is integral to human success.

Considering these points, the role of educators and/or mentors, inventors, artists, animators and musicians will remain significant in the 4IR. In many respects, they may

become even more important as many people may wish to experience life in a more personal, intimate and authentic way as they become overexposed to the presence of technology, AI and robots. However, there are no guarantees, and there is the possibility that AI could one day outpace people in every sphere. Saadia Zahidi, the World Economic Forum's Head of Education, Gender and Work has stated that the future may see up to 47 per cent of jobs being lost to automation– however, other predictions are less severe, suggesting that only nine per cent of existing jobs will be taken over by machines (in *World Economic Forum 1*, n.d.). The developed and developing worlds will be drastically affected by automation, but it is likely that the developing world will be the most adversely affected, because of the ways in which their economies are structured. For example, countries like Venezuela and India thrive on an economy driven by cheap labour and materials, and with many companies looking to automate their operations locally, the days of outsourcing labour may be coming to an end. This will almost certainly have a negative effect on the livelihoods of many working-class citizens who will be forced to leave their already low-paying jobs. Working class people in the developed world have already started being supplanted by outsourced labour and local automation, and further automation will likely add to their woes. While the 4IR stands to benefit some, many jobs will almost certainly be lost as modern technology replaces humans. This, as we have mentioned before, will be an issue that global leaders, civil society and business will need to resolve to the benefit of humanity.

As we have seen, machines will make a great deal of human work far more efficient and cost-effective. Brynjolfsson echoes this sentiment when he claims that human/machine partnerships will allow us to solve problems that we have never been able to solve before (in *World Economic Forum 1*, n.d.). We have mentioned some of the fields where the 4IR will affect the nature of existing jobs. What about entirely new jobs that the 4IR will bring? Although there will be numerous new positions and occupations, for the purposes of this textbook, we will examine a few of the more prominent and 'current' positions that people will need to fulfil in the 4IR. These include: experts in artificial intelligence, 'Internet of things' (IoT) specialists, cybersecurity experts, Internet governance professionals, social media specialists, machine learning specialists, robotics engineers, 3D printer experts, autonomous vehicle experts and blockchain specialists (Gallo, 2018). Although some of these jobs already exist, they are still evolving and becoming more widespread as a result of 4IR technologies. Each new role will be outlined and discussed in some detail to help you gain a basic understanding of them.

Experts in AI

This particular job is one that will likely become more and more prominent as the 4IR gains momentum and takes root in society. While there are already several AI specialists, companies that want to automate their operations will need one or more in-

house AI professionals to guide the implementation process and program machines accordingly. AI experts need to have a range of abilities and be skilled in several disciplines if they want to be appointed. The following list, courtesy of *Computer Science Degree Hub* (n.d.), outlines these skills and competencies in some detail:

- A strong knowledge of mathematics, including probability, statistics, algorithms, logic and algebra.
- An understanding of advanced networking and machine learning.
- Knowledge of science, especially physics, engineering and robotics.
- The ability to grasp aspects of computer science, namely programming languages and coding.
- Deep knowledge of the way people and machines think and understand information.

AI experts have to be highly degreed and know a great deal about how machines function. With the rate at which the field is developing, it is almost certain that AI experts of the future will need to be far more knowledgeable and dynamic in order to find work and earn a living. Of course, not all AI will necessarily be supremely advanced or require years of study to understand, so 'low-level' AI jobs may form a major part of the 4IR. Having said this, those experts who want to earn a lot of money and gain status in society will need to work hard and apply themselves in order to realise their ambitions. If an AI expert can identify a role or industry that has not yet been enhanced by the presence of AI, they may be able to enhance their career by adding AI to their skillset. As *Computer Science Degree Hub* (n.d.) observes, people who can translate pieces of digital information into meaningful human experiences will be able to find a career in AI. It seems that the migration toward automation has already begun (at least in developed nations), so workers who want to secure long-term employment and remain relevant in the new economy should seriously consider studying artificial intelligence, since those people who excel in this particular field could be very successful.

'Internet of things' (IoT) specialists

Before we discuss this particular career, it is important to understand what the IoT actually is. In simple terms, it is a vast network of interconnected physical objects such as cars, trucks, aircraft and home appliances that use sensors and application programming interfaces (APIs) to connect to and share things over the Internet (SAP, n.d.). Almost any device that is able to connect to and transmit data over a network can become part of the IoT, which means almost any device can be designed to join this network (TechTarget, 2018). As more and more objects become connected to the IoT, the number of interconnections increases, and the nature of these interconnections

accordingly becomes more complex. Vast amounts of data need to be analysed continually, and so systems need to be kept running at optimal levels for as long as possible. For this reason, both artificial and human intelligences are required to maintain the IoT and keep everything running smoothly.

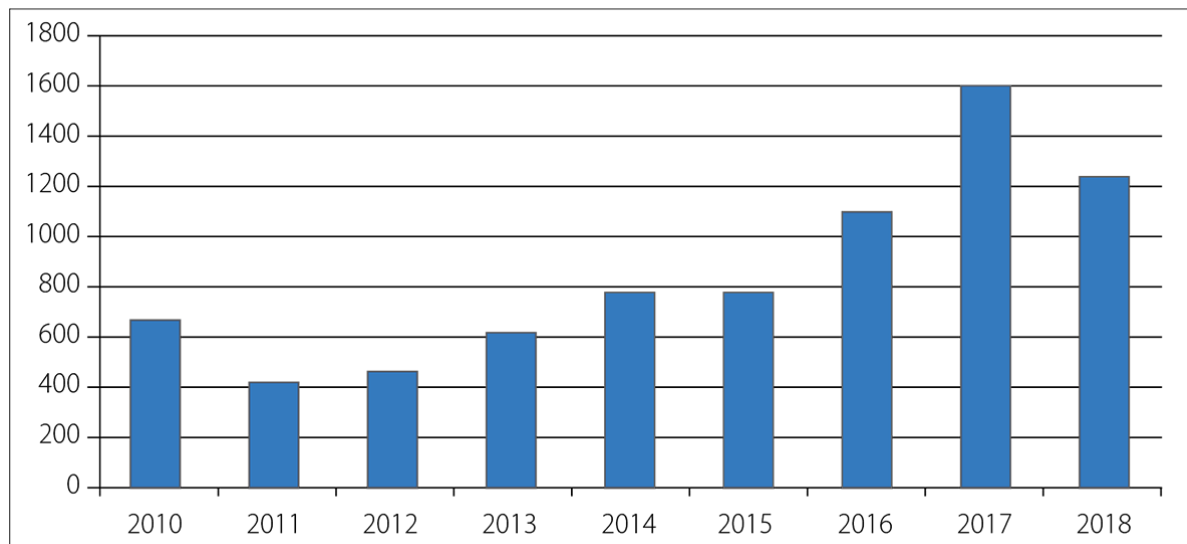
Since the 4IR will be driven by AI, we could argue that data is the main commodity that will be exchanged and developed by businesses all over the world. Devices and objects connected to the IoT therefore have a great deal of value, because they can be used to gather information and improve products and processes over time. It is possible that for every physical product, animal or vehicle that exists, there will be a digital representation or equivalent of that particular thing, which can be used to monitor the status of the object in question, in addition to objects that are similar and/or close to it. For medical researchers, marketers, AI specialists, software developers and a host of other professionals, the data that the IoT will be able to gather, analyse and correlate is invaluable. This is why people who specialise in the structure, functioning and monitoring of the IoT will be of great value to companies: they will determine just how successfully the businesses will perform in both the physical and virtual worlds.

Cybersecurity experts

While this is already a thriving industry, as more products and services are added and developed, it is likely that more people will be required to specialise in the challenging world of cybersecurity. New technologies invite new kinds of cyberterrorists and criminals. With this in mind, there is an enormous need for cybersecurity experts to keep pace with their opponents, as hackers and criminals come up with ever more intricate methods of hijacking businesses and causing anarchy on a global scale. In the same way that an AI can be programmed to keep things safe, it can also be programmed to hack into systems and attack the sensitive infrastructure of businesses and even governments. It will therefore be the responsibility of cybersecurity professionals to develop countermeasures, and anticipate the kinds of attacks that cyberterrorists may perpetrate. As a result of the growing complexity of interconnections and technology, cybersecurity experts will need to be trained in a number of areas and have an understanding of ransomware, malware, viruses, and will probably encounter entirely new types of digital threats.

As Girotra (2018) states, one of the new areas that is susceptible to attack in the 4IR is the IoT. Many new viruses and threats are migrating from traditional computers and laptops to mobile devices and the Internet of things. As new devices are added to the IoT on a daily basis, the number of vulnerable elements also increases. In the near future, practically every aspect of our lives will be connected online, from children's toys to fridges. The potential vulnerability if such a device is hacked or stolen is massive (Girotra, 2018). As mentioned earlier, the main commodity in the 4IR is data, and many companies are responsible for keeping their clients' details safe: think of banks,

insurance companies, Internet service providers and mobile phone companies. If data is leaked or stolen, customers are at risk of losing their assets. In order to understand the significance of cybercrime and why cybersecurity is becoming more and more of a concern, the following graph illustrates how many data breaches have occurred between 2010 and 2018.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Strain, 2018)Figure 1.4:
Number of data breaches, 2010–2018

As the graph reveals, there were 662 data breaches in 2010. The number of breaches then increased to 783 in 2014 and again to 1 091 in 2016. A dramatic increase in the number of data breaches occurred in 2017, with a total of 1 597. While the number of breaches decreased to 1 232 in 2018, more records were breached in each event, so the reality is that the overall damage to the privacy and data of people all over the world increased between 2017 and 2018 (Strain, 2018). Apparently, organisations and governments are not doing enough to prevent data loss and are, at present, losing the war on cyberterrorism. This is exactly why cybersecurity experts with a number of different skills and capabilities will be so necessary in the 4IR. It is unlikely that AI will be able to combat every attack from hackers, and because it is susceptible to hijacking and/or corruption, it will almost certainly be the job of humans to oversee processes and use AI, along with their own intellects, to turn the tide in the fight on cybercrime and keep sensitive data safe, secure and out of the hands of hackers.

Internet governance professionals

This topic has been debated for several years, and is dealt with in different ways by various governments and businesses. The Internet is busy growing this very second, as more content is added by users and stored on servers in locations all over the world. According to *Statista 2* (2018), the number of Internet users has increased from just over one billion people in 2005, to about 3.58 billion in 2017. This means that an average of

213 million new users have joined the Internet annually over this 12-year period. In other words, between 2005 and 2017, about 6.75 people have joined the Internet every second. Assuming that this evidently upward trend will not stop in the near future, by 2020, over four billion people will be connected to the Internet.

The Internet itself is evolving at an unprecedented rate. There are constant discussions about how Internet content should be regulated: while some say that the Internet should be a place that everyone should be allowed to explore freely, others believe that certain types of content and opinions should not be allowed to see the light of day. At present, many countries, including China, Cuba and Saudi Arabia actively limit Internet access or block many websites because they wish to control what their citizens are able to see and learn. In most Western countries, and here in South Africa, access to the Internet is not curtailed, but this could change in the future. Whatever happens in future, companies and governments will need humans to decide how the Internet should be governed, and what constitutes hate speech and anti-competitive behaviour. Furthermore, there will need to be regulations related to the use and access of data collected by websites, especially as the IoT increasingly brings interconnectivity – and therefore the ability to harvest data – into the home. We will discuss the need for this regulation in more detail in the following section.

It is simply impossible for an AI to make decisions about human issues when an AI is, in many respects, part of the issue in question. If humans have no say in how the Internet is governed and managed, then all control over it has effectively been lost. It is possible that AI will be employed to assist in governing the Internet (and is in fact already being used by many countries), but ultimately human Internet governance experts will always be required to steer us toward a better future.

Social media specialists

Social media forms an integral part of many people's lives. In many respects, it has become the most significant tool in shaping human behaviour and influencing the thought patterns of people on a global scale. The fact that so many social media users openly share so much about themselves so openly without any consideration of the potential audiences it reaches, tells us something about the values held by the majority of humans. It also shows that social media users tend to trust large centralised companies such as Facebook and YouTube that their data is safe, secure and not being used against them somehow.

The truth is that data is a major commodity and is highly valued by companies (and criminals) all over the world. We only need consider the Facebook-Cambridge Analytica data scandal that occurred early on in 2018. The British data analytics firm Cambridge Analytica harvested the data of millions of Facebook users without their consent (Martin, 2018). What makes matters worse is that both Facebook and Cambridge

Analytica knew that this had happened for two years before it was made public knowledge. As a result of their clandestine activities, the data of some 50 million Facebook users was collected and used to engineer specific messages, to target American voters in particular. The fact that such a massive event was kept secret suggests that other data hacks and misuse have occurred, and that more will happen in the future.

For this reason, cybersecurity experts will need to work with (and probably be) social media specialists in order to safeguard the data of users. With more functionality and complexity being added to social media platforms, it is safe to assume that people who specialise in social media marketing, content-creation and governance will be needed for the foreseeable future. AI is an integral part of social media, but people need to interact with other human beings in order to feel a sense of authenticity and connection. Social media is based on the interconnection and networking of human beings, along with their institutions, ideas and values. As the number of social media platforms and users grows, companies will need to cater to larger and larger audiences, and respond to their unique needs and tastes in an honest and transparent way, if they wish to remain popular and socially relevant.

Machine learning specialists

This is actually a subset of AI, but as AI itself becomes increasingly complex and varied, the need for people to specialise in machine learning and deep learning is of vital importance. In many regards, it will become more and more difficult to keep up with the rate of change and understand exactly how AI develops. This is because with current advances in machine learning, AI is now able to learn and change on its own (*World Economic Forum* 2, 2018). Machine learning today is far different from what it was in the past: while a lot of machine learning algorithms have been around for quite some time, the overall ability to apply complex mathematical calculations to big data without any manual input, and at an ever-increasing rate, is a relatively new development (SAS *Institute Inc.*, n.d.).

There are several areas where machine learning is used. The following list, courtesy of SAS *Institute Inc.* (n.d.), illustrates some of the more prominent sectors in which machine learning is employed to improve our experiences:

- The Google self-driving car is a product of machine learning and is, at present, the epitome of machine learning as a driverless vehicle is expected to autonomously navigate roads across the world.
- As mentioned in a previous section, online recommendations from companies such as Amazon and Netflix are some examples of machine learning that have already been around for several years.

- The ability to find out exactly what customers, stakeholders and the public at large think of your company or unique offering(s) via Twitter involves machine learning combined with linguistic rule creation.
- The fraud detection employed by many websites and banks is probably one of the more obvious and important areas in which machine learning is used, since it safeguards against criminal and terrorist activity.

In many respects, we have only just begun to unlock the potentials of machine learning. It is possible that planes and helicopters could one day fly themselves, and that many other basic functions, such as regulating temperature and airflow within a building, will be handled by AI and machine learning algorithms. Eventually, many functions, both virtual and physical, could become fully automated and handled by machine learning. Machine learning specialists will therefore be needed by companies to help drive the coming changes and to understand just how machines and AIs are evolving and adapting. Machine learning specialists will need to anticipate any possible anomalies and work to streamline and improve both machine and deep learning, so that optimum levels of progress and development can be achieved now and in the future.

Robotics engineers

Robots are already used in manufacturing and other industries, but several companies are working to create robots that resemble or are exactly like humans in every single way. To get an idea of some of the capacities that robots have at present, a robotics company called Boston Dynamics has developed a robot that can run like a person, and even jump great distances without slowing down or making any errors (Weisberger, 2018). The idea that an artificial humanoid could display such agility and speed is simply mindboggling, but in the future robots will likely be able to do far more than simply run, jump and roll. Another robot – called Sophia – was activated in April 2015, and was granted citizenship in Saudi Arabia in October 2017 (Gohd, 2018). The world's first robot citizen, as Sophia has become famously known, has since appeared publicly and given speeches on women's rights and ethical robot design.

The possible applications for robots, from rescue work and warfare, to surgery and menial work, are enormous. Once robot designs become integrated and put into production, there really is no telling exactly what the future will be like. Again, although robotics engineers already exist, there is a growing need for them, as companies race to perfect designs that will be rolled out internationally. Robotics engineers will need to have knowledge of machine learning, AI, psychology, physics, biology, programming and even nanotechnology. In the future, nanomachines (or 'nanites') may be used to heal and perform surgery on animals and human beings. A nanite is a nanoscopic (smaller than microscopic) machine or device (Rouse, 2011). While the technology is still being developed, companies are working consistently to make nanites and other

nanostructures a reality. Robotics engineering will almost certainly be one of the most important careers in the 4IR, and will be needed by businesses in every industry across the globe. With the rapid progress of robotics in the past few years, future advancements promise to be even more impressive and take the field into all-new territories. Robotics engineers will need to be highly qualified and experienced in various areas in order to secure the best positions in the industry.

3D printers

Although the technology is over three decades old, recent advancements have allowed 3D printing to now become a major industry. In simple terms, 3D printing allows you to build a physical object from a digital model or blueprint (*Ultimaker*, n.d.). There are already a number of companies all over South Africa and the world that are able to print a host of different items and objects for a relatively low fee. Businesses can create a design and then have this design 3D printed in other parts of the country or the world. Today it is even possible to create your own home 3D printer. You can create your own schematic or even download one off the Internet - if you have your own 3D printer, you can print the object in question yourself, and you can print it in nylon, polypropylene or many other materials. If you do not own a 3D printer, or if your printer is too small to print your chosen object, you can mail your design or take it to a company that has the facilities to handle your request.

There are several benefits to 3D printing, especially for engineers, designers and scientists who can create prototypes and physical test models quickly and cheaply (*Ultimaker*, n.d.). These printers can use a variety of different substances, and because they print layer by layer, rather than by cutting resources away, the process can use less material than in conventional manufacturing processes. 3D printers can be used to create specific and customised parts for cars, prosthetic limbs and components for electronic devices– the number of possible applications for 3D printing are seemingly endless. As technology improves, 3D printers will likely become cheaper and more efficient, and so many members of the public may soon be able to print their own models, designs and prototypes in the comfort of their own homes and offices.

Of course, larger and more complex 3D printers will likely be used in manufacturing and the creation of robots, and so designers and engineers who specialise in 3D printing will likely be in demand throughout the 4IR. According to an article by Wild (2018), South Africa has the world's largest 3D printer. It requires about R 7.5 million in titanium powder to fill it. As the article points out, 3D printing is considered 'the future of niche manufacturing', because 3D printing allows complex shapes to be created without wasting any valuable materials in the process (Wild, 2018). As a result of the speed, efficiency and low cost of 3D printing, it will become a more popular method of manufacturing as the 4IR gains momentum. 3D printer specialists will need to have knowledge of current affairs and trends, as well as expertise in engineering, design,

physics, materials and industrial applications. It is also possible that clients will want their parts and designs printed in countries where the exchange rate works in their favour, and where shipping and export duties are negligible. In this way, many developing nations may have a chance to grow their economies because they will be able to 3D print certain items for a lower cost than many of their developed counterparts

Autonomous vehicle experts

Autonomous vehicles are basically any craft that can be used to transport cargo and/or passengers, but instead of having a human driver or pilot, the vehicle is in control of itself. In fact, it is more accurate to say that a special AI runs the show: it is capable of scanning the surrounding environment, interfacing with other autonomous vehicles, planning routes and ensuring that the vehicle and its payload are kept in one piece. The main issue with creating autonomous vehicles is that the level of complexity required to make a vehicle truly safe, trustworthy and efficient is monumental. Field tests are already in progress in research centres and certain cities, but there are still some significant challenges that need to be overcome. Nevertheless, autonomous vehicles could be the answer to heavy traffic jams and congestion, road accidents and fatalities, and high levels of air pollution. Self-driving vehicles could make road travel much simpler and safer, and they could be used in virtually every space in which vehicles exist: from trains to cars and planes to ships, autonomous vehicles could reduce overall running costs and make transportation and freight operations far safer and more effective.

Having said this, new infrastructure will almost certainly need to be installed in cities and roadways across the world, in order to accommodate autonomous vehicles and realise their full potential. This also means that governments and businesses will need to buy into autonomous vehicles and market them. Once these parties are convinced of autonomous vehicles' significance and possible applications, greater investment is almost sure to follow (provided that government budgets make provisions for such 'unprecedented' developments). There may well be significant economic incentives for companies who develop and use autonomous vehicles. In the shipping industry, for example, on smaller vessels, over half of the overall costs are attributed to human crew members who need to be paid for the labour, as well as fed and kept healthy while at sea (Davies, 2018).

As self-driving cars, buses, ships or even planes become more widespread, people who have made their living as drivers and pilots may find that these skills are no longer in demand. However, creating and maintaining the network of AIs, capturing the data required for GPS and navigation systems, and rigorous testing of the AI used for these vehicles will provide new areas of employment in the 4IR. This is especially relevant in terms of safety and decision-making, as people taking these autonomous forms of

transport will want to be assured that it is safe. Engineers, designers and programmers will be vital in ensuring that the system can account for errors and accidents, applying geography, navigation, and machine learning principles to the process. Simplifying and making transport more reliable and secure may also have significant benefits for businesses in the future, ensuring deliveries of resources and the movement of employees and customers is safe, fast and reliable. Autonomous vehicles will be used for everything from transportation to deliveries and tourism to the exploration of hazardous environments. Both large and small businesses will need experts who specialise in this type of technology if they wish to stay buoyant in the future economy.

1.7 Workplace skills needed to flourish in the fourth industrial revolution

With such unprecedented changes taking place, it is clear that a number of important skills and competencies will be required by workers in order to succeed in the 4IR. A white paper created by UBS (a Swiss bank) in 2016 predicts that further polarisation (division into two opposite groups) in the labour force will occur as low-skilled jobs become automated, and it is likely that this trend will extend to middle class jobs as the 4IR gains pace, leading to wider inequality in the future (Elliott, 2016). The new generation therefore faces some significant challenges, since the world population is constantly growing and low-skilled jobs are busy disappearing. This means that employability will become more and more dependent on skills, and that only those workers with a range of abilities and competencies will remain relevant in the 4IR. It must also be noted that this in itself may not be enough – even highly-skilled and well-trained people may miss out simply because the competition is certain to be fierce. In other words, it may not be enough to simply have an undergraduate or honours degree in the future, and only those students who consistently apply themselves and achieve will be considered for mid- to high-level positions. Given the weight of the challenges ahead, what can people do in order to succeed in the new economy?

An article by Gray (2016) lists the following ten most important skills for workers in 2020, as outlined by the World Economic Forum:

1. Complex problem-solving
2. Critical thinking
3. Creativity
4. People management
5. Coordinating with others
6. Emotional intelligence (EQ)
7. Judgement and decision-making

8. Service orientation
9. Negotiation
10. Cognitive flexibility

Since these have been identified as the most important skills, we will spend some time discussing each skill in some detail in the sections that follow. We will also explain exactly why these skills are significant and how company requirements are shifting in response to the development of AI and the introduction of automated systems into industry.

1. *Complex problem-solving*

This is currently the top requirement of companies and will certainly remain in first position. Even with the inclusion of AI and automated systems, complex problems require a great deal of time and money from various stakeholders to be solved. With new ways of doing business and all-new technologies to become familiar with, it is likely that teething problems will occur. Furthermore, any errors or issues with AI or automated systems could potentially cost companies millions, so complex problems will always need to be solved, be it by machines, humans, or both. Issues like climate change, the increasing demands of an ever-growing world population, and the best way to ensure a comfortable life for all are just some examples of issues we will continue to grapple with in future. Complex problems are often impossible to fully resolve. Sometimes, solutions can themselves be the source of new challenges. It is for this very reason that people who can actually create solutions to complex problems will always be sorely needed, and so this particular workplace skill is unlikely to change at any point in the future, regardless of new technologies.

2. *Critical thinking*

In a VUCA world, critical thinkers are needed now more than ever and will almost certainly become even more important in the future. Critical thinking means taking the time to deconstruct ideas and determine their merits and potential shortcomings in an objective and systematic manner. Unfortunately, we are all influenced by our own experiences, memories, worldviews, cultures and opinions, which can make us fail to think objectively and fall into a thinking trap called 'cognitive bias'. For example, we may choose to accept as fact a statement that suits our worldview, what we think about politics, or what we would like the world to be like, without questioning whether the statement is true.

In spite of the challenges, critical thinking is important because it ensures we work against our own thinking limitations and work toward solving problems in a judicious and meaningful way. Without critical thinking, most companies are doomed to failure because they will simply be unable to deal with reality and create benefits that are

sensible and long-lasting. Critical thinking is not only important in the world of business– with fake news, digitally altered pictures and videos, misinformation spread online and via social media and the use of the Internet to infringe on intellectual property, it is more important than ever for people to master critical thinking to evaluate the truth and intent behind the information they see when navigating the 4IR world.

The first and most important factor cannot exist without the second, so each of the skills outlined by the WEF are interrelated and interdependent. This also reveals how skills need to be integrated so that solutions and ideas can be realised on a number of different levels at once. While AI may be able to do much of the thinking for us, we must still evaluate solutions derived by artificial minds and come to our own conclusions – if we fail to do this, we are at risk of abandoning independent critical thinking entirely.

3. Creativity

In 2015, creativity was ranked as only the tenth most important skill by the World Economic Forum. However, in 2020, it will become the third most important skill, as workers will need to be creative and innovative if they wish to stay ahead of their peers and be recognised by their superiors. Perhaps the most exciting (and frightening) aspect of new technology is the ability to create and invent in entirely new ways. People capable of harnessing the power of AI and other inventions will most likely prevail in the 4IR and make a name for themselves in whichever area they decide to specialise in. Creativity and critical thinking go hand in hand with the process of complex problem-solving, because without novel approaches, it is unlikely that any real solutions can be found. In other words, creativity is a core component of the previous two skills.

AI will probably be unable to replicate human creativity anytime soon, and even if it does, it will probably be constrained in certain ways. The beauty of human creativity is that it is only limited to our imagination and the technologies that are available to us. In the realms of music, art, literature and innovation, it does not seem plausible that AI will be able to rival us, at least not for the next few years. If anything, AI and relatively new industries such as 3D printing will only make us more creative, and enable us to accomplish things that would have been near impossible in the past. As Gray (2016) states, the ‘avalanche’ of new technologies, products and services means that workers will need to be highly creative in order to benefit from the changes that are currently underway. Simply knowing about these new technologies and the opportunities and challenges that will arise in the 4IR will better equip us to deal with reality and to realise our creativity. Having said this, acting on what you have learned, and doing what you can to further educate and prepare yourself, is of the utmost importance.

4. People management

There is little doubt that the management of people, be it large networks or small teams, will remain an important part of the skills workers require. When people have problems

or complaints, they usually want them to be personally tended to and taken care of by a helpful human being. Have you ever called a company and been put on hold or had to navigate toward the service you need with your phone's keypad? While it may not agitate everyone, most people do not like wasting time or being made to feel that they are not valued. People management is about managing the expectations of various stakeholders and making them feel valued and cared for. While an AI can basically perform the same function now, it does not feel particularly personal, and so it is almost certain that many consumers feel like businesses do not care about them.

Although AI may be able to assist with managing people in the future (and does to a limited degree already), it cannot do so on its own. Fully understanding the psychology and behaviour of humans is a tricky task, for humans and AI, and this makes people managements skills vital for workers in the 4IR. It is key to the ability to work in teams, meet stakeholder expectations and keep people happy, productive and healthy.

5. *Coordinating with others*

While this skill is lower down on the 2020 list than the 2015 list (Gray, 2016), it is still crucial if workers want to thrive in the 4IR. Our own individual skills and abilities can be bolstered by working with other individuals who have their own skills and competencies. In this way, we are able to compensate for our own weaknesses and achieve something significant through cooperation and collaboration. The 4IR suggests that people will need to work together if they want to stay afloat and truly succeed in the new economy. Social media experts should support and work alongside content creators. Specialists in machine learning should work to educate upcoming talents in the industry. Nanotechnologists should talk to surgeons about the intricacies of their profession. Coordinating work and undertaking tasks in a logical and systematic fashion allows for optimum levels of efficiency to be reached. Being able to prioritise and delegate tasks to certain individuals is of vital importance, especially in a world that is undergoing such incredible change. While it is a good idea to improve your own skills and grow your own competencies, you can do this while working with others as well. Making organisations work is a difficult endeavour, and means that workers need to be self-motivated and willing to encourage those who need it most. Companies that do not keep up with their competitors will be most likely to fail. Coordinating with others means communicating constantly, keeping to deadlines and being able to adapt as and when it is needed.

6. *Emotional intelligence (EQ)*

Since AI will soon be able to outpace us from an intellectual perspective, it is important that we take the time to develop our emotional intelligence and learn to empathise with the people around us. Being emotionally intelligent means that you are able to read into peoples' feelings, and relate to them more personally and meaningfully. Although there

are some AIs that are being programmed to have an emotional component, they cannot feel or experience reality as humans do – at least not yet. An emotionally intelligent person is particularly conscious of their own emotional state, and able to manage their behaviours and reactions accordingly (*Psychology Today*, n.d.).

There are four essential components of *emotional intelligence*, namely *self-awareness*, *self-management*, *social awareness* and *relationship management* (Hinds, 2017). Self-awareness is the foundation upon which all other aspects of EQ rest, and it relates to understanding how you feel about yourself and the range of emotions that you experience. Self-management relates to how well someone can control their emotions and conduct themselves in a considerate and professional manner. Social awareness means being aware of others and understanding how to modify your own behaviour, given the unique setting or situation in which you find yourself. Lastly, relationship management relates to how well you can connect with others, and how effectively you can grow and maintain a relationship with them. Understanding these components of EQ is invaluable in the workplace, and will become increasingly important as the 4IR progresses. This is because fostering relationships with key role players and fellow employees will be crucial in retaining a place in a company.

7. Judgement and decision-making

AI may be able to provide us with some impressive data and even suggest solutions to our problems, but we will still need to make the decisions and determine which solution is the best to implement given the circumstances. The trouble with many companies is that they are indecisive and do not follow one set path for a long enough period of time. While it is important to explore opportunities, eventually businesses have to follow through on their chosen path(s). This skill is directly related to critical thinking: if workers are able to think objectively and carefully about issues, then their ability to make judgements and decisions should be far better than those who are not critical, analytical thinkers. Decisions lead to actions, and it is only through the execution of plans and policies that any business is able to succeed.

Poor judgement and indecision can cause even the best of businesses to fail. For example, the South African insurance company, Momentum, was embroiled in a PR nightmare in late 2018, after they refused a R 2.4 million life claim after a man was shot dead in a hijacking in March 2017 – the reason for their refusal to pay out was because the victim of the hijacking had not disclosed that he had high blood sugar levels prior to his untimely death (Khumalo, 2018). The fallout from this was driven by social media, as policy holders complained about the company's response, threatening to cancel their decision in protest. Although Momentum eventually moved to retroactively pay the victim's widow out, the damage to their reputation had already been done. In the end, humans still need to be discerning and careful in how they approach sensitive ethical

dilemmas. For this and other reasons, judgement and decision-making will certainly be of vital importance throughout the 4IR.

8. *Service orientation*

A lack of good and consistent service can spell the end of even the most promising companies. Workers need to be service-oriented and be ready to respond to the requests of their clients at the drop of a hat. There is almost nothing more frustrating than having something you need urgently delayed or in for repair for an extended period of time. Customers expect good service, and when they pay for service, they expect for it to be provided to them. Workers who keep their customers and peers in mind have a much better chance of surviving in the 4IR than people who lack any motivation or desire to help others. If workers will have to compete with machines in terms of service, it is important that they streamline workflows and ensure that jobs are done with gusto and as expediently as possible.

9. *Negotiation*

Since we are all individuals with our own ideas, beliefs and values, it is basically guaranteed that we will always have to negotiate disagreements and division. Differences and disagreements will not disappear just because technology is improving and progressing. In fact, with the pressures of new technology, compromises will have to be made, and there is a good chance that standoffs between workers and businesses and/or citizens and governments will occur in the 4IR. The art of negotiation will be extremely important in the new economy, as new policies and procedures are debated and decided upon. There is the possibility that AI could be used as a kind of intermediary or arbiter in the near future, but many issues will need to be debated and resolved by human participants. Widespread changes will mean that many issues will need to be taken care of, and so workers will need to negotiate with their employers and peers if they want to make it in the 4IR. It is likely that new technologies will lead to a polarisation in the workforce and society as a whole, and this can lead to conflicts, unrest and perhaps even violence. Negotiation is something that must take place between two or more different viewpoints, and so common ground must be found and petty differences put aside so that meaningful progress can be made.

Negotiation goes beyond resolving conflicts between different parties. It extends to the internal monologue that many people experience on a continual basis. We are constantly being bombarded with new information and challenges, so it is up to us to negotiate the meanings and values of things. This means communicating and interacting with others, thinking carefully and coming to our own conclusions, using our own experiences and principles as a primary point of reference. Dialogues and negotiation will be common occurrences now and throughout the 4IR, so negotiation skills are absolutely necessary for workers to acquire and hone.

10. Cognitive flexibility

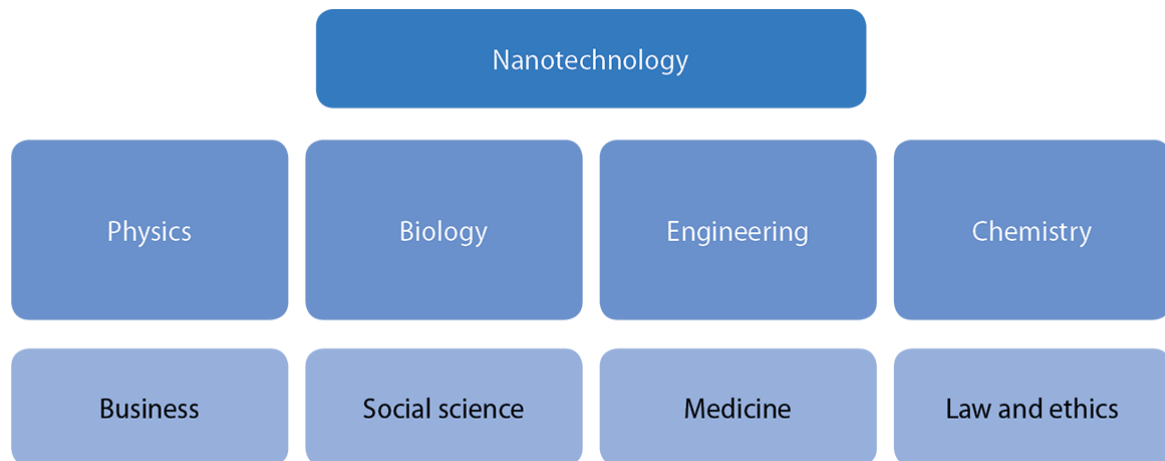
With the complexity of the future, workers will need to be able to think in novel ways and change their strategies if they find that their particular method is not playing out as they hoped it would. Lateral thinking, improvisational skills and the ability to adapt are all part of cognitive flexibility. In certain situations, it may be necessary for workers to think on their feet and come up with solutions within a few seconds or minutes. This may sound impossible to many, but with practice and exposure, the ability to improvise and adapt can increase significantly. Learning constantly, acquiring new skills, listening carefully and being able to empathise and see things from others' points of view are all also major components of cognitive flexibility. Cognitive flexibility involves switching between different tasks at any given time and being able to think about multiple concepts or solutions at once. Workers in the 4IR will probably have to fulfil a number of crucial roles and switch between them seamlessly. While it may only occupy the tenth place on the list, it is by no means an unimportant skill and could well provide workers with the edge they need in order to succeed.

While these skills are indeed critical, they are not the only skills workers should possess. These ten skills are just a guideline and should help you to make better decisions and focus on those areas you feel need the most attention. It is also possible that this list of skills will change in the coming years, so what may be important today may be less so in the near future. Whatever the case, skills acquisition and development is vital for workers across the planet.

FAST FORWARD: Jobs of the future

Nanotechnologists

Nanotechnology, which involves working on machines or machine outputs a thousand times smaller than a millimetre, has been in development for some time and is already in use in some fields. For example, billions of tiny 'nanowhiskers' have been hooked to both natural and synthetic fibres on a molecular scale, in order to make clothing and other fabrics stain-resistant (Picraux, 2018). Similarly, nanoscopic zinc oxide crystals have been used to make invisible sunscreens that block out harmful UV rays and prevent skin cancer. These are just a few of the current applications of nanotechnology, and the future holds great promise for the industry, as new advancements are being made on a regular basis. With this in mind, what skills and abilities does a person need to have if they want to become an expert in the field of nanotechnology? The following diagram shows both the primary and secondary areas of expertise any aspiring nanotechnologist will need to possess.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Dublin Institute of Technology, 2010) Figure 1.5: The primary and secondary skills required by nanotechnologists

As the table reveals, nanotechnologists need to have knowledge of physics, biology, engineering and chemistry. While these primary areas are certainly important, we could also add aspects like robotics, artificial intelligence and programming to the list of skills that overlap with the field. In terms of secondary areas of specialisation, DIT lists business, social science, medicine and law and ethics, since nanotechnology promises to affect the lives of people in a number of key areas. So, what industries will need nanotechnologists now and in the future? According to *National Nanotechnology Coordinated Infrastructure* (n.d.), nanotechnology is needed in many areas, such as electronics/semiconductor development, material science, sporting goods, biotechnology, computing, agriculture, forensics, geosciences, military and national security, medicine and pharmaceuticals.

Being able to create and manipulate things on a nanoscale means that materials can be altered and improved upon at the most fundamental level currently possible. In terms of semiconductor manufacturing, nanotechnology means that processing power is likely to improve at an exponential rate over the next few years (Picraux, 2018). Nanocoatings may be able to protect and preserve materials at a molecular level and keep them from being scratched or perishing. Nanotechnology may well be applied in solar panels to make them more efficient and ensure that the maximum possible energy output occurs on a constant basis. Many of the proposed designs and ideas that exist now will almost certainly be realised as the 4IR grows and evolves, and millions of dollars are already being invested in nanotechnology companies (*Nanalyze*, 2018). Some examples include (*Nanalyze*, 2018):

- *DigiLens Inc*: Developing nanomaterials for heads-up-displays (HUDs) and augmented reality displays; for example, to highlight cars and road signs to aid drivers.

- *Abionic*: Developing a biomedical screening device which performs the world's fastest test for sepsis (a deadly illness as a response to an infection), requiring only a drop of blood.
- *BlueWillow Biologics*: Developing treatments of respiratory and sexually transmitted diseases (STDs) that are simple and less invasive.

These companies are just a sample of start-ups and entrepreneurial ventures that are now getting their products off the ground. Along with AI and robotics, nanotechnology promises to bring many new opportunities to workers around the world. Materials that are lighter, stronger and more durable could make the construction of houses and other structures far more cost-effective and less time-consuming. New developments in medicine could make remedies more effective and less troublesome in terms of side-effects. The number of possible applications is astounding, and many of the current developments in the industry are already impressive enough. There is little doubt that with further technological advancement and investment, nanotechnology will become an ever more important part of the 4IR economy. Whether it is waste management or surgery, military-grade armour or HUDs, nanotechnology will likely come to form some part of virtually every aspect of our lives. Nanotechnology is certainly an astounding job of the future.

1.8 Summary

Whether we like it or not, the fourth industrial revolution is here, and it will bring a variety of changes. While there will be many new career opportunities for us, it is almost certain that many unskilled jobs (and even several skilled ones) will disappear in the coming years. We face a future with amazing opportunities as well as worrying volatility, uncertainty and risk. This unit has explored the historical developments that led to the 4IR, and showed some of the core ways in which the new technologies being pioneered are likely to affect global leadership and the workplace. From artificial intelligence to automated factories, self-driving cars, and 3D printing, the workplace of the future will be highly supported by sophisticated machines.

Because of this, the workplace of the future will require new skills to handle the coming uncertainties, challenges and opportunities provided by technology. We have looked at the core skills like creativity, critical thinking and problem solving which will define the role of humans in the workplace alongside artificial intelligences and in new fields enabled by new technologies.

Chapter 2

2.1 Introduction

Thabang works for a small South African company, where he is well known for his ability to communicate with people in his team and assist in resolving any disputes that arise. Thabang has always enjoyed working with computers and has been teaching himself about programming in his spare time. Although Thabang has some useful skills and abilities, he does not occupy a senior position within the organisation yet. A few months ago, the company was bought by an overseas organisation that decided to implement a new artificial intelligence (AI) solution to make the company more profitable. Unfortunately, the company can no longer afford to keep all of their staff members because their expenses are simply too high, even with the new AI solution, and so several employees are going to be retrenched. However, as a result of Thabang's unique knowledge and people skills, he is excluded from the retrenchment process and remains with the company. In fact, his knowledge of programming and his people skills mean he is promoted to a new position where he will help to upskill his fellow workers, and oversee the implementation of an AI that is built specifically for the South African context.

Think about all the ways in which South Africa has developed to use technology, from cell phones and smartphones, to satellite TV and fibre optic Internet. Emails, social media, and instant messaging help us to connect in our personal and professional lives. As the fourth industrial revolution (4IR) progresses, the unique history and context of South Africa, and other Southern African countries, presents unique challenges to implementing new technologies. This unit will ask questions about the potential opportunities and challenges that Africa, and especially South Africa, will encounter going into the high-technology future. Africa is a continent with rich resources that could be deployed to overcome some of the challenges facing the continent already. However, some of these challenges pose risks to the development of an African 4IR:

- Many African countries face low levels of education, with *UNESCO Institute of Statistics* (n.d.) reporting that sub-Saharan Africa has the highest rates of education exclusion. Over 20 per cent of children between the ages of six and 11 years old, 33 per cent of adolescents between 12 and 14 years old, and 60 per cent of teenagers between 15 and 17 are not in school in the region.
- Unemployment is also very high in many African countries. Although the average unemployment rate in sub-Saharan Africa has decreased slightly, from 7.7 per cent to 7.3 per cent between 1991 and 2017, in some countries it has worsened;

for example, in South Africa the unemployment rate has grown from 20.1 per cent to 27.3 per cent over this period (*The World Bank* 1, 2018).

As the first unit in this book highlighted, the 4IR will see many jobs becoming automated and directed by AI, and so this will almost certainly have a negative effect on employment levels in Africa and abroad. There are many potential benefits that new technology could bring to South Africa, but with our already high unemployment rates, low education rates and weakened economy, there are many obstacles for the country to overcome in order to make good use of 4IR technology. An article by Khumalo (2018) cites the poor domestic job market, high public debt and deteriorating finances as being the main culprits behind South Africa's low levels of economic growth. The trouble is that ratings agencies such as S&P Global have already downgraded the country's credit rating to junk status, effectively making it a high-risk option for many foreign investors (*Business Tech* 1, 2018). While this may make the future seem grim, if major reforms are pursued by the government, and education, job creation and employment become a priority, it is very possible that South Africa could be a champion in the 4IR economy. As we will see in this unit, there are already South African businesses pioneering AI solutions.

Regardless of the country's ultimate destination, it is important for Africans and South Africans to prepare for the monumental changes that the 4IR is sure to bring. By taking the necessary steps to familiarise yourself with the possible consequences of the 4IR, you can equip yourself with the knowledge required to succeed in an economy in a time of rapid change and technological development.

After studying this unit, you should be able to:

- argue for South Africa's need for a learning revolution, in order to thrive in the fourth industrial revolution;
- identify the potential risks and benefits of the fourth industrial revolution for Africa, and discuss how these risks and benefits apply to South Africa in particular; and
- predict the impact of the fourth industrial revolution on the world of work in South Africa.

2.2 Key terms and definitions

The following key terms and definitions are applicable to this unit:

- *Automation*: A situation where certain work processes are handled in whole or part by AIs and/or robots.

- *Futurist*: Someone who examines current trends to make determinations about the future.
- *Inequality*: A situation in which wealth and income is unequally distributed within a certain country.
- *Innovation*: Something that is highly unique as well as feasible, viable and desirable.
- *Premature deindustrialisation*: Migrating to a service-based economy, rather than an economy driven by factories and goods-production.
- *Triple threat*: Inequality, unemployment and poverty are considered South Africa's 'triple threat'.
- *UBI*: A universal basic income (UBI) is a small salary paid by the government to all citizens of a country.
- *SCARF*: An acronym that stands for status, certainty, autonomy, relatedness and fairness.
- *Soft skills*: Emotional and social skills that allow someone to connect meaningfully with others.
- *STEM*: Science, technology, engineering and mathematics (STEM) are critical in the 4IR economy.

• **2.3 Risks and benefits of the fourth industrial revolution (4IR): African and South African impacts**

Since the 4IR is still in its early stages of development, it is necessary to make some predictions about the possible effects it will have on Africa and, more specifically, South Africa. With a population of around 1.3 billion people, Africa is home to about 16.6 per cent of the planet's inhabitants (*Worldometers* 1, 2019). Africa is a resource-rich continent, but the vast majority of this wealth does not go to benefitting the people of the continent. Much of the wealth of the continent goes to foreign lenders and multinational corporations, but the outflow of wealth is not only through legal trade – around \$ 6.8 billion leaves the continent due to fraud, corruption or illegal dealings (Curtis and Jones, 2017: 4). Unfortunately, corrupt leaders, a lack of infrastructure and education, and various other factors are also part of the problem, but the underlying issue is that much of Africa's wealth does not remain in Africa, and of the portion that does, very little finds its way into the hands of the masses. For this reason, many Africans are unable to meet their basic needs, and cannot access many of the opportunities that people in the developed and developing worlds take for granted. As a

result, many African people remain disempowered, relegated to unskilled labour and hard physical work. This brings us to another important aspect that needs to be considered when discussing Africa, namely *inequality*. The following table shows the ten most unequal countries in terms of income distribution for their most recent GINI coefficient statistics. This statistic measures the inequality between the richest and the poorest in a country, giving a sense of the gap between the two on a scale of zero to one, or a percentage (zero per cent to 100 per cent). The closer to 100 per cent in the following table, the greater the inequality between the richest and poorest. It is important to note that these figures come from various surveys at different times, some from 2015, but others can come from much earlier.

Country	GINI coefficient (as a percentage)
South Africa	63
Namibia	61
Botswana	61
Zambia	57
Central African Republic (CAR)	56
Lesotho	54
Eswatini (formerly Swaziland)	52
Brazil	51
Colombia	51
Panama	51

(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from World Bank in Beaubien, 2018)

Table 2.1: The ten countries with the highest levels of income distribution inequality as per their latest statistics

While the top ten most unequal countries change almost every year, the top three, namely South Africa, Namibia and Botswana, have remained in their respective

positions for many years. Alarming, South Africa is the most economically unequal country on the planet, and if anything, inequality has become worse since South Africa became a democracy in 1994 (Head, 2018). The situation in South Africa is undoubtedly a dire one, and so reforms are required at a number of levels so that poverty, unemployment and inequality can be effectively reduced. The *World Bank 2* (2018) lists the following factors necessary in order to eradicate inequality:

- *Skills:* South Africa's inequality has, in recent years, been increasingly driven by developments in the labour market that require skills that the country's poor presently lack, and so it is vital to ensure unskilled workers have access to training.
- *Competition:* At present, there is a lack of competition in South Africa's economy, with many large corporations holding a monopoly or oligopoly (control by a wealthy elite) over certain industries and economic activities. Therefore, promoting competition and small business development can reduce inequality, unemployment and impoverishment.
- *Policy:* Policy uncertainty – or the lack of clarity among businesses, investors and citizens around government's plans for the future – drives fear and anxiety around socio-economic and political issues. Government should provide more assurance and guarantees around economic and political policy to reassure citizens, businesses and investors.
- *Migration:* While it is vital to empower locals, attracting skilled migrants from other countries through incentives can bring South Africa key skills and knowledge to catalyse economic growth and business development, creating jobs for locals and migrants. The country could certainly do more to attract skilled migrants who are catalysts for economic growth and development.
- *Spatial integration and public transport:* Spatial integration (improving connectivity between different spaces or places) through public transport development will aid poorer South Africans in getting to and from work, reducing travel costs, enabling them to take new employment opportunities and improving quality of life.

The *World Bank 2* (2018) also points out that, although progress is slow, there have been some improvements in education for the poor, who are beginning to access new opportunities and gain a greater share of the wealth that is afforded to more skilled workers. However, with the advent of the 4IR, this progress may be far too little. With high levels of poverty, unemployment and inequality, South Africa needs to take decisive actions quickly and close the numerous gaps that prevent it from becoming truly prosperous for all.

Although South Africa is at the top of the list of unequal countries, there are several African and South American countries with significantly high levels of inequality that also need to take steps to address their issues if they wish to participate and be a driving force in the 4IR. It must also be added that while South Africa may be unprepared for the 4IR in many respects, there are other African countries that are arguably even worse off. For instance, conflicts in places like the Democratic Republic of Congo (DRC), South Sudan and Burundi have had a negative impact on economic growth and development in these countries (Bhalla, 2018). While inequality, poverty, conflict and other issues are major issues plaguing African countries, there are other factors that have to be considered. According to futurist and business strategist Craig Wing, most of Africa is still in the agrarian age (in Tredger, 2018). This means that most of Africa is still largely under-developed, with most of its inhabitants engaging in some kind of agricultural activity in order to survive. Consequently, there is a growing need for an education revolution here in South Africa, and the rest of Africa. Dyantyi (2018) states that a technological revolution is coming and that South Africans need to take steps in order to prepare for it. The only way that this is going to happen is if more people are exposed to computer technologies and applications, and are allowed to familiarise themselves with everything from basic data processing to programming and the intricacies of AI. Sadly, with such widespread inequality, unemployment and poverty, many South Africans are unable to learn about the fundamentals of technology because they cannot afford or make use of such technology. It must also be noted that many African countries, such as Ethiopia, Uganda and Tanzania, restrict the ways in which their citizens can use the Internet and technology, significantly reducing their opportunities for work, education and critical thinking (Matfess and Smith, 2018). However, with an increase in smartphone usage and affordable data plans, 2018 saw a 20 per cent increase in the number of African Internet users (Mumbere, 2018).

The opportunities and risks for the 4IR

What then of the possible benefits and risks of the 4IR, in light of Africa and South Africa's respective (and interlinked) situations? Since Africa is a continent with 56 countries, it is short-sighted to believe that any scenario will play out in the same way in each one. There are so many different factors that can be considered, and to list them all may take a textbook of its own. Although South Africa has many challenges to overcome, there are people who truly believe that the 4IR will drive economic growth in the country and bring about a new age of prosperity for all South Africans.

One opportunity relates to African youth (Dagada, 2018). Compared to many other continents, the median age (the middlemost age between the oldest and youngest people) in most African counties falls between 15 and 21, whereas in developed nations like Canada and much of Europe, the median age is between 40 and 52 (Allison, 2017). Since African youth can be expected to work for longer, they will benefit from skills and

training going into the 4IR, and it is likely that, as other populations shrink, businesses will open African offices or factories with local employees (Dagada, 2018). However, as we have seen in the previous unit, there is equally a risk that manufacturing and low- to semi-skilled jobs may be replaced by automation, limiting the extent of this opportunity to fields of certain kinds of unskilled and skilled work not easily handled by machines and AI.

Another advantage Africa has is its large amount of arable (farmable) land, around half the arable land worldwide (Dagada, 2018). While Wing marks Africa's agrarian economies as a limitation (in Tredger, 2018), the growing demand for food around the world may allow another key route for African development. As of 2019, there are approximately 7.68 billion people on planet Earth (*Worldometers 2*, 2019) and the world's population will likely reach nine billion people around 2037, ten billion by 2055 and 11 billion by 2088 (*Worldometers 2*, 2019). It is clear that resources will become scarcer and scarcer as the population continues to grow on a constant basis, and by 2050, the demand for food may be 60 per cent higher than in 2018 (Hincks, 2018). In addition to this, factors such as climate change, water shortage, urbanisation and the degradation of soil quality will shrink the amount of available arable land that can be used in food production. In other words, if decisive action is not taken now, the price of food will simply be too high for the majority of humans. Since Africa is home to the greatest proportion of arable land on the planet, investment in and optimisation of African farming ventures is virtually guaranteed. The main issue with smaller African farms and subsistence farmers is that they are not technologically developed enough, and do not produce high enough yields to make them commercially viable. This is why direct investment and socio-economic development of such farms is not only necessary for the well-being and survival of all of Africa, but also for the rest of the world.

While agricultural production in Africa will remain critical, it is only one broad area that the 4IR will have some kind of (hopefully positive) impact on. It is difficult to predict exactly how the 4IR will affect all of Africa, but again, since the continent is comprised of different nations with different problems, cultures and values, there will inevitably be winners and losers. Knott-Craig (2018) lists thirteen ways in which the 4IR is likely to affect South Africa in particular:

- *People will behave better:* As more devices are connected to the Internet of things (IoT) and more ways to be connected are opened online, we are becoming part of a 'mega-society', which in some ways is like a small town where everyone knows everyone else and what they are doing and saying. This connectivity can encourage tolerant and better behaviour, but comes with a risk of invasions of privacy.

- *We will connect faster:* With the number of Internet users in South Africa increasing on a daily basis, and with Internet connections becoming faster and cheaper, there is little doubt that the 4IR will enable us to connect and network with others more efficiently and intimately. With the development of social media platforms like Instagram, Twitter and WhatsApp, we are already able to do business and socialise with one another much more easily than we were able to before.
- *Education for everyone everywhere:* One of the major benefits of such interconnection and the development of technology is that vast quantities of information are available for little to no cost. Additionally, mobile devices and cloud computing allow people to access this information in novel ways from virtually any area with connectivity. Education will be less limited by geography or status, and so most Africans will be able to participate in the digital world regardless of where or who they are.
- *Greater safety and security:* New technologies and more extensive surveillance mean that more communities will be safer, since it will become more and more difficult to commit crimes without being identified and apprehended. Of course, surveillance and security will only apply in cities and towns for the foreseeable future, but this may become more prominent in rural areas down the line. There are risks, however, that authoritarian or oppressive governments could use this surveillance against ordinary citizens.
- *A more democratic society:* New technologies can allow for different kinds of democratic participation for more people. For example, consider the ways in which the Internet has facilitated debates and discussion around events in South Africa, and around the world. From hashtag protests like #RhodesMustFall or #MeToo, to the online scrutiny of politicians, the 4IR is already enabling ordinary citizens to share their views and become better informed. Knott-Craig (2018) cites the 2017 ANC National Congress where Jacob Zuma lost to Cyril Ramaphosa as an example of the power of the Internet and information sharing to strengthen democracy.
- *A healthier population:* Not every South African has access to a doctor or clinic to receive good medical care. Many South Africans who can afford to do so already wear devices that can track their heart rate, health and sleeping patterns. In the future, we may all have access to devices that can monitor our health and help doctors or nurses to ensure we receive the care we need, diagnose illnesses faster, and ensure treatments are more successful. This could especially help those in remote areas who may not have easy access to a GP.

- *Smarter, better travel:* Many of South Africa's cities have major infrastructural issues. These range from traffic jams and congestion to high numbers of car accidents. Motor vehicle accidents in South Africa have killed over 135 000 people since 2008, about 13 500 people every year (Herald Live, 2018). This figure is unacceptably high and costs the government billions every single year. Self-driving vehicles will be able to monitor the external environment and abide by the rules of the road at all times, eradicating human error and preventing many accidents from occurring.
- *Less wasteful living:* 4IR technology will allow us to be far more efficient and resource-conscious. For example, an AI will be able to control devices within your home and turn them on only when they are required. If there are any leaks or electrical issues, the AI will alert you and advise you to take action before a major event takes place. Items that are past their sell-by date will be disposed of, and your refrigerator will order only the items that you need.
- *International shopping:* As mentioned in the previous point, many of our devices will simply order things for us when we need them, and with algorithms that analyse buying habits, retailers will be able to make accurate predictions about what we will buy in the near future; for example, identifying whether a family is expecting a baby and connecting them to useful items, from eco-friendly nappies to children's toys or clothes, that can be bought online from anywhere in the world.
- *A distribution revolution:* Thanks to the development of both electric alternatives and biofuels, 4IR technology such as drones will be able to cross great distances for a fraction of what it costs today. It will not matter whether the item you want is from Dubai or London, you will be able to order anything you want from virtually anywhere, at a much more pleasing and affordable price.
- *Participating in a sharing economy:* The IoT will allow us to share just about anything we want with others, and decide what we want in return. Whether it is a power tool or bicycle, the IoT will allow us to track rented items in real time, thereby making people more honest and forthcoming when it comes to returning the things they have rented. A society that is able to share items makes economic sense since items are only acquired on a needs basis, rather than bought and left unused for extended periods of time. From bicycle-sharing in many countries, to Airbnb and MySmartFarm (an app for sharing information with small-scale farmers), there are already examples of this form of economy tapping into South African ideas of ubuntu.
- *Access to finance:* Smartphones and similar devices are bringing about new ways for us to make exchanges without having any physical cash on hand, and

tailoring solutions for people with limited access to traditional banking. Cryptocurrencies (specific online currencies not tied to banks or centralised institutions) may be used one day to help create fast and secure global trade outside the control of banks and corporations. In the near future, street vendors will be able to accept secure payments without having to worry about their earnings being stolen from them, as it becomes easier for everyone to trade in a cashless economy. In essence, transactions will be safer, faster and easier to track and manage.

- *Inflation will fall:* 4IR technology is removing friction when it comes to making financial transactions (such as bank charges), and making people more accountable by reducing fraud and corruption. It is also lowering marginal costs significantly, since virtual courses and products can basically be shared with thousands or even millions of people. Renewable energy is making the new economy more sustainable, and so our reliance on fossil fuels should decrease in the coming years. This means that we will waste less, and so prices on food and energy should drop as a result.

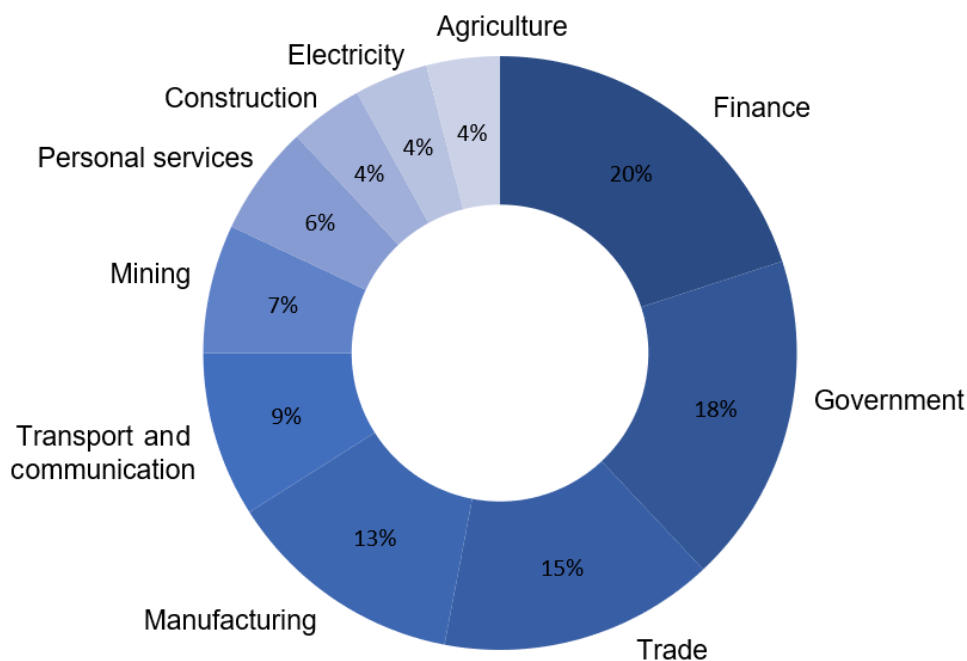
Based on Knott-Craig's (2018) appraisal, it seems that the 4IR will have a number of different impacts on South Africa and Africa as a whole. His outlook seems generally positive, but there are risks for the continent that need to be taken into account. *SAIIA* (n.d.) points out several risks for Africa and South Africa, and suggests that the 4IR may be detrimental if not properly instituted and managed. Firstly, many African countries moved away from manufacturing activities and into the provision of services long before more developed countries did. This is what Dani Rodrik calls 'premature deindustrialisation' (in *SAIIA*, n.d.). While developed nations have seen a decrease in the share that manufacturing businesses have in their economies for quite some time, the same has been happening in developing nations. This is unexpected, since manufacturing is still seen as one of the main ways in which to modernise and absorb unskilled workers into the economy. Unfortunately, many factories that were set up in African nations after they gained independence are deteriorating rapidly. For this reason, joblessness may increase in Africa as the population continues to grow at a substantial rate. The next section will delve deeper into the future of work in South Africa, and will analyse the various benefits and risks in greater detail.

2.4 The impact of the fourth industrial revolution (4IR) on the future of work in South Africa

From the previous unit, we can see that the 4IR will affect businesses all over the world, including South Africa. Technological advancement generally tends to disrupt societies and cultures because it fundamentally alters the way that things are done. For example,

smartphones and wireless Internet have allowed us to communicate with one another via voice and text for little to no cost, and so the way we interact and handle transactions has fundamentally changed.

In order to better understand the possible impacts of the 4IR on the South African economy and workforce, it is necessary to get an idea of what each sector contributes to the total gross domestic product (GDP) of the country. The following graph illustrates the percentage contribution of each major economic sector to South Africa's GDP for the second quarter of 2018.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Statistics South Africa, 2018) Figure 2.1: Percentage contribution of each major economic sector to South Africa's GDP, Q2 of 2018

This figure reveals that the finance sector is the largest contributor to South Africa's GDP, constituting about one-fifth of the country's GDP for this period. Government comes in second place with 18 per cent, while trade comes in third, accounting for 15 per cent of South Africa's total GDP. The percentages provided in this figure also reveal that manufacturing is one of the largest sectors in South Africa's economy, contributing 13 per cent for the second quarter of 2018. Since manufacturing is a significantly large sector, it follows that numerous South Africans work within this particular industry. Automation will likely bring about disruptive changes in manufacturing, and therefore

many workers stand to lose their jobs as the 4IR unfolds. Other sectors will be affected by automation as well, most notably mining and agriculture, so the coming changes to work processes will have a major impact on the livelihoods and futures of many South African citizens.

According to data from the *Department of Trade and Industry* (n.d.), South Africa is ranked between 46th and 75th place in the world, using various metrics, in terms of its overall 'readiness' for the 4IR. In other words, while South Africa is hardly the worst country with regard to preparedness, it is still far from the best. The *Department of Trade and Industry* (n.d.) also highlights some of the major threats the 4IR brings to the country:

- In 2019, South Africa's Internet speeds are not particularly high. This seems to be a major problem throughout most of Africa, and it must be noted that South Africa offers the highest Internet speeds in Africa at 9.93 Mbps (megabits per second) (Ihekwoaba, 2018). However, this is 45.2 Mbps behind Singapore, the world leader in terms of broadband Internet connectivity. It may not seem like a major issue at present, but faster Internet connectivity allows for greater participation in the digital economy. For example, in order to hold video conferences and perform other data-heavy tasks, a high-speed connection is required. The faster data can be transmitted, the more efficiently and effectively work can be done. For this reason, South Africa lags behind many countries in the world, and so many of our citizens are excluded from more data-intensive types of work because they simply do not have the Internet speeds that people from other countries do. If South Africa wants to be taken seriously in the 4IR and open up doorways for its people, it must offer higher Internet speeds as soon as possible.
- While it has been mentioned before, it cannot be overemphasised that the current school system is not preparing the majority of students for the future of work. South Africa needs more skilled labourers and must move away from unskilled manual labour. The education system needs to be overhauled so that the necessary changes can be made and the syllabus updated to better meet the requirements of the 4IR. All schools need to meet the basic needs of their students. More technical subjects should ideally be made part of the syllabus, and basic skills relevant to the digital sphere should be included in the everyday lives of students at primary and secondary levels. Programming, coding, digital literacy and a knowledge of big data and the IoT should be more prominent, as knowledge of these will be vital throughout the duration of the 4IR.
- When it comes to primary (extraction), secondary (manufacturing) and tertiary industries (services), these distinctions will likely fall away as 4IR industries will blur the lines between them more and more. In addition to reducing the number

of workers required in industry, industries will become more 'integrated' as numerous activities fall under one umbrella. In many ways the 4IR is about companies becoming more self-sufficient and less reliant on suppliers, consultants and middlemen, so the risks to the current economy are indeed significant and will require addressing from all parties involved in the conversation.

These concerns are some of the more pressing ones that need to be considered sooner rather than later. However, the *Department of Trade and Industry* (n.d.) also mentions the following risks:

- Certain businesses and industrial processes may be 'reshored' – i.e. they may be moved to foreign countries that are more prepared for the 4IR. Global value chains are shifting, and they will move to those territories that have made provisions for the 4IR. If South Africa does not step up and provide adequate infrastructure that is conducive to new industry, it may be left behind.
- The 4IR may indeed provide opportunities for a select few highly skilled individuals, but it seems that inequality and exclusion is set to increase, making the poor more dependent on the government and charitable organisations for their survival. An increase in inequality could increase levels of crime and lead to civil unrest on a massive scale, but this all depends on how government responds in the situation.
- One of the major threats of the 4IR is that it may create a 'winner-takes-all' economy where one or a handful of companies dominate production and create high barriers to entry. In other words, small businesses and entrepreneurs will have to be well-positioned in order to participate in the 4IR, and any that are underfunded or simply lack the knowledge and resources may simply be left out.
- The 4IR will have a major impact on virtually all spheres of society, so without regulations and policies in place to cater for the disruptive changes it will bring, there is no telling what scenario will play out.

As technology has improved worldwide, the need for skilled workers has increased while the need for unskilled workers has decreased. This is because many industries have become highly mechanised, and so many processes are handled by machines rather than human beings. As we saw previously, the 4IR will likely reduce the number of people in the workplace in both unskilled and semi-skilled roles. Considering that many African countries have a substantial number of uneducated people who hold no special qualifications, the 4IR could be disastrous for many African workers. Estimates suggest that South Africa could lose 4.5 million jobs to automation (Daniel, 2018). We have discussed the fact that machines can often do tasks more efficiently than workers. While the initial costs are high, many businesses turn to machines which do not strike

or get sick, and can perform at a faster rate for less monthly expense in order to keep up globally. This is especially true for mining, manufacturing and agriculture, three sectors that account for most of the jobs in South Africa (Daniel, 2018). Daniel reports that major mining operations, such as Implats Distinctly Platinum, Lonmin and Gold Fields, need to retrench thousands of employees to turn a profit (Daniel, 2018). It is clear that automation is already threatening job security in some of South Africa's core sectors.

A key balance on behalf of workers in South Africa is the existence of trade unions, and just as the 4IR will influence the workplace, it will also impact the power and relevancy of these unions (Kaggwa, 2018). Kaggwa notes that each industrial revolution brought significant improvements for a select group of people, but also increased the gap between the rich and the poor. With such high levels of inequality in South Africa and other African countries, it is difficult to argue with this view. From his estimation, which matches some of the views in the previous unit, the 4IR may only serve to increase the space between the wealthy and the poor. Trade unions have used protests to fight job loss and low wages, such as the significant protests by the South African Federation of Trade Unions in May 2018 over minimum wages that remained too low to live on (*Times Live*, 2018). However, these protests may also be a further incentive for companies to automate; Daniel argues that a machine cannot down tools and strike, and that the frequent disruptions of protests by trade unions reduce the profitability of South African companies, encouraging them to automate (Daniel, 2018).

Former secretary general of the Union of Mineworkers, Frans Baleni (2018), quotes Dr John Tibane: 'if you don't manage change, change will mismanage you'. This statement applies directly to South Africa in terms of the 4IR. Trade unions, businesses and governments need to collaborate and work to facilitate change, but since there are conflicts of interest between trade unions and employers, and because automation seems inevitable (and even necessary), negotiations may be difficult. As a former mine worker himself, Baleni also points out that mining and other forms of unskilled labour are often physically strenuous (Baleni, 2018). No worker enjoys hard labour, especially constant and excessive hard labour which can lead to future health problems. If there is an easier or more efficient way to do things, workers will almost certainly employ these methods to lighten their loads. However, if an easier way to do things removes the worker completely, this is not desirable either. This has been the primary characteristic of all of the industrial revolutions, namely that jobs which once belonged to humans or animals begin to be performed by machines. Of course, industrial revolutions have also created jobs or reframed what existing jobs entail, but as we have advanced as a society, so too have the requirements for economic participation become more stringent and exclusive. As the 4IR unfolds, solutions to the challenge of ensuring fairness and participation in a new economy need to be created, and quickly. With this in mind, let us consider what strategies are being put in place to position South Africa for the 4IR.

Campbell (2017) states that South Africa is not prepared for the 4IR, but he also states that it is not badly underprepared. He indicates that some of the main weaknesses that the country faces include rising costs of production, relatively poor levels of productivity, a lack of specialised and highly skilled workers, and ambiguous or uncertain government policy. However, in November 2018, a multisectoral discussion was held which may have the potential to move South Africa toward a policy plan with regard to the 4IR (Musgrave, 2018). Around 40 people – who represented the government, universities and public research institutions – gathered at BCX, an information and communications technology (ICT) company in Pretoria, to discuss a way forward for South Africa in the 4IR. Although this gathering saw some significant progress, this is not the first time such a gathering has taken place (Musgrave, 2018). In addition, while many key stakeholders were present at the meeting, there is no mention of any trade union members or any other persons who may represent the interests of the working class and unskilled labourers. Perhaps the only person at the meeting who may have represented the wills and concerns of workers was the representative for the Department of Trade and Industry. For this reason, we must question just how committed government and other stakeholders are to dealing with the coming job losses and including such people in the conversation. It is definitely good that discussions and deliberations about the 4IR are taking place, but without equal and judicious representation, the goals and policies outlined at such meetings may be null and void. Time is quickly running out for such roundtable discussions and deliberations to be held, and for solutions to be found. Universal inclusion will be necessary if government wishes to circumvent disaster and alleviate the fears and concerns of the working class. If drastic economic transformation is to take place, it should favour the majority of society.

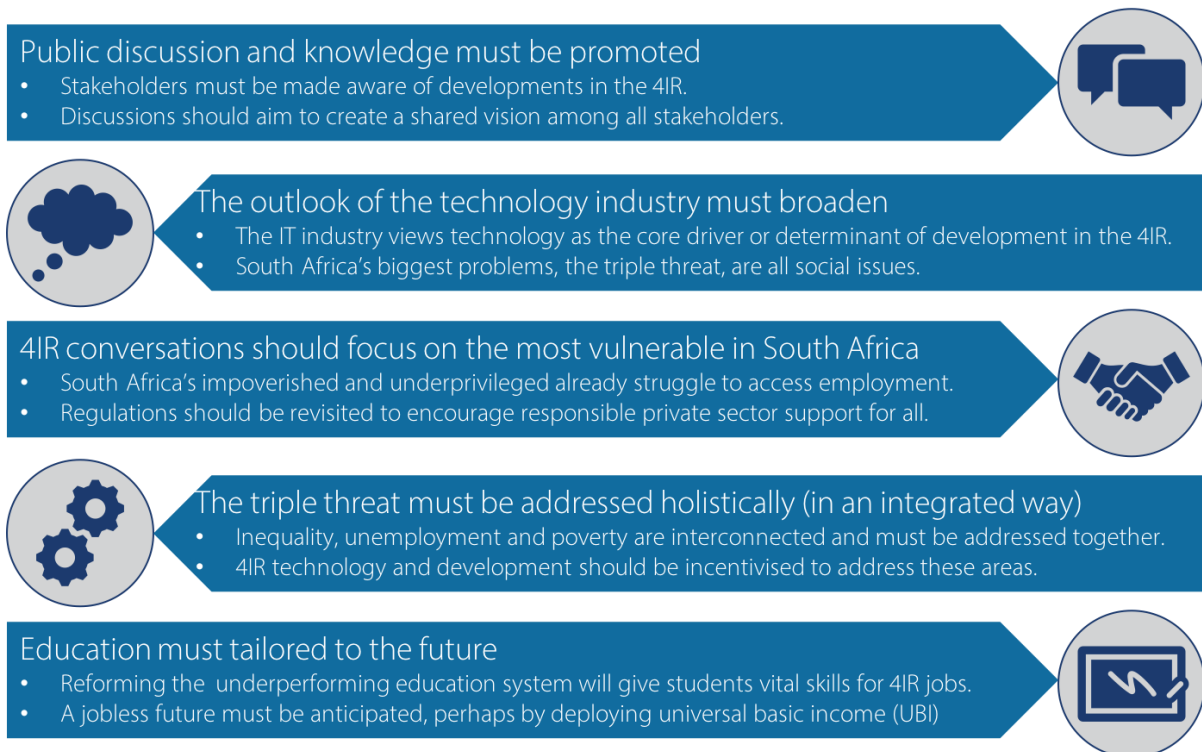
Another key area that will be affected by the 4IR is transport. We have already seen how new technology has disrupted the transport industry in the form of ride-sharing apps like Uber and Bolt. These companies have globally affected the metered taxi industry, lowering prices for consumers and removing the barrier to entry for drivers who simply need a car of their own to participate. In South Africa, the taxi industry employs over 600 000 people, and there are well over 180 000 minibus taxis on South Africa's roads (SANTACO in *TETA*, n.d.). Consider the violence that erupted at the end of 2017 when the cars of two Uber drivers were petrol bombed by metered taxi drivers (Burke, 2017). When we consider the likely arrival of self-driving cars, drivers of taxis and trucks may have their livelihoods severely affected. While self-driving cars may initially be too expensive for the average South African, in future, even the minibus taxi industry might be affected by automation. Of course, new jobs will be created, but they will be far fewer and for skilled people with degrees and years of experience. Beyond self-driving cars and trucks, planes that can fly and land themselves, as well as boats that can steer themselves, may displace pilots, navigators and entire crews. The transport sector in

South Africa (and other countries) could see millions of jobs shed in the 4IR as once-human tasks are carried out by autonomous AIs. The projections cited by Daniel (2018) may therefore be conservative, and the number of unemployed could soon increase dramatically.

Given these concerns, the South African government will need to make it attractive for businesses and foreign organisations to invest in the country, by instituting policies that encourage investment and incentivise 4IR industries. Policy alone will not suffice, and so investing in education and upskilling workers from all parts of the economy will be necessary if widespread unemployment and poverty is to be successfully diminished or eliminated. According to *SAKAN* (n.d.), the presiding ANC government has three options in terms of how to respond to and facilitate the 4IR:

1. Fully adopt and institute 4IR technologies and harness them for their massive productivity, accepting that this may lead to high levels of unemployment, at least in the short term
2. Favour traditional jobs and methodologies in order to secure national stability, but become ever more uncompetitive as 4IR capable companies outpace us in almost every sphere imaginable
3. Make use of 4IR technologies by adapting them to deal directly with each elements of South Africa's 'triple threat': unemployment, inequality and poverty

The first two options are essentially unfeasible, because while they succeed massively in one area, they fail entirely in others. The third option therefore makes the most sense in the South African context. It would be a problem to institute 4IR technologies without considering how they should be used. While widespread and feverish adoption of such technology will have some major benefits, there is little doubt that a great deal of social unrest would arise in the process. 4IR technologies have the potential to truly revolutionise the way society functions and to bring benefits to people from virtually every walk of life. If applied responsibly and in the correct ways, government and businesses alike could attain a 'best of both worlds' reality, in which workers are empowered and new technologies are introduced. *SAKAN* (n.d.) outlines five challenges that must be met in order to attain this theorised reality – these are outlined in the following figure.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from SAKAN, n.d.) Figure 2.2: Five challenges South Africa must meet for success in the 4IR

If these challenges can be met and overcome, then there is the possibility that South Africa will thrive or at least benefit more from the 4IR. The coming change is inevitable, so there is no point in running from it, as this will only leave the country worse off in the long run. The idea of a universal basic income (UBI) is indeed compelling, and the fourth richest man in Africa, Johann Rupert, is in favour of the introduction of UBI (McFarland, 2017). Rupert is the chairman of Richemont, a Swiss luxury goods company, and Remgro Limited, a South African investment holding organisation. He believes that people need time to adjust and upskill themselves in order to fit into the new economy, so without some kind of stipend or basic salary, many will be left out of the equation and the triple threat will grow exponentially. A UBI allows all people a certain amount of money to live on, whether or not they are working, rather than income only going to people who work or receive government grants. While a UBI may sound like a good idea in theory, there are some problems with the model. Amadeo (2018) outlines some of the benefits and the risks of a UBI; the following tables outlines these pros and cons and explains them in some detail.

Pros	Cons
A UBI would allow workers to wait for a better job or to negotiate a better salary – workers could upskill or even stop working in order to care for a relative or a friend if they had no other options.	If people received a UBI, inflation would increase. Many would immediately spend their salaries, driving up demand and increasing prices, therefore negating the benefits of a UBI in the long run.
Problems related to existing welfare programmes would disappear. If people receiving welfare make too much money, they do not qualify for certain benefits, so a UBI would even the playing field and eliminate structural inequality.	In all likelihood, a basic income high enough to eliminate poverty would be too expensive for the government. There are too few taxpayers in South Africa and too many people in need for a UBI to truly be viable on a national and long-term basis.
Welfare programmes are generally complicated and time-consuming for both administrators and recipients. A basic income would simplify things and significantly cut down on bureaucratic costs.	If the UBI is not high enough, it is unlikely that any real difference will be made in the lives of the poor in any case. A few hundred or thousand rand may do little to truly change their circumstances.
Money paid to everyone would also cut down on costly income-verification paperwork and processing time. The benefits for government lie in the simplicity and	A UBI may actually encourage people to stop working as they would see work as ‘optional’. This would worsen an already decreasing labour-force participation rate and decrease human productivity.

straightforwardness of a UBI programme.	
If everyone earned a salary, this would provide them with much needed 'ballast' during times of recession, and give them the ability to participate in the economy to a lesser or greater degree.	If the UBI exceeds or equals the minimum wage rate, then there will be no incentive to work at all. Either the minimum wage would have to increase or the UBI would have to be substantially lower.

(Source: *EDGE Learning Media (Pty) Ltd, 2019; adapted from Amadeo, 2018*)

Table 2.2: Pros and cons of a universal basic income (UBI) in South Africa

There are certainly a number of good reasons why a UBI should be instituted in South Africa, but of course there are also several counterpoints that need to be taken into consideration. Although a UBI may displace welfare programmes and even save the government money in certain areas, the UBI should be enough for people to live above the poverty line. For instance, if 30 million people received R 500 a month as a basic wage, it would cost the government R 15 billion a month, which equates to R 180 billion a year. R 500 is certainly well below the poverty line, so spending would likely need to be at least double the abovementioned figure to make the UBI truly viable. Only certain people would qualify for the UBI, but it would likely cost the South African government a tremendous amount. Whatever the future holds, action is required right now.

2.5 What young South Africans need to learn to navigate the fourth industrial revolution workplace successfully

In light of everything we have learned so far, what can young South Africans do in order to excel in the 4IR? Having a host of different skills will be of the utmost importance in the 4IR, but there are certain skills and capacities that will be highly sought after by companies in the 4IR. The most important lesson South African youths need to learn is that change is inevitable, and that change is occurring far more rapidly than any of us may actually realise. This makes the task of learning that much more demanding and difficult, since what is required of us increases on an exponential basis. Additionally, there may be skills and capacities that are necessary now, but will become partially or completely unnecessary in the 4IR. Let's look at some of the areas young South Africans can focus on for the 4IR.

Soft skills	Developing important interpersonal skills and empathising with others
Exponential learning	Learning and training constantly in order to augment existing skills
Focal points	Knowing what to focus on and avoiding skills that will become redundant
STEM subjects	Science, technology, engineering and mathematics are crucial in the 4IR
AI and robots	Knowing how to automate functions and work alongside AI and robots
Innovation	Creating and following through on feasible, desirable and viable ideas
Ideation	Coming up with good ideas by being creative and thinking dynamically
Initiative	Being motivated and taking the necessary steps in order to succeed
Privacy and security	Understanding how our rights are changing, and safeguarding our data
Collaborating and networking	Liaising with and harnessing the skill sets and expertise of others

(Source: *EDGE Learning Media (Pty) Ltd, 2019*)

Table 2.3: Ten skills and competencies for young South Africans to develop in the 4IR

These ten areas are not necessarily the only ones that young South Africans will need in the 4IR, but they should serve as a general guideline and provide a foundation upon which to build other competencies and abilities. It must be noted that many of these concepts are interrelated in some way. For example, innovation relies greatly on the capacities of ideation and initiative. Similarly, AI and robots and STEM subjects are very closely related, and in order to learn and develop holistic skills, it will be necessary to collaborate and network with various entities around South Africa and the world. We will now discuss some of these areas in further detail, highlighting how they connect to areas for young South Africans to focus on to prepare for a 4IR workplace.

Soft skills for a human workplace

In the previous unit we discussed how AI will change the way things are done, not only in unskilled work being automated but also many skilled fields, from sorting through legal cases to interpreting financial data. Many fields that have very data-heavy entry-level jobs, such as accounting, may find AI or robots taking over some or all of their core tasks – with predictions that AI may perform 40 per cent of all accounting work by 2020, for example (*Accounting Weekly*, 2018). However, that does not mean these jobs will disappear. Instead, it places importance on a new set of skills, called ‘soft skills’. If we consider the technical or practical skills of, say, recording entries for bookkeeping to be ‘hard skills’, then soft skills are skills like critical thinking, communication, empathy, listening and other interpersonal skills (Doyle, 2019). These are skills that will be vital in the workplace, even in the 4IR, and are therefore a valuable area of focus for a workplace where many hard skills can be performed by AI. Although robots and AI can carry out many tasks, they cannot interface as earnestly and intimately with customers as other human beings can. Doyle (2019) points out that soft skills are of particular importance in customer-service or front-facing jobs, and we saw in the previous unit that customer service and satisfaction are key areas for success in the 4IR. In fact, soft skills are very important when dealing with anyone, whether it is a family member, friend, co-worker, supplier or customer. In other words, the 4IR will demand people who are empathetic, caring and human-oriented, with excellent communication skills.

Focusing on education and learning

Going into the 4IR, South African education will need to be revolutionised. Naidu (2018) states that two-thirds of South African children who are currently in primary school will one day be employed in jobs that do not even exist yet. How do we prepare for a future that we do not fully know about yet? How do young people prepare themselves for jobs that are only theorised or not even imagined yet? Most universities are doing an adequate job of preparing students for the 4IR, but many schools, colleges and universities still lack the infrastructure and equipment required to truly train young people and make South Africa truly competitive in the 4IR (Naidu, 2018). Young people need to be exposed to technology from an early age and learn how to interact with and perform functions on these devices. This means more than simply playing games or watching videos on an iPad or smartphone. Students should be taught the basics of programming at primary school level, and even be tasked with writing elementary programs for projects and exams. Science and computer classes should discuss the history of artificial intelligence and explain the evolution of machine learning over the decades. History classes should include the socio-economic development of the world from the first industrial revolution to what it is now.

An additional challenge is that South Africa’s apartheid-era education system and a legacy of post-apartheid corruption and fraud have left numerous schools without even

the most basic facilities required for traditional education to take place. This means that the challenges for South Africa's government and people are significant, since a population that cannot learn and grow will almost certainly falter in the coming years. The truth is that children and young people need to be highly literate in a number of areas, and if they want to learn programming languages and learn how to write algorithms, they need to have advanced language and mathematical skills. Schools cannot continue to use antiquated traditional methods; they should focus on subjects that are crucial to the development of South Africa and its people. The overall structure of and approach to education needs to be altered to accommodate 4IR technologies and prepare students for the workplace of the future.

An area of learning that educators and young South Africans must focus on is the cluster of STEM skills, or science, technology, engineering and mathematics. These fields are going to be key for the country's development in the near future (Firth, 2018). As Firth argues, the focus of the education system should not be getting learners a matric certificate; rather, the system should focus on helping students to get a job (Firth, 2018). Unfortunately, these core skill areas for the future show a worsening pass rate in recent years, even with a pass mark of 30 per cent. This means that, even with one of the lowest pass marks in the world, the failure rate of students in STEM subjects in South Africa is too high, and should become a major priority for government for raising standards of education and empowering students for the future (Firth, 2018). For individual young South Africans, this should be seen as a call to focus on these areas which could lead to success in the 4IR.

One reason that these areas need to be emphasised is that young South Africans who are currently employed or seeking employment may also find themselves in work situations where some functions are handled by an AI or computer system. This means they will need to get used to working with and alongside technology and AI for a wide range of careers. In any profession where there is a lot of number work or vast quantities of data to sift through, an AI is almost sure to be involved now or at some point in the near future. The joy of having computers perform repetitive and menial tasks for us is that we can free up our time to do other work or improve our skills in other departments – from graphic designers having automated help retouching hundreds of images, to researchers assisted by intelligent algorithms helping them to find resources, some jobs may already have tasks automated to help people focus on other areas. The 4IR economy will be one of knowledge and skill, so young South Africans need to be developed in a number of different areas and come to terms with those aspects of life that robots are unable to. In other words, we need to focus on and grow those parts of our identities that make us uniquely human, and grow our capacities so that we are able to be more effective in every aspect of our lives.

Gladstone (2018) states that corporations will need to provide constant training for their employees in order to stay ahead of their competitors. Simply learning skills at school and university and attaining a degree will not be enough. The 4IR demands people who are lifelong learners and are capable of adapting and growing continually. While there are several factors behind exponential and ongoing change in the 4IR, the three primary driving forces behind this are represented in the following figure.



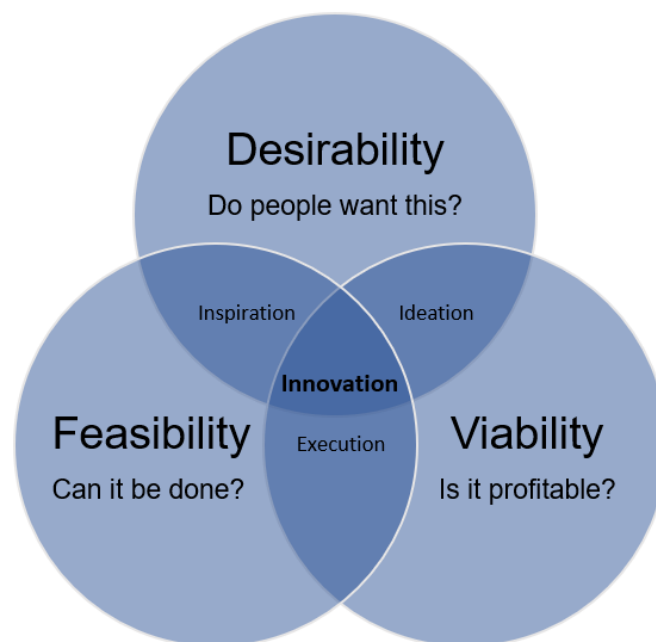
(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Gladstone, 2018)Figure 2.3: The three driving factors behind exponential change in the 4IR

The first major driving factor is acceleration, and this refers directly to the rate at which society, the economy and technology are changing. Not only is change coming rapidly, but the speed at which it is occurring is increasing, so the need to acquire new skills becomes more important with each passing moment. The second driving factor is digitisation, meaning that many processes that were once performed by humans in the real world are now being moved into digital spaces for AI and other systems to deal with certain functions, or at least to make the jobs of human beings significantly easier. Digitisation is not unique to the 4IR and has actually been occurring since the third industrial revolution. This shows how the 4IR has built on the previous revolution and is, in many respects, an evolution of the previous digital revolution. People constantly need to learn and relearn how to negotiate certain functions – this goes hand in hand with technological development. For example, at one point, emails could only be sent from computers or laptops. Thanks to smartphone technology, we are now able to send emails from our personal devices and can do so using mobile data or a Wi-Fi connection. In a few short years, the ways in which we do things now will very likely seem antiquated and out of date, as current technology and digitisation will certainly be replaced by new technological advancements. The beauty of digitisation is that it is able to transcend the barriers of time and space, and allows us to communicate with one another in real time (via WhatsApp, video call or phone call) or even asynchronously (in emails or offline messages). It also eliminates the need for a physical vehicle to transport or space to store information. Digitisation has been around for many years, but it is likely to become even more significant and present in the coming years.

Our education system should be geared, therefore, to encourage learners to improve their soft skills, master vital STEM subjects and be prepared for a workplace where collaboration with AI is almost inevitable.

Innovation for a South African 4IR

If there is one skill in particular that young South Africans need to acquire and hone, it is the ability to be innovative. The 4IR will require people in the job market to be innovative and use existing technology in ways that others have not thought of yet. Innovation is a rather complicated topic, but it has a close relationship with, and is essentially dependent on, invention. In simple terms, innovation is a process in which an idea or invention is turned into a product or service that creates a unique kind of value that end users are willing to pay for (*Business Dictionary*, n.d.). Although this is a convenient summary of what innovation is, it does not fully explain the process, as there are several other factors that need to be considered. Although it may sound pretty straightforward, innovating often takes considerable time, money and energy, and the process of taking a concept and turning it into something that is financially viable is no simple task. In order to better understand this, it is necessary to consider the three principal components of innovation, and understand how they give rise to innovation. The following Venn diagram from The Framework Bank (2016) illustrates these components and how they rely on one another in order for innovation to occur.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from The Framework Bank, 2016) Figure 2.4: The three primary components of innovation

In order for something to be considered innovative, it has to be something that is *desired*, *feasible* and *viable*. This means that a product or service must be something

that people actually want to pay for and use to fulfil a particular function. If there is no real use for something, then there really is no need to make it. People must desire the product or service and have a real need to use or possess it. In terms of *feasibility*, the product or service must make sense and be easy enough to produce and make use of. If the technological capability of the business cannot reasonably or reliably make the idea work, it is unfeasible. Similarly, it will not be feasible if a product takes a long time to charge, performs unreliably or releases substances that harm users and the environment. Finally, *viability* relates to the financial aspects of the product or service in question. If an item is too expensive, few people may buy it, but if it is too cheap, then the producer misses out on profits. If an item is too costly to produce or a service is too time-consuming, then it cannot really be considered financially viable.

All three of these elements have to be present if something is to be deemed an innovation. If an idea is feasible and viable, but not desirable, then only the execution aspect of a product or service is present, and so it falls short of true innovation. If something is desirable and feasible, but not viable, then the thing or idea in question can be considered inspired or even something with potential, but again, not innovative. If something is desirable and viable, but not feasible, then the idea is good, but without the technical capability, it is not particularly innovative.

Managing the risks of the 4IR

We have already discussed how the 4IR will bring both opportunities and challenges to South Africa, and this means that managing the risks associated with the 4IR is important for success. Some of these risks are automation and the potential loss of privacy and security from technology.

We have discussed in this unit and the previous one how automation offers significant benefits, but may also greatly reduce the demand for jobs. As AI improves through the 4IR, we are likely to see technology become better and cheaper, enabling more repetitive and menial tasks to be handled by robots and AI, from controlling the lights and temperature in an office building to processing, analysing and reporting back on vast quantities of data. This will make life and many jobs much easier, but it may also make it difficult for young South Africans to get jobs. As we have seen, many entry-level jobs are at risk from AI and automation (*CB Insights*, 2017). It is likely that competition of jobs may increase as a result of automation, emphasising again the importance of education and continual learning and upskilling for employees in the workplace. If the risks of automation are managed, and individuals ensure they develop and deploy new skills needed in the changing workplace, it is possible that the benefits of automation could be significant. One possible result is that people may spend more time on human interactions and cooperation because AI and automation take away menial and time-consuming tasks, and this could lead to a more humane 4IR where people work together with machines in more fulfilling and potentially higher-paying roles. Achieving

this, however, requires significant preparation on the part of government, businesses and all South Africans.

Another area of risk that young South Africans will need to navigate is the shrinking concept of privacy due to technology, and the security risks that come with this. From social media to cell phone navigation, it is easier than ever for us to share our private information, and for businesses and organisations to access and use it. Our data is a key commodity, as it is bought and sold by businesses looking to learn as much about consumers as possible, and to leverage this knowledge to gain an advantage over their competitors (*Trailhead*, 2019). Here are just a few of the ‘invasions’ of privacy that are already occurring in the world:

- Sites like Facebook and search engines like Google track your activity and use the sites and information you are accessing to direct advertisements and other content toward you while you surf the Internet.
- Smartphones and tablets track your location, allowing applications you download to recommend places to eat or stores to visit – although most devices allow for location services to be turned on and off.
- Online retailers like Amazon and eBay will analyse your past purchases and recommend other related products or even offer special coupons, in order to promote more sales and generate more revenue.

In the previous unit, we touched on the use and misuse of data collected from Facebook by the company, Cambridge Analytica, to target voters and influence the 2016 US presidential election. This is just one of the reasons why invasions of privacy in the 4IR can be significant. It is also important to remember that the information people share online can reach a wide audience, and the negative consequences of sharing personal information or views can be dire. In recent years, people who have shared posts with racist language have been called out on social media, leading to criminal charges and a loss of income. There is also a danger in sharing private, intimate details online, as others can misuse this information for criminal ends, from stalking to cybercrime. It will be vital for South Africans to learn how to use new technologies in a way that is safe, private, legal and responsible.

Using 4IR technology to collaborate

4IR RSA (n.d.) is an online community forum that helps individuals and businesses learn more about the 4IR and what they can do to prepare for it. One of the main reasons why this online resource exists is so that collaboration and participation can be encouraged across South Africa. Now more than ever, young South Africans will need to learn how to collaborate with one another and industry experts to bring about realities that are beneficial to society as a whole. While it is important to focus on personal skills,

teamwork and collaboration also need to be fostered, both online and in physical spaces. *4IR RSA* (n.d.) talks about the importance of creating a ‘business ecosystem’, and highlights the following points:

- By facilitating collaboration, more options can be created, resulting in better outcomes in the process. Achievements can be realised on a higher level to create greater benefits for all participants.
- Engaging in community discussions and making more expertise and knowledge available to contributors means that innovation can be accelerated and encouraged: problem-solvers, experts and specialists can offer their insights, opinions and facilities in order to drive innovation and avoid any hiccups.
- Technology can be harnessed to access resources and create a network of entities that work in conjunction with one another to develop practical solutions and assist those with limited capacities.
- A community forum means that knowledge acquisition and learning can be grown exponentially – by sharing ideas and grouping creative people together, greater things can be realised more rapidly.

An excellent way to learn about and deal with the 4IR is to join in discussions and participate in think tanks so that skills can be developed in line with the latest developments. Again, avoiding the topic of the 4IR is counterintuitive, and being apathetic about it will not bring any rewards to anyone. Young South Africans cannot afford to sit idly by while the world changes around them at an ever-increasing pace. Innovation, creativity and participation rely on a strong sense of initiative, and this is arguably what the youth of today need to learn the most: how to be motivated and take the necessary steps to create meaningful, long-lasting change in their own lives and the lives of those around them. There is no time to lose because changes are occurring as we speak, whether we choose to acknowledge it or not. *4IR RSA* (n.d.) is a useful platform for young South Africans, as simply networking and liaising with others is a great way to find out more about current affairs and future developments, and to stay ahead of the crowd.

Failure to consider one or more of the guidelines discussed in this section does not mean that you will not succeed in the 4IR, but by keeping all of these aspects in mind, you should be better equipped to deal with the challenges that will be presented to you. Even if you do manage to observe all of these aspects, success is still not guaranteed. There are obviously situations and eventualities that cannot be predicted or avoided, so if there is one final ‘eleventh’ skill that young South Africans should consider, it is the ability to think on your feet and improvise. When faced with limited options and resources, use what you have at your disposal and take action to make the most of your specific situation. You may not achieve quite what you hoped, but you may do much

better than you anticipated – staying active, remaining focused and responding intelligently will be key in the 4IR. Young South Africans also need to be patient with themselves. Excellence takes time and perseverance, and even when you feel disheartened or unmotivated, persist and continue to work – with dedication and practice, you can succeed in the 4IR.

2.6 The future of artificial intelligence in South Africa

The 4IR and the use of AI are already underway, and our ability to adapt and innovate using new technology will be what sets us apart from our international competitors. South Africa may not be prepared yet for this AI-powered future, for the following reasons (Sithole, 2018):

- *Capital investment must increase:* Businesses and government must be more proactive in investing in AI and machine learning ventures.
- *Planning and consideration must occur:* Without considerable planning on how to apply new technology, why it should be applied, where it will be used and by whom, it will be difficult to set and reach intended goals to maximise our gain and minimise risks and negative consequences.
- *Policies must first be in place:* Existing policies must be adapted and new ones developed to prepare us for and safeguard against negative economic fallout from deploying AI and machine learning.

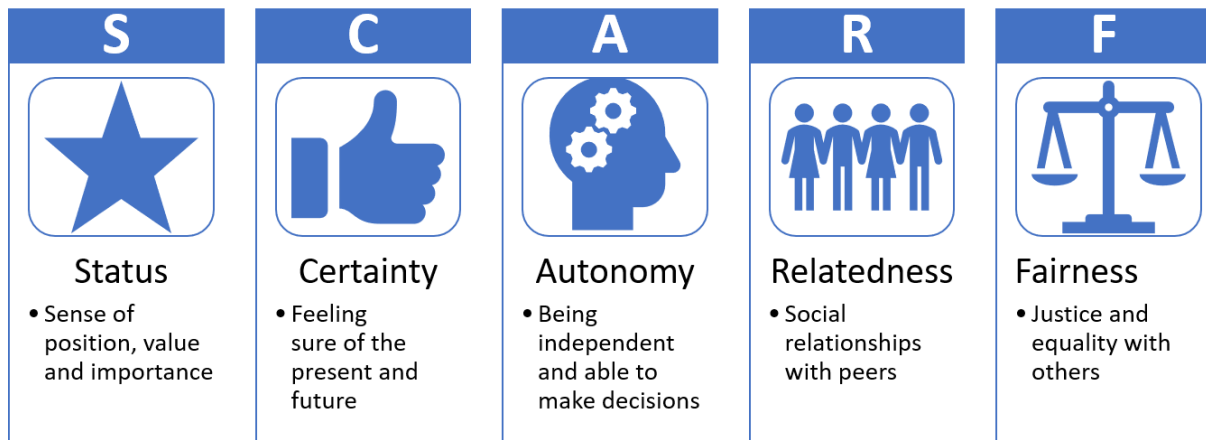
As mentioned earlier, there are conflicting reports about South Africa's level of technological adoption, so more thorough research is required before any real conclusions can be reached and solutions offered. Whether we are good at adopting new technology or not, we need to take the necessary steps to stay ahead of other countries and remain competitive in the 4IR.

Jardine (2018) cites research by Accenture and the Gordon Institute of Business Science (GIBS), which states that AI has the potential to double the growth rate of the South African economy and increase profitability by around 38 per cent by 2035. While this certainly sounds promising, we must take South Africa's unique socio-economic status into account. AI and automation have already started displacing jobs in certain industries, and will see more workers in various roles losing their jobs in the coming years. There is little doubt that many South Africans are nervous about the 4IR and AI, and citizens from other countries are almost certainly experiencing similar feelings. There are also many South Africans who have not heard of the 4IR, and it is likely that many do not even know that there have been four industrial revolutions to date. If we are to overcome longstanding social issues and rise up as a nation, we need to ensure that

all South African citizens are included in (or at least aware of) the 4IR and the changes it will bring with it.

This is why subjects that deal with AI and machine learning need to be introduced at primary school level across the country. Of course, many South African schools are severely limited and lack even the most fundamental infrastructure to facilitate traditional learning, which is a major obstacle to overcome before a curriculum suited to the 4IR can be introduced. While introducing AI and/or robots into the learning environments of students at school and at home would be a good step forward, as children could become accustomed to and aware of such technology from a very early age, there are clearly more basic obstacles to be overcome before this can be done for the many students struggling in the current education system. Making provisions for such changes is of paramount importance, but at present there is little being done to rectify longstanding problems, never mind newer challenges.

Another area we need to consider for the use of AI in the 4IR is the way it may impact our social lives and the ways we think of ourselves and one another. Humans are social beings, which means we like to feel as though we belong in a group that values us (Jardine, 2018). Technology can often impact our sense of belonging; for example, consider the feeling of being blocked on a social media site, or having a stranger treat you poorly in the comments of a website. On an institutional level, too, AI and automation may have profound effects on us. Think about how many people may be involved in the daily operation of a university: lecturers, research assistants, administrative staff, student advisors, maintenance, grounds and cleaning staff and more. This represents a wide cross-section of South African society and brings people together for the common goal of education. Introducing an artificial intelligence may begin to replace some of those people, and this raises questions about how we value certain jobs, and whether the future of many South Africans will be secure in the 4IR. This is why many people see AI as a threat to natural human relationships. We have to consider the social experience of humans when discussing AI, and so for this reason, we have to give some thought to the five factors that are integral to us as social beings. The following figure illustrates these five factors and summarises them using a simple acronym.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Jardine, 2018)Figure 2.5:
The five domains of the human social experience

Each term can be understood as (Jardine, 2018):

- **Status:** The presence of AI in a workplace has an effect on an employee's status, as their value is brought into question. Of course, AI may actually serve to increase the status of certain employees, but in other cases it will replace them altogether and likely strip them of their roles.
- **Certainty:** This is important in the human experience and is a source of comfort and stability for just about every person living on Earth. AI is already causing disruptive changes – both internationally and here in South Africa – and considering that we are living in a volatile, uncertain, complex and ambiguous (VUCA) world, there is a growing lack of certainty which affects the mindsets and security of people all over the world.
- **Autonomy:** This relates to how much freedom and control we believe we have in our lives. AI threatens our autonomy as we may become partially or totally reliant on AI to help us perform work and make a living. If we feel like we have no options or that we are resigned to a particular reality, anger, depression, resentment and disillusionment may arise.
- **Relatedness:** AI may serve as a bridge between different people or even replace certain people in the value chain, fundamentally severing many longstanding

professional relationships between people. Many employees become friends and work alongside one another, and so when one of them is removed from the equation, productivity and job satisfaction may be negatively affected.

- *Fairness*: This is the idea that people are treated equally and that one perspective is not necessarily more valuable than another. If there is a lack of recourse or choice as a result of AI and machine learning in businesses and other organisations, then the very concept of fairness may disappear.

It is clear that AI and 4IR technology will need to be carefully implemented. Many workers whose sense of status, certainty, autonomy, work relationships and fairness is threatened may resist the introduction of these disruptive technologies. While AI may be an inevitability, the assumption that there will be little or no fight against such technology is remiss. Fighting against the introduction of technology is ultimately counterintuitive, as it will simply delay what is virtually guaranteed to come to pass. However, this will mean little to the numerous employees who may be unable or unwilling to adapt and overcome their limitations to achieve in the 4IR. AI is being introduced into South Africa, but it may be a long time before it is fully accepted by South African citizens. The South African context is particularly challenging to all stakeholders and will require tremendous effort.

Business Tech 2 (2018) asserts that AI will likely hit South Africa harder than most countries. One of the main reasons for this is that the people who build AIs and create algorithms may imbue their creations with their own biases, misconceptions and worldviews. For example, the AI built into voice recognition software often has difficulty understanding people who speak with certain accents. In this way, the usefulness of such software is limited to people who are similar to or the same as the creators of the software. Most AI algorithms are created by companies in developed countries, and so these algorithms are almost always built for use within those contexts. This means that the relevance and efficacy of AI is often significantly lower in developing nations, since the nature of the underlying algorithms may not be ideal for the features of those societies (*Business Tech 2*, 2018). This is why South Africans need to take existing technology and become personally involved in the development of AI algorithms, so that they can be made suitable for our unique requirements and concerns. AI will have to be adapted if South Africans are to overcome the challenges presented by the 4IR. If we are able to 'take ownership' of AI and apply it in such a way that it can help us overcome the problems we face as a nation, then we stand to benefit tremendously from the 4IR. The knowledge and expertise needed to develop and appreciate AI will have to be earned, and we will all need to have some kind of understanding of what AI can accomplish if we really want to harness its full potential. South Africa may even have some insights and knowledge that will be of use to people in nations all over the

world, so if we become particularly adept at creating smart, affordable and intuitive AIs, the potential benefits are indisputably immense.

AI in South Africa

Although there is much ground to be covered, there are a handful of South African companies that are already working to refine their expertise and create products and services that are highly innovative, easy to access and affordable for South African consumers and businesses alike. Lourie (2017) identifies four South African AI start-ups that are striving to establish operations that are relevant and profitable in the 4IR. We will briefly discuss each of these South African start-ups and provide some explanation about their unique offerings (Lourie, 2017; and *Aerobotics*, 2018):

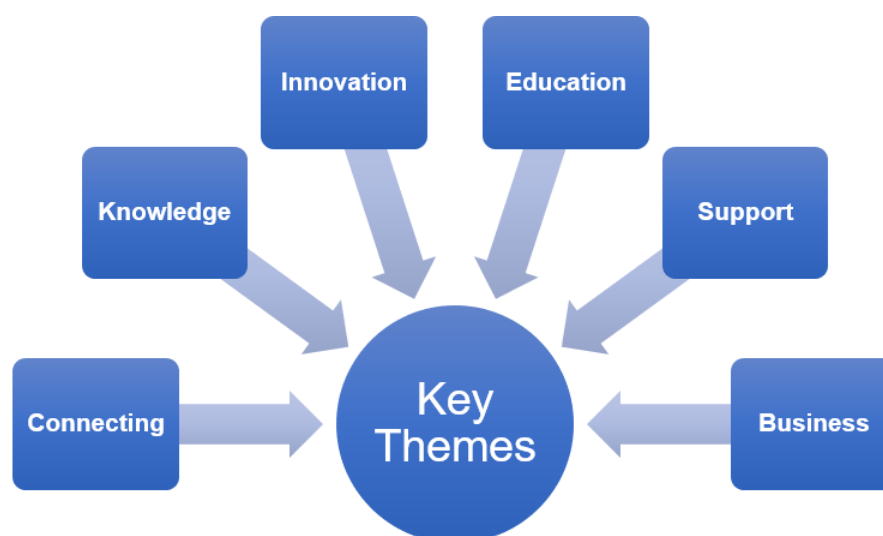
- *Xineoh*: This company specialises in machine learning and using mathematical models and algorithms for advertising. Their algorithms use data in a way that is similar to how Amazon and Netflix make recommendations to users based on their previous online behaviour. Xineoh's algorithm takes available data from users' interactions with products to make predictions and draw general conclusions about what a person might be interested in. The company reports that their algorithm has helped to create more than \$ 30 million combined revenue for all their clients across the world.
- *CLEVVA*: This company aims to use technology to make people more effective in the workplace, rather than replacing them. Their current project uses virtual advisors to assist sales and technical consultants to be more efficient, uniting AI and human workers to achieve more.
- *DataProphet*: This start-up is a machine learning specialist with an African focus, creating algorithms for businesses of different sizes. These various algorithms allow businesses to improve the quality of their products and services, and run their operations more effectively and securely. By employing advanced algorithms, companies can identify, prioritise and chase important leads, develop more effective schedules and prevent fraud and corruption before it can take root.
- *Aerobotics*: This company specialises in using autonomous drones for various agricultural, logistical and mining industry operations. For example, smallholders and farmers can manage their operations through the use of drones, as these drones can monitor crops and notify farmers about risks, follow set routes for in-depth analysis, monitor the size and overall health of trees, and help with research tasks like planning irrigation probe placement, collecting soil samples or even applying fertilisers by themselves. Their technology has been used in South Africa, the UK and Australia.

While these start-ups are all making their mark, larger South African companies are also looking to improve their operations. Vilar (2019) claims that South African bank Absa has developed an AI that automates account switching for its clients. This is a ‘world first’ according to Jacques Barkhuizen, the CIO of Absa (Vilar, 2019). The process of switching your beneficiaries from an external bank to Absa takes less than seven minutes and is fully automated, meaning that no manual input is required once the process is started.

According to *TechCentral* (2018), Ethos, a major private equity investment company, has created a fund with a target of R 1 billion to invest in AI businesses here in South Africa. The company already claims to have secured R 600 million and concluded two investments, one in Channel VAS and the other in Vertice MedTech Group. There is clearly significant interest in 4IR technology and AI from both local organisations and foreign entities. For many, it is not a matter of ‘if’ but rather ‘when’ AI and machine learning will really start to take root in modern societies.

Building an AI culture in South Africa

The AI Expo Africa takes place annually at the Century City Conference Centre in Cape Town (*AI Expo Africa 1*, n.d.). The inaugural event took place in September 2018, and due to the success of the exhibition, the 2019 event will run from 4–5 September. According to their website, AI Expo Africa is the largest AI and data science community event in Africa. The exhibition focuses primarily on businesses and what they can do to improve their operations and increase their profitability and sustainability using 4IR technology. With over 2 000 community members, 732 delegates and 43 sponsors, AI Expo Africa promises to deliver incisive content and insights into the rapidly changing world that we live in. The 2019 conference will focus on six specific themes that are especially relevant in the world of business. The following figure illustrates these themes.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from AI Expo Africa 2, n.d.) Figure 2.6: The six key themes of the AI Expo Africa community

As illustrated in the figure, the six key themes of the AI Expo Africa community are as follows:

1. *Connecting*: This refers to networking and liaising with AI experts, business owners, influencers, investors, service providers and a host of other individuals who are part of or contribute to the AI and data science community (*AI Expo Africa 2*, n.d.).
2. *Knowledge*: This refers to the sharing of information and insights regarding the current and future state of AI in Africa and abroad.
3. *Innovation*: This is represented and upheld by 'The Innovation Cafe' which houses and nurtures the AI start-up ecosystem.
4. *Education*: This is all about how people and businesses can learn about disruptive AI trends and maximise their understanding of 4IR technologies.
5. *Support*: This encompasses the activities and roles that government, investors, training and skills development will play in making AI more pervasive and effective in Africa.
6. *Business*: This refers to making AI profitable and providing both local and foreign businesses with intelligent, affordable and effective AI and machine learning solutions.

At present, many of the AI and machine learning solutions that South African companies are employing are fairly rudimentary, and in time it is likely that these solutions will become more effective and innovative. Whether they specialise in AI or outsource AI specialists, South African companies are going to have to get used to the fact that their processes, especially data-heavy ones, are obsolete if they do not employ some kind of AI at some point in the value chain. Essentially, AI can be employed in a number of different areas, if not all areas, of a business. The real benefit of a few South African businesses adopting AI solutions is that others will have to follow suit if they wish to remain competitive. In this way, we can conclude that adoption is almost certainly the main driver behind innovation and creativity: a few companies adopt AI, and then a few more until many are adopting AI-based solutions. It is inevitable that just about every company, no matter how big or small, will have to include some kind of AI in order to optimise their business and increase their overall profitability.

There are certainly moves to get South Africa on the map in terms of AI and 4IR technology, and in the future, both Cape Town and Johannesburg could become two major hubs for AI and the rollout of new technology in Africa (Chinner, 2018). As of late 2018, Vian Chinner, the CEO of Xineoh, asserts that Africa has brought very little to the

table in terms of global AI and machine learning solutions. He highlights the fact that San Francisco in the USA and London in the UK are at the forefront of AI innovation. The two main factors behind the success of these cities are:

- the existence and development of a culture of mutual trust; and
- a propensity for risk-taking behaviour.

Chinner (2018) also highlights the fact that San Francisco and London are 'home to some of the best universities in the world'. For these reasons, it is important for Africans and South Africans to foster similar values in order to drive AI development and adoption forward. Chinner states that, apart from Xineoh's machine learning solution, the capacities and policies needed to bring about a thriving hub for AI in Africa simply do not exist yet. As a result, Chinner (2018) suggests that South Africans take one of the following two routes to build these capacities:

1. *Bring in skills from Silicon Valley:* Importing experts, innovators and industry leaders from Silicon Valley (a region of California, USA where many technology start-ups and companies began and are based), or at least having them consult with local businesses and government, will make a significant impact on the local AI and data science industry, and will likely stimulate development on a national level.
2. *Put infrastructure in place to create our own:* Creating policies and incentives to encourage the development of new infrastructure that will be conducive to AI solutions and technical skills development would certainly go a long way in driving South Africa's own 4IR and making it a major global player.

Either of these routes should be feasible, and there is no reason why they cannot be combined in order to create further benefits for all parties concerned. The more that can be done to encourage the 4IR and AI in Africa, the better. Although Cape Town seems to be the AI capital of South Africa at present, Chinner (2018) believes that Johannesburg may well become a force to be reckoned with in the 4IR. The main reasons for Chinner selecting Johannesburg is because it has a long history of risk-taking, and is home to a significant number of top universities. He also states that Xineoh has found that businesses and other organisations based in Johannesburg are generally more open to and enthusiastic about implementing AI solutions. Having said this, there are very few true data scientists in South Africa, which means the creation and implementation of home-grown AI-based solutions is highly unlikely at present. This is why it is important to consider the first route suggested by Chinner (2018), in order to encourage AI solutions and development to take root in South Africa, at least in the short term. Tapping into the skills and knowledge of industry leaders is highly beneficial and could be a major game changer for South Africa. Locals with limited or no knowledge can learn entirely new skills, and those with some knowledge can improve

upon what they already know. People from other countries can also bring fresh insights and energy, as new ideas are formulated.

It is too early to tell exactly what the future of AI in South Africa will be, but there are definitely a handful of companies that provide the country with hope. Having said this, there are certainly a number of possible events that could serve to curtail the rollout of AI solutions in South Africa. There are numerous socio-economic issues that South Africa has not adequately dealt with, and the longer poverty, inequality, lack of service delivery and unemployment persist, the more likely it is that the country will face further hardships. If government and businesses can find ways to use AI and machine learning to address these issues and stop corruption and fraud, then the potential benefits could be massive. However, if AI, data science and other 4IR technologies are not carefully managed and instituted, existing problems could be exacerbated, making the situation worse for everyone. In time, South Africa could well become the heart of 4IR development and advancement for the entire African continent.

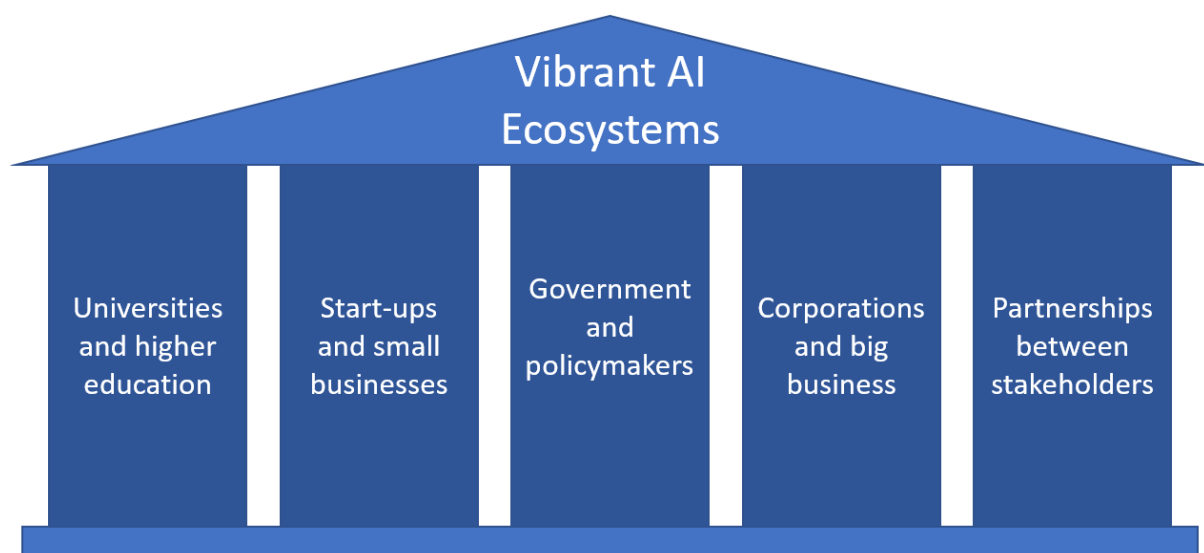
FAST FORWARD: Gearing up for local artificial intelligence – Are we ready?

We have discussed AI and machine learning in great depth in this unit, but even though there are some promising developments, the question remains: are we ready for local artificial intelligence? In an analysis that appears on *EE Publishers* (2017), many companies are weighed down and held back because they are stuck using traditional systems and technologies, legacy business models and corporate business structures. On top of this, South African companies are heavily invested in infrastructure that is basically obsolete. Many of these businesses employ traditional approaches and have large numbers of workers who are unaware of, or unready for, the 4IR and AI. According to research undertaken by Accenture, almost four out of five South African executives agree that they will need to make significant investments in AI-based solutions, in order to remain competitive and relevant in the 4IR (*EE Publishers*, 2017). When it comes to actually planning to invest over the next three years, however, only around one third of their businesses are ready to spend money on AI projects. It would seem that there are some attitudes, concerns and other factors that are holding South Africa back, and in a roundtable discussion held by Accenture and GIBS Business School in July 2017, a number of pressing issues were highlighted (Schoeman et al., n.d.: 10):

- *Data quality:* A lack of well-defined and reliable data poses a threat to the development of AI and could negatively affect South Africa with regard to its growth prospects and socio-economic advancement.
- *Data privacy:* Data security and data privacy are not particularly high on the list of priorities for most South Africans, let alone businesses, and compromised or unsecure data may make the development of AI solutions very difficult; privacy and security concerns must be addressed to encourage AI solutions.

- *Workforce readiness:* Many South Africans are unaware of the 4IR and the coming changes to how work is done. For this reason, the majority of South African workers are ill-equipped and unready to deal with the realities that they will be faced with in the near future.
- *Re-skilling:* There are not enough programmes and incentives in place to truly bring South Africa into a world where robots and AI will be part of just about every aspect of life. Re-skilling often means upskilling, and for many, this is simply an impossibility since they lack many fundamental skills that must be built upon consistently, if more advanced or long-lasting skills acquisition is to take place within a business.
- *Potential job losses:* Companies and individual workers are worried about the potential effects of firing or retrenching workers. Mass layoffs could cause chaos and bring the country to its knees. The trouble is that AI and 4IR technology is here to stay, and job losses are almost certainly unavoidable.

Other issues facing South Africa include greater competency in data science for organisations as well as the need for clear government policies to encourage AI adoption and development (*EE Publishers, 2017*). Perhaps the most important step for South Africa to take is to create infrastructure to encourage the formation of a vibrant AI ecosystem. Having studied numerous countries where AI is developing well, Accenture's research specifies five pillars upon which great AI ecosystems (the networks or systems that encourage and allow AI-based projects to flourish) rest. These are illustrated in the following figure.



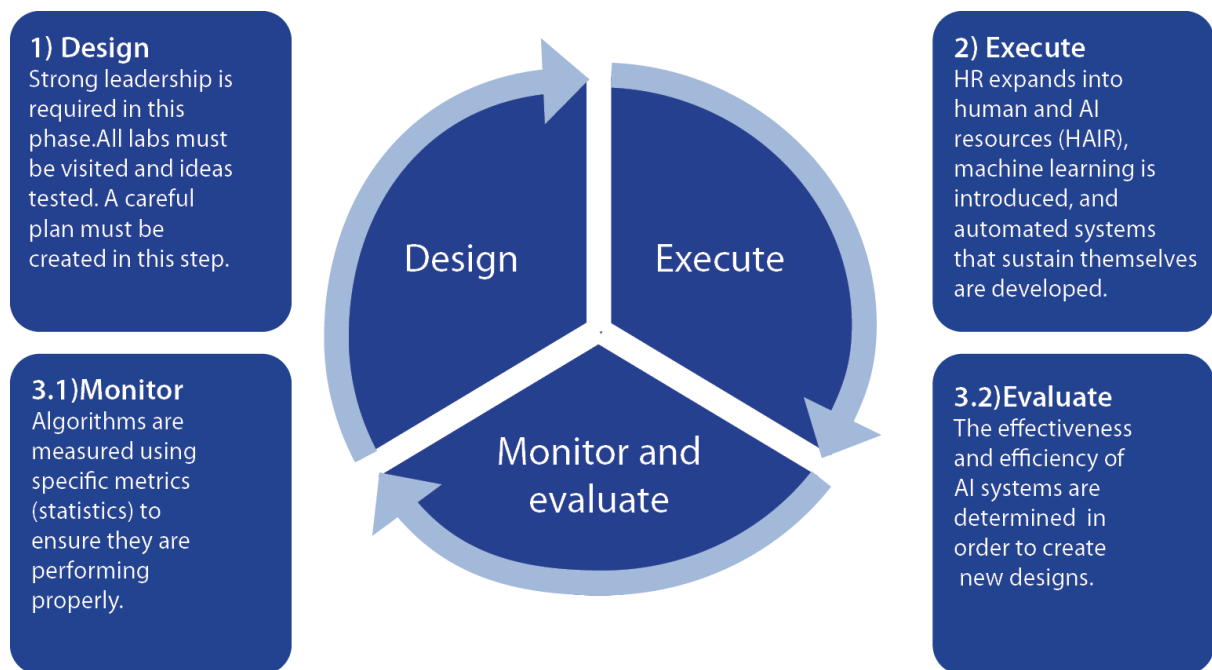
(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from *EE Publishers, 2017*) Figure 2.7: The five pillars underlying or supporting vibrant AI ecosystems

As we have already mentioned, it is necessary to have well-developed and engaged universities and other dynamic tech start-ups to create an environment for AI

development (EE Publishers, 2017). Naturally, large corporations are also a necessary part of the equation, as they have the power and resources required to institute AI solutions and develop customised solutions to suit their particular needs. Policymakers are arguably the central and most important pillar, as they ultimately determine how the other pillars will operate, form and network with one another and other entities. Finally, multi-stakeholder partnerships will allow a greater network to come into being and for an AI ecosystem to form across South Africa. At present, South Africa has all of these components in some measure, but there are definitely issues with regard to policy, as well as the number of start-ups and the nature and quality of university graduates.

Aside from creating a vibrant AI ecosystem, South Africa must find a way to translate AI investment into AI-driven growth. The only way for a company to increase its artificial intelligence quotient (AIQ) is to ensure that they are proactive in the areas of collaboration and invention (EE Publishers, 2017). The potential of AI to fundamentally alter a company's growth prospects will remain largely untapped if collaboration between businesses and the development of new technology is not fostered. Investing in AI means doing more than simply throwing money into such technology – such investments must be carefully directed and linked with other important areas if any meaningful improvement is to be realised. Practising responsible AI will also be of paramount importance for policymakers and businesses. Failing to deal with the possible fallouts of AI implementation with regard to job losses and an increase in inequality could bring South Africa to its knees if people begin to revolt. If the change process is not carefully managed, then there is the strong possibility that calamity could ensue. Sensitive issues require carefully thought-out solutions, and without swift and responsible action, South Africa's 4IR prospects may never be fully or even partially realised.

If South African companies want to adequately prepare themselves and initiate AI solutions, they will need to follow the three steps illustrated in the following figure.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from EE Publishers, 2017) Figure 2.8: The cyclical three-step process for AI solution implementation

Inevitably, South African companies and related organisations will have to be proactive in preparing themselves and the country for the 4IR. There are certainly many good signs for South Africa, but the rate and level of change is not yet adequate enough to facilitate the transition from a 3IR economy to a 4IR one. Therefore, when answering the question, ‘Is South Africa ready for local artificial intelligence?’ the answer is, ‘somewhat’. We are certainly not as prepared as we could be, but we are not completely unprepared. The answer lies somewhere between ‘yes’ and ‘no’, but without some improvements and further investment in AI solutions from all parties concerned, the scale could quickly tip toward ‘no’. Essentially, South Africa could go either way, but if plans and strategies are not followed up and brought into the execution phase (and executed well), then the 4IR and the prospect of local AI will become more and more daunting in the near future.

2.7 Summary

In this unit, we identified some of the risks that the 4IR poses to South Africa and its people as well as some of the benefits that automation could bring to local businesses and government institutions. We also examined some of the possible scenarios with regard to the future of work in South Africa and how the 4IR could change our socio-economic situation for better or for worse. We also made a strong case for a learning revolution in South Africa, so that we can prepare a national workforce that is highly

skilled in both soft and hard skills. We also determined that South Africa is not particularly well-equipped for the 4IR, but it is also far from unprepared. Further investment in AI and a shift in mentality toward patterns of excellence, skills acquisition and a high proficiency in STEM subjects will be necessary in order to make South Africa an attractive place for foreign businesses to invest and to increase the country's growth prospects. A culture of lifelong learning, innovation and ideation will need to be fostered across the board to build a generation of South Africans who are capable of thriving in a world of constant uncertainty, disruptive change and volatility.

Chapter 3

3.1 Introduction

Imagine the following scenario. You have just graduated from university, and you are about to start your very first job. On your first day, you walk into the building and are presented with a pristine, chic workplace that is open-plan, and has a range of greenery, tastefully scattered throughout the area. The space includes separate 'creative spaces' for teams to brainstorm in, and there is an abundance of technology available. From your very first glance, it is likely that you will be impressed, and may feel ready to get to work straight away. It is for this exact reason that modern companies spend a lot of time and effort making the workplace comfortable and appealing; it motivates employees and can lead to greater satisfaction.

Workplaces today function very differently from any you would find 100 years ago. It is safe to say that they are much more comfortable, colourful and efficient. Space and time efficiency in particular are visible advances in the way that 21st-century offices operate, as we have moved from using physical filing cabinets to digital storage, and many devices now perform more than one function. There has been a general trend toward open-plan designs, technological integration, and building natural 'green' spaces within the office, in order to improve employee well-being. These are aspects of work that you will need to understand in order to get the best out of your career; or, if you are a business owner one day, to make sure that you have the best information available to make your office an appealing workspace. We will look at each of these elements in detail, as we learn to navigate through 21st-century workplaces successfully.

After studying this unit, you should be able to:

- outline the 21st-century office in terms of design, people, projects and technology;
- discuss the types of interactions that the 21st-century office invites, with reference to workplace diversity, virtual workspaces and flexible work environments;
- explain the importance of effective ergonomic set-up, cognitive spaces and wellness spaces in the workplace; and
- apply knowledge gained on 21st-century workplace organisation, interaction, and cognitive and wellness spaces, to South African case study material.

3.2 Key terms and definitions

The following key terms and definitions are applicable to this unit:

- *Cubicle office design*: A traditional office layout of individual cubicles with their own walls, separating employees from one another.
- *Desk pods*: Desks are arranged in close proximity of one another, so as to encourage group work.
- *Digital citizenship*: The online presence and interactions that every person has, and the online trail they leave, which includes search history, photographs, emails or comments on posts on social media.
- *Ergonomics*: The study of equipment and individuals in the workplace to ensure that individuals are interacting with equipment in a manner that is safe and not harmful to their health and safety.
- *Local area network (LAN)*: A system used within an organisation for the ease of sharing information.
- *Modular seating*: This allows for an office space to be easily changed, merely by moving furniture.
- *Productivity styles*: The various ways in which employees feel they are able to work best.
- *Sociotechnical environment*: The management of the workplace in such a way that both the benefits of technology and the needs of employees who work with the technology are considered.
- *Virtual workspaces*: Rather than a physical building where people come to work, this refers to a greater online presence of people working remotely from all over the world, rather than from one dedicated workspace.

- *Wide area network (WAN)*: This occurs when a number of different LANs connect to different networks via the Internet.

3.3 What does the 21st-century office look like?

Throughout history, people have come together to do business, even before the use of designated office spaces. In fact, the understanding of an ‘office’ as a place of work is a modern one, as it once referred to a particular position that an individual could hold (*Morgan Lovell*, n.d.). We still use this idea to some extent, by referring to the ‘office of president’. This refers to the job itself, and not the place where the president works. One of the earliest organised office spaces was created in the early 18th century, by the East India Company, which conducted trade and martial operations in Asia. Their main workspace shifted between various private homes, until finally occupying a property on Leadenhall Street in England, which was called the East India House (*Morgan Lovell*, n.d.). Due to high volume of trade agreements and transactions that were being recorded between England and India, the company generated massive piles of paperwork that needed sorting. This meant that they needed an efficient space to work in, and thus, one of the earlier forms of the office was born.

Using this example, we may think of an office as any space that is used to perform work. However, they have become much more than this. An office is specifically designed for the type of work that is being performed. The design used can depend on various factors, including:

- the space required to complete the work;
- the speed at which work is completed;
- the quality of work produced;
- the efficiency of time and effort;
- the comfort and needs of employees; and
- the technology needed to perform work.

There have been many attempts at balancing these factors, so that a business can be the most productive it can be, while still economising on the space and time that it gives employees to perform their work. One particularly well-known attempt at this is ‘cubical farming’, which was implemented during the 1960s. Hundreds of employees were placed into small cubical spaces, intended to focus their time and effort completely on work. However, this limited other interactions, especially with their co-workers (*Morgan Lovell*, n.d.). The effectiveness of this design is still currently debated, and there are some emerging approaches that rely on open and employee-centred designs which have gained traction in modern businesses.

Currently, one of the most popular trends in the design of office spaces is the open-plan layout. This has been found to encourage participation, teamwork and collaboration between employees, in the sense that there is literally (and figuratively) no separation between individuals, thus promoting a higher level of teamwork. The modern workplace also greatly emphasises technology in that, more than ever, employees are working in virtual teams. There is also a current reliance on social media, which essentially sets apart those who will succeed from those who will fail, based on how they can utilise their social media to interact with both employees and customers.

Various approaches to office design will be discussed in the following section.

3.4 Workspace organisation

Managers and business owners are constantly looking for ways to balance productivity with comfort, in order to ensure that employees are working hard, and that they feel good about their work while they do so. The physical workspace is a big part of this, and the layout and design of the workplace has many implications for the efficiency and satisfaction of employees. When the physical office space is built or chosen, the owners of the business must carefully consider how they will divide the space up. Introducing a layout without knowing the particular needs and optimal working conditions for each employee can be a difficult task. Sometimes, it is best to opt for a flexible layout that can be adjusted according to the changing employee needs.

3.4.1 Office design

For many years, the standard approach to office layout was the cubicle system. Single cubicles or individual offices have traditionally been used to focus employees on their work, and to limit any distractions that might occur. However, this layout can also lead employees to feel lonely and disconnected, as they are boxed into a small space and unable to see or easily communicate with other people. This, effectively, can be counterproductive to teamwork, as it prevents employees from communicating openly and frequently with one another.

As you can see in the following figure, each employee is limited by the wall-space of their cubicle, and faced away from their co-workers, so that they are required to do their work without interacting with others. They will not be distracted by noise or movement, but they also have very little space to move.

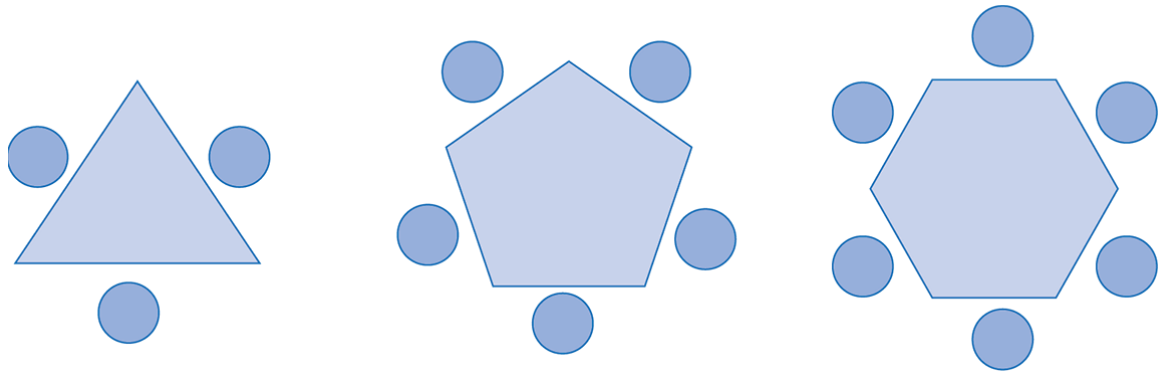


(Source: Shutterstock.com)Figure 3.1: Traditional cubicle separation

While cubicles have some benefits, they may also have negative effects on employee comfort and job satisfaction. It is important to note that, for some people who are easily distracted or find it difficult to manage their time, a cubicle may work rather well.

However, as previously mentioned, there are also now more modern approaches to office design that typically focus on a more open-plan layout. The main types of open-plan layouts include desk pods and modular seating.

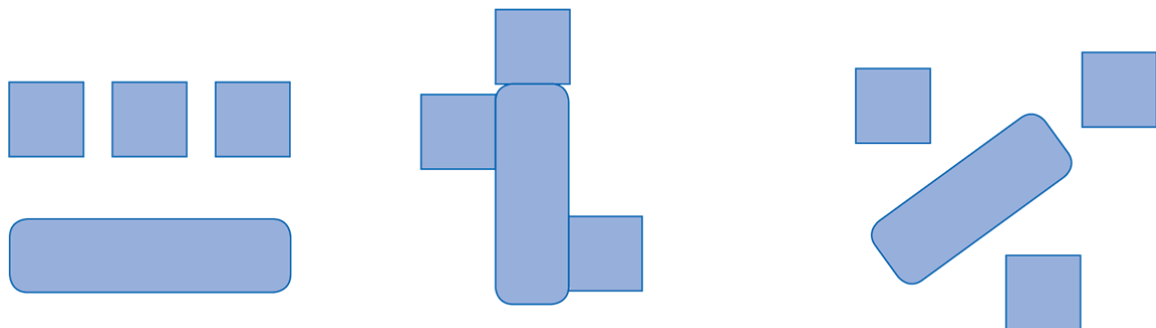
Desk pods have been shown to encourage greater collaboration between employees, as they offer an open alternative to separation by cubicles; while employees are still allocated to particular desks, the layout is far more communal, as employees are not blocked off from one another. Desk pods do not often have physical divisions, but can be divided by frosted glass or another light material that does not completely cut the employees off from one another. This allows employees to speak to one another easily, particularly if they are working together on a project, or often need to ask for advice. An example of this style of seating is shown in the following figure. Each circle represents where an employee would be seated.



(Source: EDGE Learning Media (Pty) Ltd, 2019)Figure 3.2: Desk pods

Modular seating has grown in popularity in recent years, as this can open up a working environment visually, and provide more physical space (*Fuze Business Interiors*, n.d.). This will make both employees and visitors feel more at ease, and will create a hospitable atmosphere.

Modular seating allows for better ease of movement and customisation than traditional office furniture, such as couches. Imagine, for example, that you need to host a promotional party for one of your products. Using the office lobby or reception space may not be possible if the seating is permanent, but modular seating can be moved aside to create floor space. This is illustrated in the following figure.



(Source: EDGE Learning Media (Pty) Ltd, 2019)Figure 3.3: Modular seating

The concept behind modular seating is to allow businesses to change the layout of a room without having to buy new furniture. This approach saves both money and time, as it better prepares the workplace for continually changing needs.

One contemporary example of a well-known business that makes use of an open-plan setting is Google. The company is renowned for their wacky office setting, interesting decorations and wide-open spaces that promote collaboration and group work. This makes the space feel more open and relaxed, which is vital for a company that relies so heavily on creativity and teamwork. The following figure shows the type of look that open-plan offices like Google may have.



(Source: Shutterstock.com)Figure 3.4: An open-plan office design

Beyond open-plan office spaces, there are more trends in office design that continue to inspire businesses to change the look of their offices. One of these is to create ‘green spaces’, or to incorporate the ‘outside in’ approach, which essentially brings nature into the workplace, by decorating the space with plants and greenery. This can create a calming and comfortable space, as plants not only bring colour into the room, but also help to freshen the air and improve oxygen circulation (*Fuze Business Interiors*, n.d.).

According to research conducted by Nieuwenhuis et al. (2014), there are three main benefits that plants can have in the workplace:

1. Plants positively influence ‘the climate of the working and living environment’ (Nieuwenhuis et al., 2014).
2. A green, planted environment reflects the natural world, and thereby supports human physiology.
3. Employees are inclined to view a green working space as one which enhances their well-being.

Plants influence the working environment positively by cleaning out impurities in the air during their natural process of producing oxygen. A working environment that reflects the natural world allows our minds and bodies to feel more at ease in natural spaces. Therefore, by adding natural spaces to the workplace, employees will feel more comfortable. Finally, it has been proven that employees feel that their employer cares

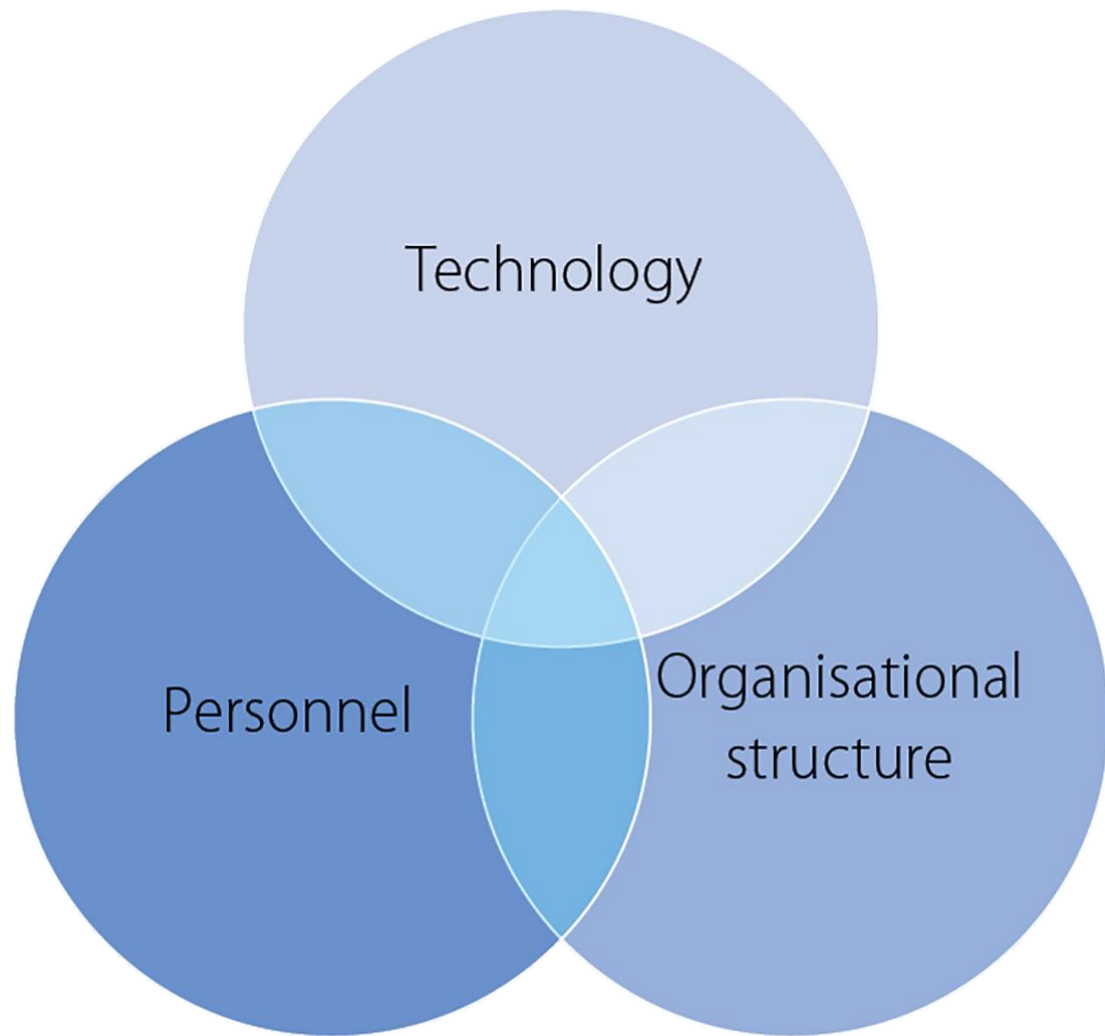
more, if an effort is made to furnish the space within which they work. This enhances their sense of well-being as a result (Nieuwenhuis et al., 2014: 200–201). Simply seeing the effort that management makes for their employees' comfort at work is enough to make employees feel better in the workplace.

3.4.1.1 *The psychology of the open-plan office*

While we have listed some of the benefits of the open-plan office designs, there are some concerns that must also be considered. An experimental study conducted by Smith-Jackson et al. (2016: 810) identified three major areas that accounted for differences in employee satisfaction in the open-plan office:

1. *Satisfaction with privacy/acoustics*: In an open-plan office, it can often be difficult to have private conversations or perform simple tasks without other people being aware of your actions. As a result, employees can sometimes become frustrated by the lack of personal space. For example, if someone would like to send a confidential email to their boss or another employee, they may have to do so outside of the office space, in case someone is able to read the message they are sending. This can waste time or cause feelings of discomfort among employees.
2. *Satisfaction with ventilation/temperature*: The air quality and temperature in the office should be managed so as to create a comfortable and constant environment that suits the majority of employees. This can be done by installing heaters and air-conditioning units, to make sure that the environment is suitable across different seasonal changes. The quality of air can also be improved, by increasing the number of plants in the office.
3. *Satisfaction with lighting*: Poor lighting conditions can put strain on employees' eyes, either because the light is too bright or too dim. Some people have sensitive eyes and require less direct light than others, and might choose to sit in a darker area of the office. Other people may prefer a lot of direct light, and would perhaps make use of an extra desk lamp.

An open-plan office creates what is called a sociotechnical environment, which involves the employees, the leadership, the devices that are used to perform work, and the relationships that explain how each of these factors are interlinked. A sociotechnical environment recognises the value of technology in the workplace, but also places great importance on human resources, and the way in which they operate the technology. This interconnectivity is represented in the following Venn diagram.



(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Smith-Jackson et al., 2016: 809) Figure 3.5: The sociotechnical environment

This shows that the sociotechnical environment involves the needs of personnel, the needs of the organisation, and the availability of technology. The interaction of personnel with technology is a reciprocal one, which means that they each benefit from interacting with one another. This could happen where employees are using a certain computer program or piece of equipment, and by doing so, their job is easier to perform. The employees would also provide feedback about how the technology works for them, and where it could be improved. This is a reciprocal relationship, as both the choice of technology and employee productivity is improved.

There is an interdependent relationship between the organisational sector and technology, which refers to the fact that the type of organisational structure determines the type of technology needed. At the same time, the available technology (whether constrained by cost, space or time limitations) will determine whether certain organisational structures are viable. For example, if a small company wants to install a

very large computer server on their premises, they may not have the space or the funds to do so. It would also not be practical, due to the low number of expected users. We will now look at three related areas, namely *people*, *projects* and *technology*.

3.4.2 People

The organisation of people in the workplace is usually determined on a functional basis. This means that you would consider the factors that influence how individuals work, and which combination of factors would be best for their particular position. The needs of employees and their particular strengths would also be considered. Examples of the suitable working environment in accordance with this are presented in the following tables.

Employee needs	Solution
Light sensitivity	You may consider placing the employee in a darker space.
Noise sensitivity	The employee can be placed in a quieter environment, such as a private office.
Easily distracted	Employees who become too easily distracted may need to be placed in an office where they can work alone without distractions – particularly those in higher positions.
Asthma / respiratory difficulties	The employee should be seated where there is a good, fresh air supply.

(Source: EDGE Learning Media (Pty) Ltd, 2019)

Table 3.1: Examples of employee needs and solutions

Employee strengths	Work environment
Works well in a team	Such an employee will work best at desk pods, where they can easily work with others.
Works well alone	This type of employee should be placed in a private office space if possible.

Works well in an open environment	The employee will work best in an open-plan layout, where they can interact with others.
Works well in an enclosed space	The employee should work in a private cubicle or private office space.

(Source: *EDGE Learning Media (Pty) Ltd, 2019*)

Table 3.2: Examples of employee strengths and suitable working environments

3.4.3 Projects

A project is a specific piece of work that has a distinct start and finish, and requires a particular team of people with the right skills to bring it to completion. A project team will often be created by selecting existing employees at the business who have the appropriate skills. These team members will then collaborate with one another, in order to complete the project. To ensure that the project team can work together without interrupting others, some seating changes might occur, or there may be a specially designated meeting room for the team to use.

Projects need to be carefully coordinated; for this purpose, there should be designated meeting spaces, which depend on the nature of the work. These spaces could be physical, such as an office or a boardroom, or they could be digital, such as a conference call or a team Internet chat room. Using programs such as Skype for Business or RingCentral Office will allow teams to connect with many different members across the globe. Note that the speed and clarity of the connections to other countries and areas will depend on the quality of the Internet connection on both sides.

It is also possible to bring in external help to a project team through outsourcing. If a temporary employee is brought in from outside of the business for the duration of the project, short-term seating changes may be made for this individual, to ensure that they are able to work closely with their team. The team may also bring in consultants from within the business; for example, if the team needs an expert from the IT department to provide input about how they can present the project online, they would be able to ‘borrow’ that employee for a meeting.

3.4.4 Technology

Modern technology still performs the same functions as older technologies, but in more imaginative and revolutionary ways. Communication has been streamlined so that businesses can speak directly with their consumers through social media, and handle

complicated support calls with the assistance of photographs and video content. Clients can be contacted with video conference calls or given virtual tours of the workplace without needing to travel, and email can be used to send official correspondence. Even employees benefit from the advances in communication through electronic media; there are several programs that allow users to create chat groups and resource sharing groups, which can keep everybody up to date with the latest news and changes in the company. Employees can even e-commute to work, which is when someone logs on to their work system from home or another location; for example, being able to work on the actual business server, while travelling to a foreign country. This also presents employees with an increased level of flexibility within work.

Documentation is another area that has been completely revolutionised by the introduction of electronic storage and media. Advances in storage mean that we can now keep vast amounts of data on our computers, which saves a significant amount of physical space. Files are automatically indexed when stored on a hard drive, which makes them easy to find and access using the computer's search function (provided you know the correct file name). One of the most efficient ways to transfer large amounts of information is via the Internet, through services like Google Drive, the Cloud or Dropbox. You can now store up to 100 million pages, which is the equivalent of four terabytes (TB), as opposed to a traditional filing system which can only store 1.5 gigabytes (GB) or 40 000 pages.

However, with all of these technological advances comes responsibility: the responsibility to be more digitally literate. As offices convert to using predominantly electronic devices and media, we must continually learn how to use these devices and media more effectively. In the past, a basic understanding of email and word processing would probably have been sufficient to claim that one is digitally literate. Today, however, the list is a bit longer, and people are encouraged to be literate in more than one form of media. For example, a working knowledge of the Microsoft Office suite and Google Docs and Sheets is a must. Then there are also the multitude of devices to consider. Being able to use a desktop computer, tablet and mobile phone are essential, as is a basic knowledge of how to perform certain tasks on both the Apple iOS and Windows-based operating systems. Tasks will vary between businesses, but good digital literacy is key to navigating the modern workplace.

3.5 What interaction does the 21st-century office invite?

The way that employees interact with their workspace can depend on a few variables, including their natural work style, how familiar they are with the virtual workspace they are in, and whether the working environment itself is flexible. The 21st-century office will bring you into contact with people who approach work, and work relationships,

rather differently. You might come across someone who does not think like you do. Rather than seeing this as a challenge, you can use it to your advantage.

A diverse team that has many work styles is stronger, because of the different areas they focus on. You need people who focus on data, details, expression and the strategic ideas – and very few people can cover all of these bases by themselves. This is why it is an advantage to recognise the work style that individuals have, so that they can be encouraged to work to their strengths, and not to fit themselves into a style that does not come to them naturally.

3.5.1 Workplace diversity: Different work styles

Everybody has different qualities, which means that they will approach their work in very different ways. The quality and speed of work may not only depend on the organisational layout or the needs of the employee; they could, in fact, be more fundamentally built in. Our general work style is seen as a reflection of how we make sense of the world, more than a skill that we learn and develop. In this way, building a team of diverse working styles can be easier to do when you hire a new employee, rather than trying to change the work style of an existing employee (Tate [a], 2015). This is because each person's work style is developed within themselves, and is formed by many influences as they grow and learn.

3.5.1.1 Productivity styles

We each develop a particular approach to work, and this can be influenced by our home life and schooling, as well as our general temperament. According to Tate [b] (2015), there are four main types of employees that can be defined:

1. *Prioritiser*: Logical, analytical and data-oriented. The prioritiser will function best when given tasks that they can quantify, and that have a logical start and finish. They are good at keeping track of what needs to be done by listing tasks, and have a keen eye for working to deadlines.
2. *Planner*: Organised, plan-focused and detail-oriented. The planner will be able to block their time by task, ensuring that they spend the appropriate amount of time on each job. This enables them to focus on the finer details of each project they work on.
3. *Arranger*: Supportive, expressive and emotionally-oriented. The arranger will be able to express ideas in ways that others can understand, and provide useful

support to their colleagues by offering their opinion. When planning a project, they will often first consider who needs to be involved, or who they work well with, in order to get the job done.

4. *Visualiser*: Strategic, integrative and idea-oriented. The visualiser will be able to take unrelated ideas and link them together, which makes them very important for reaching consensus between others in meetings. They have a tendency to think about how certain projects will fit into the overall vision of the business itself, and how this can be maximised.

Depending on your particular work style, you may be better suited to certain approaches to work and not to others. If you know that you are a planner, for example, you will want to have set deadlines and clear tasks that can be organised into work blocks. A prioritiser, on the other hand, would be more comfortable working on tasks that have a required element of analysis. It is not uncommon for people to have a combination of these work styles; you could be an arranger and a visualiser who excels at both integrating ideas and expressing them in a way that others can easily understand.

3.5.1.2 Productivity style assessments

In order to manage or coach employees according to their work style, you must first determine what their natural work style is (Tate [b], 2015). The four-element work style assessment on the following website will score you based on your answers regarding prioritisation, planning, arranging and visualising (Tate [c], 2015):

hbr.org/2015/01/assessment-whats-your-personal-productivity-style

This assessment will provide insight into your work style, by classifying you into one of the four main employee styles. Understanding this about yourself is important, as it helps you to understand exactly what the optimal way for you to work is.

There are also other assessments that can provide insight about how an employee thinks and how they would likely approach their job, such as the Myers-Briggs Type Indicator (MBTI). This test has 16 outcomes. Among these, the following elements of an individual's personality are measured (Owens, 2015):

- Extraversion versus Introversion
- Intuition versus Sensing
- Thinking versus Feeling
- Judging versus Perceiving

To briefly elaborate, extraverts are individuals who are more outgoing and sociable, while introverts are more withdrawn, and can become easily worn out from too much

social interaction. Intuition refers to the extent to which an individual relies on their gut feel and on faith alone, as compared to someone who relies more on sensing, which would indicate that this individual depends more on the facts and the physical evidence.

Those who rank higher on thinking are individuals who are practical and reasonable, and rely mainly on logic. Those who rank higher on feeling are more emotional and driven by their emotions. Finally, those who rank high on judging are more orderly and organised, as compared to those who rank high on perceiving, who are more flexible, and do not mind uncertainty.

The MBTI is one of the most comprehensive personality tests developed in recent history, though it can be difficult and time-consuming to conduct a test of this magnitude on each and every employee.

The 'Big Five' personality test is another measure that can help you to understand how you work with others. This is similar to the MBTI, in that it also measures the level of extraversion. However, over and above this, it also measures openness, level of agreeableness, conscientiousness, and finally, level of neuroticism.

Openness refers to the extent to which an individual is open-minded and open to new things. Agreeableness is the extent to which an individual is flexible and prepared to go along with what others ask of them. Conscientiousness is how disciplined and organised an individual is, and neuroticism is how anxious and emotionally unstable an individual is (Owens, 2015).

Based on the scores you receive for a Big Five personality test, you will be able to tell how you react to others. For example, you might find that you score highly on agreeableness, and identify that, sometimes, taking on too many tasks in order to help others is slowing you down. A more targeted test can be developed by the company themselves, to see how potential (and even current) employees would react in realistic situations that might occur in the business. These are called situational judgment tests (STJs), and can be very useful for testing an employee's approach to work under pressure, or their critical problem-solving skills (Owens, 2015).

Coaching employees according to their work style means that they will be encouraged to use their strengths to perform as effectively as they can. Providing employees with tasks that work to their strengths will make them feel more competent in their work, and this will also increase their job satisfaction, because the work will make intuitive sense to them. They will not have to put in extra effort to understand how they are supposed to do their job, because they will already know how to approach it. Where projects are concerned, it is vital that a team is made up of people with various work styles, so that the workload is balanced across different skills and approaches (Tate [b], 2015). This

will ensure that each aspect of the work is addressed by someone who will take their tasks on with confidence.

3.5.2 Virtual workspaces: The workplace is no longer a destination

Companies are connected across the world by virtual workspaces that allow them to coordinate and share information. Near instantaneous sharing has become the norm for companies that operate over large geographic spaces. Contract workers and freelancers can work for companies all around the globe, provided that they have a suitable Internet connection.

Freelancing sites such as Upwork connect individual workers to companies who require their skill set. Even full-time employees benefit from having the ability to commute to work remotely via the Internet, if they need to work from home. This is especially useful for people who live quite a distance from their physical workplace.

For example, if a website publishes global news every day, they might hire freelance journalists in many different countries to submit articles that they may want to publish on their site. The journalists would work in their own country, on their own time, and submit the work to the website when it is needed. The journalist may never actually meet their employers, but they will be able to communicate their needs to one another well enough to maintain a working relationship via the Internet. It costs the company less to outsource the job than it would to set up a media office in every country, and it gives the freelancer the freedom to work at their own pace and take on more than one task at a time.

3.5.2.1 *Transparent virtual workspaces: Sharing information*

The sharing of information in modern businesses is a vital aspect of successful communication. Without effective communication systems, companies can lose thousands of rands every month, due to loss of time, misunderstandings, or incorrectly performed work. Increasing the transparency of information in both communication and file sharing is the way to combat these losses before they happen. When a decision is made about any given issue, it is now easy to include members of staff who are not in the room at the same time, by creating virtual workspaces. If there are people missing from the office when important notices go out, they can be updated via email. This improves the transparency of the business, by allowing employees to catch up on important administrative issues at their own pace, and to make sure that they are not left out of the loop.

All sorts of meetings, deadlines, and other correspondences can be orchestrated online, and employees can be reminded of their commitments hourly, daily or monthly. What sort of measurable benefit can this have for a company? According to Johnson

(2017), transparency can have four major benefits to the inner workings of business processes:

1. *Better decision-making*: Information should be made readily available to employees, so that when decisions must be made, everybody is up to date, and is prepared for casting their vote or making their voice heard. Decisions can be made more quickly, and potentially, more voices can be represented, if decision-making processes are carried out through a digital platform.
2. *Transparency promotes trust*: The more employees know about the work they are doing, the more they trust one another. For example, recording minutes (organised notes) in a meeting can keep track of any employee contributions or concerns. Employees will feel that their voices matter, and that they will be given credit where their ideas succeed, or feedback where their ideas do not succeed.
3. *Collaboration and synergy*: When people know how their work affects others, and where they can utilise the time and skills of others, it can create an atmosphere of synergy. For example, when you can see that your job needs to be completed before your co-worker can perform their task, this might become a motivating factor to meet your deadline. This is because you know that your co-worker will be put under extra pressure if you do not do your own work.
4. *Integrated workflows build value*: Representing the current work that employees are doing on a visual platform, such as a workflow, can make the process of work easier to understand. Each employee will be able to see how a project is made up of various tasks, and when tasks will need to be completed, in order to deliver a project on time. Due dates assigned to members of staff, and any other important information, will be recorded on this workflow.

Virtual workspaces are areas where multiple people can contribute at the same time, such as on online document programs like Google Docs, Microsoft Word Online and Zoho Docs. Each user is given an identifying name and colour on the document, so that their contribution can be tracked. A document history is also kept so that each edit is recorded, in case changes need to be rolled back.

Even school systems across the country are using programs such as the d6 School Communicator, which is an application that can periodically update parents about the latest news. Updates can be sent as a general message across the entire school, but resources can also be divided up between each grade and class if need be. This easily accessible information allows parents to be more involved in their child's schooling, from their own workspaces.

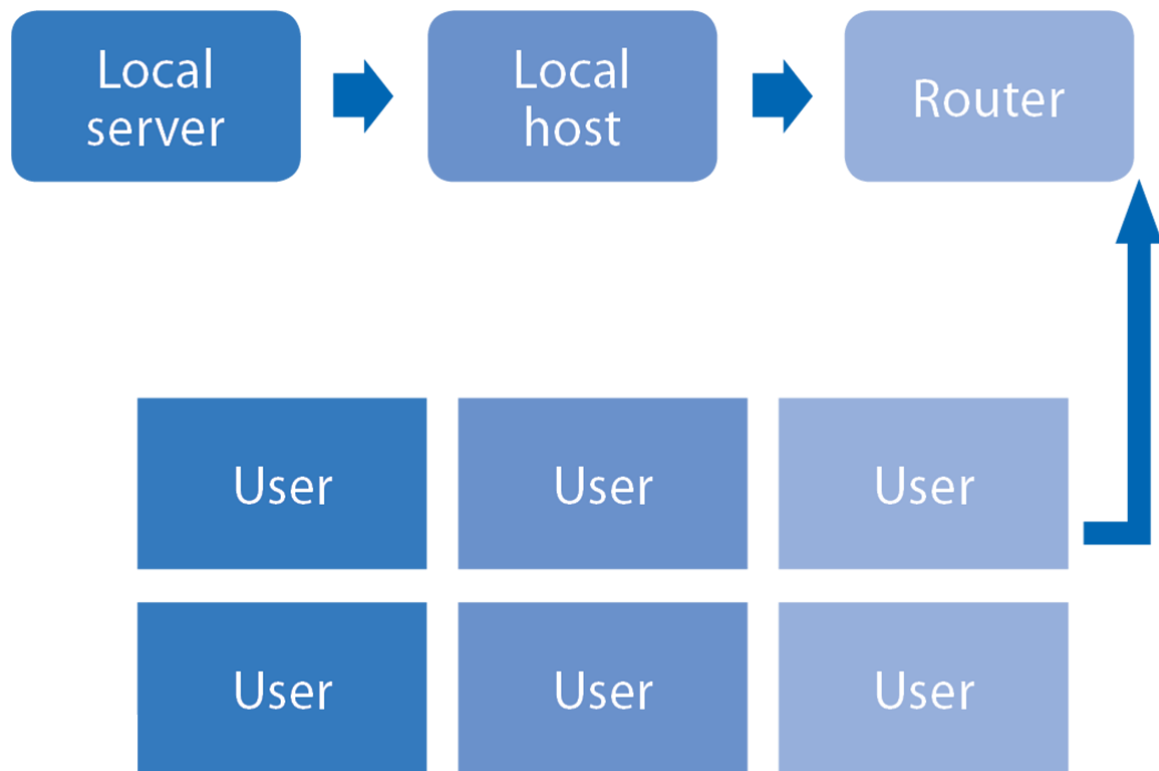
3.5.2.2 Networking

In order to share information between computers without using an external storage device, such as a USB drive, it is most common for businesses to have an internal network. This network connects various devices, so that they can communicate and transfer information among them. A network can be connected by cables, or wirelessly, but there will always be a router that manages the flow of data traffic. We will now look at a few network configurations that are common to the workplace, and at connecting to the work network remotely.

Local area network (LAN)

A local area network (LAN) at a business will usually have four core components:

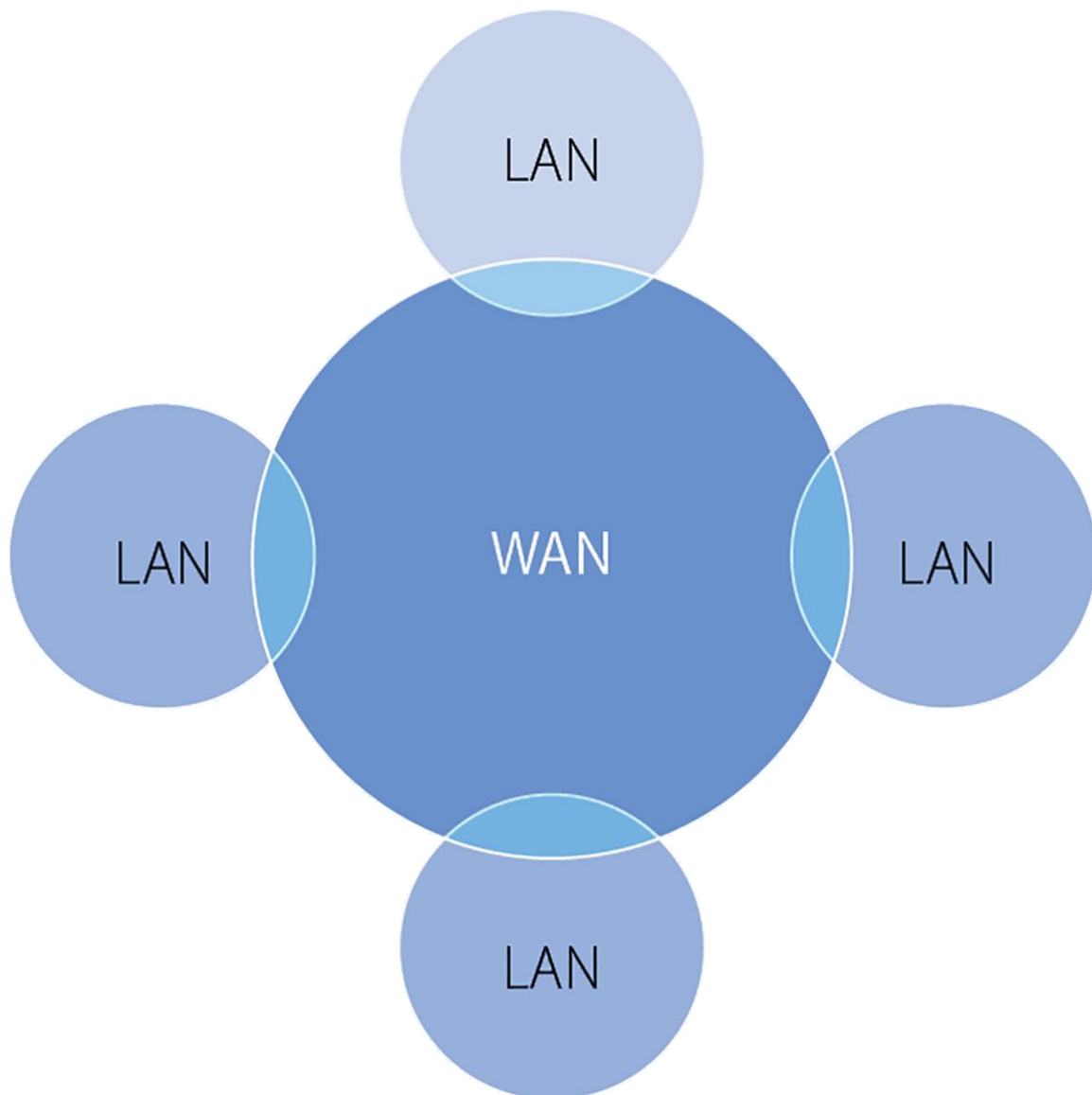
1. *Server:* A server usually consists of a number of hard drives that contain all of the data produced by the business. The server is accessed by the host, and users must each be assigned permission by the host when they want to add or remove content from the server itself. Important servers will have a second set of hard drives called 'mirrors'. These are backup copies of the original hard drives, in case one or more of the drives fail.
2. *Host:* A local host manages the permissions and connections that are made between the router and the server. Any new users that need to be added, or special conditions that need to be set (e.g. allowing a user to view, but not add or remove files), is managed by the local host.
3. *Router:* A router manages the data traffic between the host and the users, connecting them either physically via cable, or wirelessly by Wi-Fi. The router will assign a unique Internet Protocol (IP) identity to each device that connects to it, in order to make sure that there are no clashes during transfer. The IP address is usually found in the form of a number code (e.g. 192.168.100.10), and this identity is used even if the device is not connected to the Internet.
4. *User:* A user is someone who works on their device (i.e. a desktop, laptop or even mobile device), and accesses the server by connecting to the host via the LAN connection. Users may be able to see one another on the network if their network visibility is turned on, or they may only see the main server in their network locations.



(Source: EDGE Learning Media (Pty) Ltd, 2019)Figure 3.6: Local area network (LAN)

Wide area network (WAN)

A wide area network (WAN) connects several local networks together, through a common exchange point. The largest example of a WAN is the Internet itself, which connects millions of users to one another through both wired and wireless connections. Content shared on the Internet and other WANs might seem to exist only in digital space, but each set of data must be stored on a physical server at some place or another.



(Source: EDGE Learning Media (Pty) Ltd, 2019) Figure 3.7: Wide area network (WAN)

The figure represents a WAN that we could imagine between four different businesses, or four branches of the same business across different provinces. Each of the networks would have an internal LAN, which would then connect to the other networks via the Internet. The number of connections made to the Internet is exponentially larger than this example; in 2017, there were a recorded 3.57 billion Internet users (Statista, 2017). As you can imagine, managing the data traffic across different websites with this many Internet users can be a very challenging task.

Cloud storage

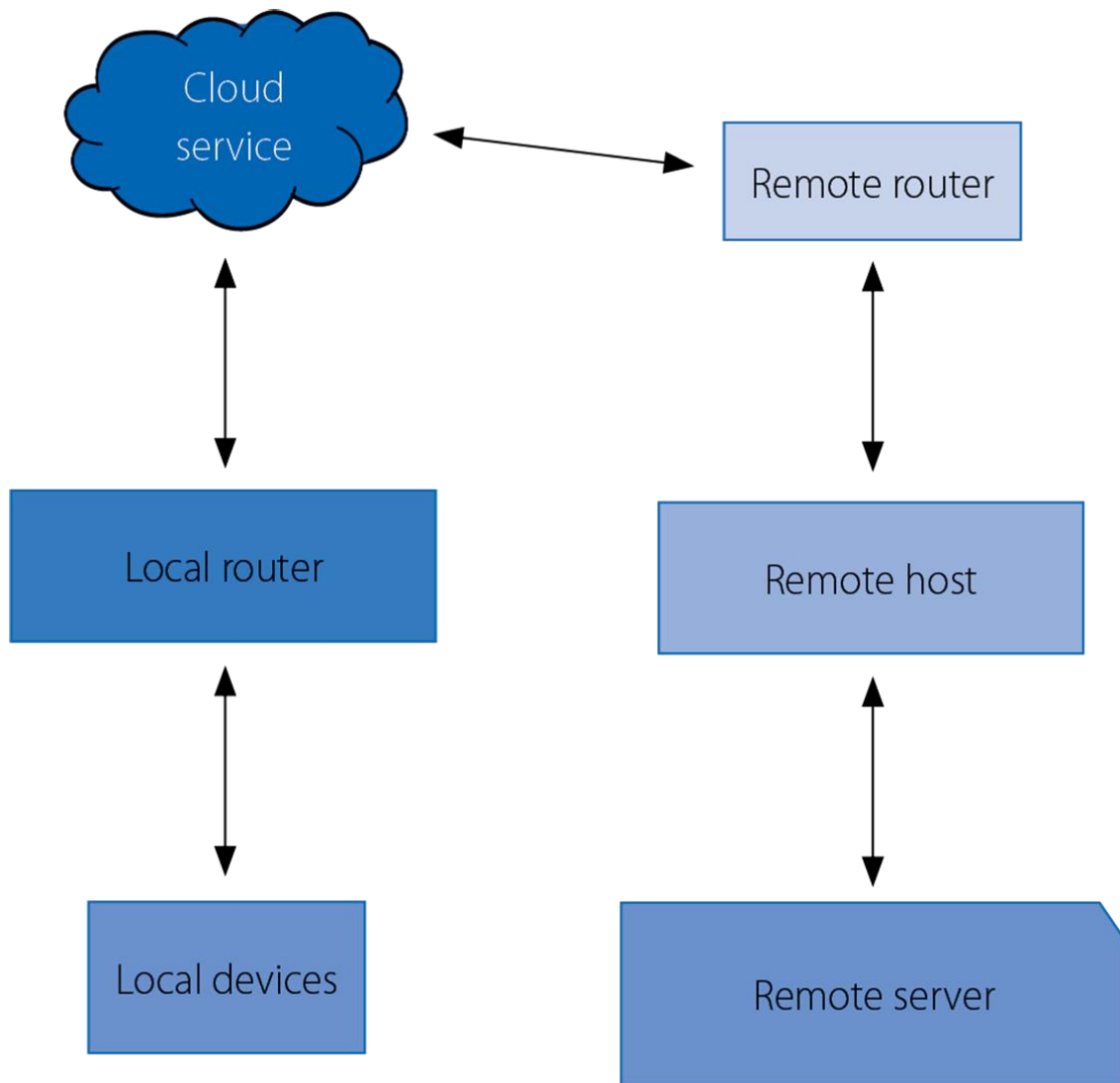
Cloud storage is a service that some businesses provide on a WAN. The 'cloud' refers to the invisible space that is created for users to store their data, and is often represented with an image of a cloud, because we think of it as something that exists 'above' the

existing Internet services. In reality, cloud storage services like Google Drive use the Internet to link users to their data servers. This means that the service is not truly 'invisible', because all data has to exist on a physical storage device somewhere. The reason we think of the cloud as invisible is because we only interact with it on the Internet, and do not get to see the physical servers that keep our data safe. Your data is backed up in large databases and servers that are maintained in warehouses, such as the Google server room. The following image is an example of such a server room.



(Source: Shutterstock.com)Figure 3.8: Example of a server room

All cloud storage services rely on the Internet or some other WAN to function, and users can only interact with their data through their connection to the WAN. This means that one of the limitations to cloud storage is Internet connection. If you have a bad Internet connection, you might experience slow transfer speeds or an unstable connection, which could occasionally cut off your access to your files. In areas with bad Internet infrastructure, this can be particularly difficult to deal with. A few of the most popular free cloud storage services include Dropbox, Google Drive, MEGA and Microsoft OneDrive. There are also premium services, which offer larger storage spaces, and more sharing and editing options; however, these often come with a cost, varying from approximately R 300 (50 GB) to R 1 200 (5 TB) per user, per year (Microsoft, n.d.). Different types of work that can demand large amounts of storage space include video editing, audio recording, or photography post production work.



(Source: EDGE Learning Media (Pty) Ltd, 2019)Figure 3.9: Cloud network

In the case of a virtual private network (VPN), a user will connect to their business's local network through the Internet. Similar to a cloud network, the VPN allows users to access a server remotely, but in this case, the server remains in the physical workplace. The private aspect to this connection is that it will require a username and password to access, so that only registered users with permission are able to access the server. The data itself can also be encrypted across a private connection, which means that it is transferred within a special format that cannot be read by people who are not the intended recipient of the message. VPN connections are often used when sensitive data needs to be accessed, such as financial statements or payroll records.

3.5.2.3 Digital citizenship at work

Digital citizenship refers to your online presence, and how you are linked to others by that presence. Your online presence exists publicly in the sense that, once you have set

up profiles on social media or other sites, you can be found through most search engines. Privately, your online presence exists in the particular connections, such as direct messages, commentary and sharing, that occurs between you and other people or businesses. Social media is the biggest example of where digital citizenship can be seen in action. Facebook, Twitter and LinkedIn are all popular social media sites that allow you to create a personal profile, where you can indicate where you work, and what your qualifications are. You will typically also upload a photograph of yourself so that people can recognise you. However, for reasons varying between privacy, preference or otherwise, some people choose not to upload a photograph of themselves. A social profile is one part of your digital citizenship that people can see, without needing to search too deeply.

However, you will likely also share content that you find interesting, make comments on posts and have discussions with others. These interactions represent you in a very concrete way. Comments made on social media can be seen for years, even after you have forgotten that they are there. These residual pieces of information are known as your digital shadow, which tells a story about you. Whether that story is true or not is left up to the people who are seeing the digital shadow (*Me and my Shadow*, n.d.).

According to Luna (2015), there are nine Ps that are important to remember when interacting with content online, in order to practise safe usage, and to limit your digital shadow. Each of these will be discussed in detail.

1. *Password*

Longer passwords are more difficult to copy or steal, so it is a good idea to plan your passwords carefully. You may also encounter sites that require you to use symbols, numbers, and upper- and lower-case letters to strengthen your password. It is advisable that you use a different password for every account you create; otherwise, if someone manages to steal your password for one of the accounts, they will be able to access all of your accounts. There are password management applications that can assist you in developing and keeping track of complicated passwords, such as LastPass, Dashlane or RoboForm. You will have to remember one password to log in to the application, which will then grant you access to all of your passwords for your other sites. These password management applications have browser integration too, which means that you can log in to websites at the click of a button.

2. *Privacy*

It is a good idea to keep your personal information away from people who do not need to know it. This includes details such as your physical address, identification number and phone number. You may need to share these details privately, but it is safer to avoid making them visible on your personal profiles. Some Internet users may ask you to provide your personal details under fraudulent circumstances; for example, by

pretending to be your bank and asking for your pin number. This is called phishing (pronounced 'fishing'). A simple rule to follow is to avoid sharing private details across the Internet, unless the person you are communicating with is able to prove their identity to you first.

3. *Personal Information*

Similar to your privacy and passwords, it is good practice to avoid sharing personal information about yourself that may lead others to know your location, or what your passwords may be (such as a pet's name or your parents' names). This is because personal information is often used to form passwords, or as a means to identify oneself to friends and family. If someone is able to provide intimate details about your life, they may be able to impersonate you.

4. *Photographs*

While this may be similar to privacy, it forms a separate entry, because photographs can contain specific information and can be shared by multiple parties. Apart from potentially embarrassing pictures of your private life, photographs can also contain information that people can use to track your movements; for example, your car licence plate, landmarks in the background, the people you are with and more. Geotags (an identification of your location in the picture) and 'check-ins' are also ways that your photographs can broadcast information to others. For personal use, it is advisable to turn these features off.

5. *Property*

The content that you find on the Internet comes from many sources, some of which may be protected by copyright law. Copyright is the legal right of the content producer (an artist, designer, writer etc.) to be associated with their work as the original creator. Some content may be covered under limited use copyright licences, or be completely free under a 'creative commons licence'. Other content may have been sold to a specific company by the original creator, which would mean that the company owns the rights to the property and the original creator no longer does. You should determine whether you are allowed to copy an image for your own uses by the type of copyright it is protected under. If you are a content producer, you can perform routine checks using Google images or a regular Google search to see whether your work is being misused by others, and request that they stop doing so.

6. *Permission*

When you share content, or use it in a public space, you must obtain permission from the holder of the copyright. If you approach the creator of the content and they say that you are not allowed to use their work, then you should stop doing so until you have permission. If you do use someone else's work without permission, this is seen as

plagiarism, and could even constitute copyright infringement if you intend to profit off of the content, or associate yourself with it for some other gain. There are two kinds of plagiarism that could be committed: fraud and theft. Fraud refers to the failure to mention that the content is not your own. Theft refers to presenting someone else's property as your own, and standing to benefit financially or otherwise from it. Citing your sources is a good practice in both your personal and professional life.

7. Protection

Viruses, malicious software (malware) and spy software (spyware) are commonly hidden within files that can be downloaded from unsafe websites. To prevent yourself, and others, from being affected by this, it is necessary to use either background or live browsing protection for your computer. Not only can devices download viruses from the Internet, they can also copy them across USB drives, CDs, DVDs and mobile devices. Scanning every device that you plug into your computer will save you from transferring viruses between them. Some examples of viruses include a file infector virus, which corrupts file data; a 'browser hijacker', which only allows you to open a pre-set page; or a resident virus, which occupies a portion of your computer's memory, and has the ability to move to other sectors of your computer once detected (Young Upstarts, 2016).

8. Professionalism

Online etiquette, known as 'netiquette', is a practice that will keep your digital citizenship card clean. It involves using proper academic language, citing your sources, and practising respect and patience with others. If you happen to get into a disagreement with somebody online, you should remind yourself that your responses represent who you are, and that others may be reading them. This will help you to craft a respectful response that will not misrepresent your message or upset others unnecessarily.

9. Personal brand

The last item on this list is you. As the information that you post about yourself online stays in your digital shadow or is shared publicly, this becomes your personal brand. Your personal brand is something that you will want to curate rather carefully, so that other people see you in the right light. If someone views a comment that you made in anger, or an embarrassing photograph of a night out, this may tarnish your reputation, no matter how long ago that content was created.

Remote commuting forms a further part of digital citizenship, and involves signing in to the internal work network from home or another space. It is called remote commuting because you will still be able to interact with your co-workers, even though you will not be with them in the same physical room. This can be done through private digital

organisational platforms, such as Slack, Azendoo, Atlassian or Bitrix24, which each allow you to send messages, pictures, create project groups and more.

3.5.3 The flexible work environment

Employers are opening up to the idea of adjusting working conditions so that they are more convenient for their employees, instead of focusing purely on the output or profit that they make. This includes allowing their staff a bit of leeway where work hours and workspaces are concerned. Some employees might be given a flexible work schedule that allows them to arrive within a certain two- or three-hour period; for example, 07h00–09h00, provided that they then work their allotted hours each day. Other employees who live far from the office might be offered the chance to commute remotely from home, provided they can track their hours while they work.

Shift work is another way to handle flexibility, if the nature of the work allows for it. For example, if a company's offices are open 24 hours a day, seven days a week, employees may be permitted to work in their hours, as and when they can – provided that their minimum number of contracted hours are met each day. This is especially important in international offices and technical support sectors that may require constant, around-the-clock maintenance for their clients. If employees are employed in different countries (e.g. if a call centre needs people who work in different time zones), these employees would work in shifts.

Another aspect of flexibility is in the arrangement of the personal workspace itself, where employees could be encouraged to make the space their own. This could be as simple as allowing employees to work with headphones on if their working environment is usually rather loud, and it can extend as far as the design changes we have already discussed.

3.5.3.1 Taking control of one's own workspace

There are various things that can be done to make a workspace feel like it belongs to you, or at least that it is your own space where you can feel comfortable and productive. We will discuss both the workplace itself, and the possibility of working from home, as these will have different requirements.

When at the workplace

In the workplace, you could adjust the light, airflow and noise to your preference. This will likely have to happen as a negotiation between yourself and your co-workers, as each of us has a different preference for hot or cold, whether we breathe air-conditioned air or not, and how much ambient (atmospheric) noise becomes distracting. Having some familiar objects on your desk, such as a small pot plant or a

photograph, can help to make the space feel personalised. Using noise-cancelling headphones to limit ambient office noise can help you to focus in a noisy environment.

When at home

On top of the concerns that you might have in the workplace, you could choose a place that you designate for work and clear it of distractions. As you will likely be close to all of your comforts, including entertainment, it is advisable to designate an area that is only used to do your work. This way, each time you sit down to work – or stand, if you have a standing desk – you will be conditioning your brain and body to associate that area with work. The key to working at home is to get into a regular routine, and a work area will help you to do this.

3.6 Planning employee-friendly workspaces

Work can be a stressful place to spend long periods of time, so it is important for companies to promote healthy habits, and provide spaces that are comfortable for people to work in. Employee wellness is a focal point of many businesses in the 21st century. A number of different practices and amenities are encouraged in order to promote wellness in the workplace, including (*Morgan Lovell, n.d.*):

- regular catch-ups or contact with others;
- movement during the day;
- flexible hours;
- fun environments;
- office plants;
- increased natural light; and
- giving employees recognition for work done well.

Not only should physical and mental conditions at work be improved, but health incentives can also be put in place to encourage the personal health goals of the employees. Not only will this make the employees feel better when they are at work, but it will allow them to follow their personal health goals, which may be difficult to meet while holding a full-time job. Some of the health incentives offered by companies include (*Morgan Lovell, n.d.*):

- free fruit;

- ‘stop smoking’ groups;
- gym membership deals;
- stress management; and
- cycle spaces.

3.6.1 Ergonomic set-up

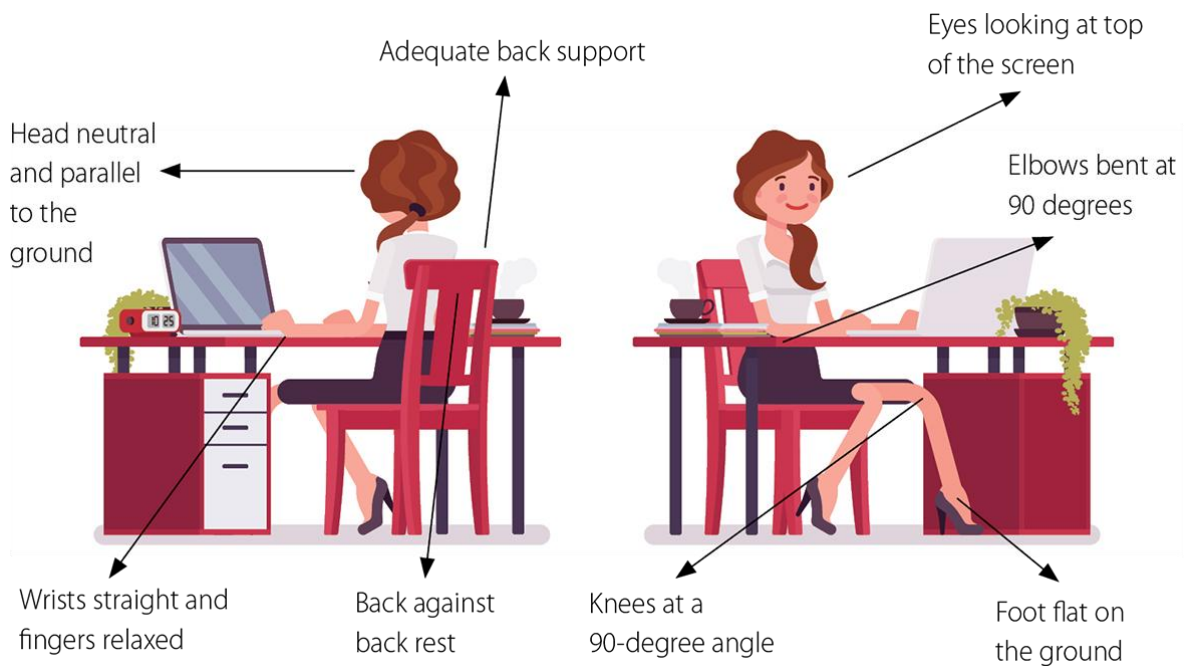
You may have come across the word ‘ergonomic’ when shopping for furniture or office equipment, but what exactly is ‘ergonomics’? Ergonomics includes the ‘theory, principles, data and methods to design in order to optimize human well-being and overall system performance’ (*Ergonomics Society of South Africa*, n.d.). This means that the tools and equipment we use, specifically those we use for long periods of time, should be designed to work with the human body instead of against it. An example here could be the difference between a simple wooden office chair, which has a straight back, and an ergonomically designed composite (plastic that has been upholstered) office chair that has a curved back with extra lumbar support. If you spend a long time sitting on the wooden office chair, you may experience pain in your lower back, because there is no specific back support provided. In the composite office chair, you will feel the effects of sitting down for long periods of time far less, and this is part of the reason why ergonomics exists. It includes health measures that protect your body from harm. We will now consider how seated and standing work differ in ergonomic needs, and the influence of blue light from computer screens.

Seated work

There are some fundamental rules that you should follow when working while seated at a desk, so that your joints and muscles do not take unnecessary strain when you work. These are particularly important to keep in mind if you work in an environment where you will be sitting for long periods of time. According to *Damon Soraya* (n.d.), when seated in front of a computer, the following positioning is recommended:

- The top of the monitor should be at eye level.
- Your chin should be parallel (in line) with the ground.
- Elbows should be bent at 90 degrees, without an armrest.
- Your lower back should be supported by the backrest.
- Knees and hips should be at 90 degrees to one another.
- Feet should be flat on the ground.
- Fingers should be relaxed, and wrists should be straight.

These positions are illustrated in the following figure.

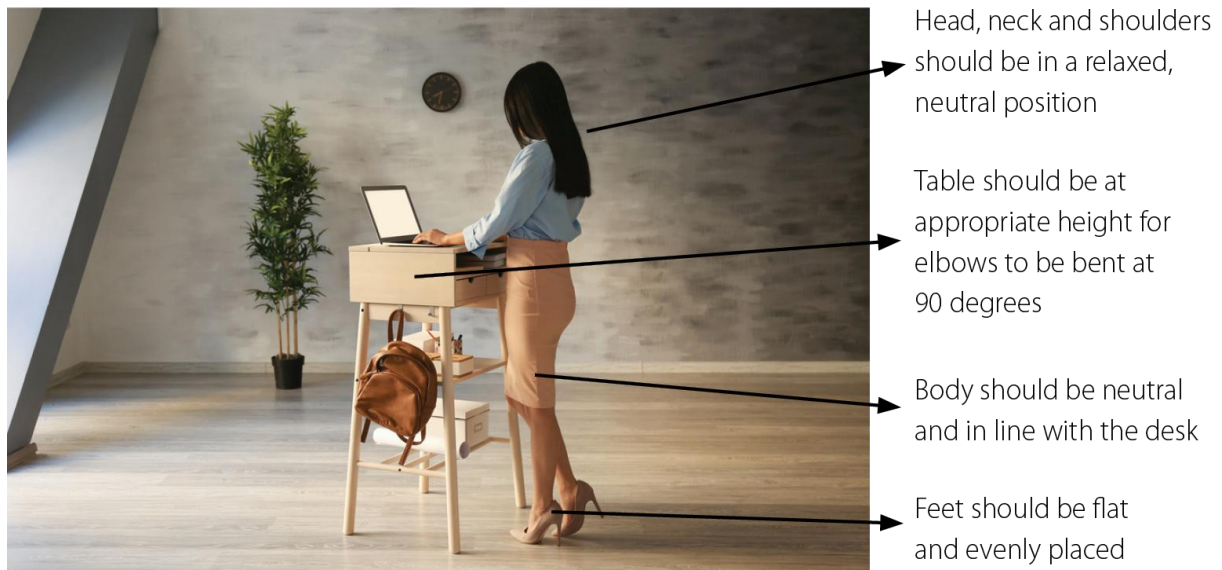


(Source: EDGE Learning Media (Pty) Ltd, 2019; and Shutterstock.com) Figure 3.10: Ergonomic sitting arrangement

Standing work

With standing work, there are slightly different rules, though some measures may overlap. The line of gravity (at 90 degrees to your body when you are standing) is the reference point for how you should hold your body when working. You should be standing straight up along this line, and not leaning forward or backward. Other principles that you should adhere to when working at a standing desk include the following:

- Hold a relaxed posture to limit strain on your neck or shoulders.
- The table height should be in line with your elbows, between 90–120 degrees.
- Your knees, hips, shoulders and spine should all follow the line of gravity.
- Frequent changes between sitting and standing can strain your body unnecessarily.
- Keep your feet flat on the floor, and wear comfortable footwear.



(Source: EDGE Learning Media (Pty) Ltd, 2019; and Shutterstock.com) Figure 3.11: Ergonomic standing arrangement

Tools and machinery

When working with heavy equipment, you should always use the largest muscle to perform the most work (*Damon Soraya*, n.d.). This ensures that your body is distributing the force of the work across the muscles that can handle it, rather than potentially overloading a smaller muscle or joint. For this reason, relying on your back muscles to do the heavy lifting when you bend over to pick up an item is not recommended. Instead, you should use your legs do the majority of the lifting, as they have far larger muscles than those in your back.

Computers

Most computers these days are connected to liquid crystal display (LCD) or light-emitting diode (LED) screens, both of which emit blue light that can damage your eyes by straining them. Limiting your time in front of computer screens, including mobile devices, is an important way to reduce your exposure to this blue light. It is also possible to filter this blue light with specially designed protective glasses. This is useful for people who are unable to perform their job without using a computer for several hours each day.

3.6.2 Creating cognitive spaces at work

Cognitive spaces are the areas in the office that you can use to brainstorm ideas, share skills with your colleagues, or simply take a moment to collect your thoughts. These are often colourful and vibrant rooms that encourage creative thinking, where your brain can take a short rest from looking at a computer screen or pouring over documents.

These areas can be used as a 'think tank', which is where a group of people get together to generate ideas. The concept behind a think tank is that the power of many minds is better than the power of just one, and that by bouncing ideas from one person to another, the end result will be much better than what an individual would otherwise have achieved. Think tanks are often used to solve problems, and to innovate business processes.

Workshops are also cognitive spaces where one or two employees take the chance to teach others about a certain skill or knowledge set that they would like to pass on. This can be done in the form of a formal lecture, an interactive activity, or a mixture of both. Workshops help employees to broaden their skills and knowledge, which will help them to feel more confident in their abilities when performing their jobs.

3.6.3 Creating wellness spaces at work

While a cognitive space can be used for rest or for work, wellness spaces are completely separate from the work area. A wellness space covers both physical and mental well-being, and can take the form of a gym or exercise space, a lush garden area that employees can relax in, or even a simple games room where people can enjoy challenging one another to a round of table tennis. Wellness spaces ensure that employees are able to separate themselves from work for a short break, in order to 'recharge their batteries'; in other words, to recover from the effort it takes to put in hard work for hours each day. Some businesses may even have a canteen or restaurant on the premises, which will allow employees to get refreshments while they relax.

FAST FORWARD: 21st-century workplace organisation, interaction and cognitive and wellness spaces

Afrihost was founded in 2000 by Gian Visser, Brendan Armstrong and Peter Meintjes. The company has since grown to become one of the leading providers of Internet access and hosting services in South Africa. Their business model focuses on customer care above all else, which has earned them a large market share in a short space of time.

Their new head offices were unveiled in 2015, located in the Sandton suburb of Johannesburg. They now occupy a spacious building with a modern workspace and many different facilities for employees to enjoy. Pay particular attention to the layout and design of the rooms as you read further. You should be able to recognise various elements that have been covered in this unit.



(Source: Shutterstock.com)Figure 3.12: An open-plan workplace

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(Source: Shutterstock.com)Figure 3.13: A glass-walled boardroom

The company boardroom is outfitted with a large television, so that conference calls or presentations can be made with ease. There is also an air-conditioning unit on the ceiling of the room, which provides temperature control that would allow employees to

cool down the office on hot days. The space is enclosed with glass so that the outdoors can be seen easily, and there are various potted plants arranged alongside the inside of these windows. An example of this kind of set-up is shown in the previous figure.



(Source: Shutterstock.com)Figure 3.14: A multifunctional lounge area

There are various spaces on the property that are outfitted for comfortable relaxation. These spaces are also media-integrated via televisions and wireless Internet. The space also has a lounging area, similar to the area shown in the figure. The lounge includes comfortable seating, a minibar fridge, a kitchenette with a coffee machine, large windows, and various little plants. The facilities here make the space multifunctional, as it can be used for relaxation or creative work, or even as a cognitive space to take a minute to think.



(Source: Shutterstock.com)Figure 3.15: On-site gym facilities

The office also has an on-site a gym, which is available to staff at all times, with a personal trainer on hand to help the employees achieve their health goals and to use the equipment properly. This is convenient for employees who want to maintain their health, but do not have a lot of time outside of work to travel to a gym.

(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from MyBroadband, 2015)

3.7 Summary

This unit discussed the 21st-century workspace, with regard to the actual physical design of the office, the managing of employees in the modern workplace, and the handling of projects and technology. It also discussed the change in interactions between people in the modern workplace, referring also to employee diversity, virtual workspaces and flexible working environments. Furthermore, ergonomics has been discussed, and examples of an ergonomic sitting position and an ergonomic standing position have been provided. The value of cognitive spaces and wellness spaces were also covered. Finally, a case study examined all the elements of the modern workplace in context.

Chapter 4

4.1 Introduction

Both school and university can equip you with knowledge and skills relevant to your prospective career. However, the skill set required for securing the job is one that you will need to develop separately. You need to understand how to become an appealing applicant to prospective employers. The most successful individuals are those who combine their skills with self-confidence and street smarts. As you draw close to the end of your studies and get ready to enter the working world, the skills required to secure your dream job will be essential to acquire. This unit discusses the most vital skills needed to secure a job and the steps you can take to become the most appealing job applicant possible.

The goal of every college or university graduate is to be gainfully employed. Of course, we all strive for our ideal jobs, but sometimes, getting there involves work experiences that are less than ideal. In most instances, we must slowly work our way toward the ideal job. Getting an interim job that leads us to our destination is therefore as important as getting our perfect job.

The unemployment statistics in South Africa today are a cause for concern. In November 2016, Statistics South Africa released a report that showed that South Africa had an unemployment rate of 27.1 per cent (Peyper, 2016). This highlights how difficult it is becoming for graduates to find employment.

As a student whose goal is to become a functioning and self-sufficient citizen, it is important for you to have a full understanding of what is required to take advantage of available job opportunities. The current South African job market is highly competitive, meaning that you need to be more skilled and better prepared than your peers to secure a job. The guidelines offered in this unit will equip you with some fundamental strategies to increase your chances of finding employment.

After studying this unit, you should be able to:

- conduct a basic job search;
- create a professional online identity;
- create a professional curriculum vitae (CV);
- write a professional profile summary;

- write an effective cover letter; and
- demonstrate a basic understanding of the interview process.

4.2 Key terms and definitions

The following key terms and definitions are applicable to this unit:

- *Behavioural questions*: Questions asked in an interview that require an employee to provide insight into their prior behaviour, as an indication of their future behaviour.
- *Broad-Based Black Economic Empowerment*: A South African legislative policy that specifies that individuals who are previously disadvantaged should be given the opportunity for upliftment. This influences the hiring policies of companies, and the chances of applicants to secure jobs.
- *Employment Equity*: Another South African legislative policy that directs that all individuals should be given equal opportunities in the workplace, and that there should be equal representation of all racial groups in the workplace.
- *Government Gazette*: A newspaper produced by the government that lists jobs currently available in government.
- *Hypothetical questions*: Questions asked in an interview that present an individual with a made-up scenario. The individual is assessed based on their response to the challenge presented in the scenario.
- *Networking*: Socialising within the business world. Through networking, business people get to know one another and try to establish more contacts. It can take place in person or online.
- *Online identity*: The online image you create of yourself through the photographs and content you post, which may be seen by a potential employer.
- *Panel interviews*: A panel interview occurs when an individual is questioned by several individuals rather than just by one. The panel may include members of management, colleagues or a recruiter. Such an interview is often held so that more than one opinion can be obtained.
- *Salary negotiation*: The process during which the employee and employer discuss the employee's salary and agree on the amount that the employee will be paid.

- *Teleconference interview:* An interview in which the interviewer and interviewee are typically far apart. This is facilitated using technologies such as Skype or Facetime, over the Internet.

• 4.3 Job search basics

- Jobs do not simply get offered to you when you complete your studies. You need to market yourself and make yourself known to would-be employers. When your applications are unsuccessful, searching for jobs can be frustrating and disheartening. Often the problem lies in presenting yourself badly. The following job search basics are important to master. Doing so will allow you to establish a profile that enables you to be proactive, and to take the initiative in finding a suitable job.

• 4.3.1 Creating a professional online identity

Technology is growing at an exponential rate. This makes it increasingly important for you to have a professional online identity. But what does this mean? It means that you need to portray yourself as a professional to people in the global working and business world, particularly to those looking to hire. You need to consider how you want to be viewed by these people, and what you want them to see (and not see).

Your personal identity makes you unique: it distinguishes you from everyone else. Your professional identity should do the same. Although it is natural to want your online identity to be formal and to tick all of the boxes typically expected of jobseekers, you need to show potential employers what sets you apart from other candidates. Think of your online identity as your ‘digital footprint’.

A professional online identity is comprised of the different profiles you have of yourself on the Internet, including social media sites. Potential employers often find candidates on professional social networking sites such as LinkedIn and Bizcommunity. Employers also tend to look at potential employees’ personal profiles on social media sites, particularly on Facebook – a site where we reveal so much about ourselves, often without thinking about what we are posting. Much of what we do online can be seen by anyone, so it is important to always be cautious and aware of what we put on our online profiles. While it is normal to have pictures with friends and family on your personal Facebook account, it is important that you remain mindful that all the pictures you post should be respectable. Individuals should also avoid posting scandalous or offensive posts on Facebook.

What constitutes your professional identity? As a guide, ask yourself the following questions:

- What are three words that best describe me?
- What formal education do I have?

- Have I completed any extra certifications?
- Have I engaged or participated in any outreach programmes or community projects?
- Do I work well in a group or do I work better on my own?
- What experience do I have in various sectors?
- What interests do I have aside from my work?

This information is important for your online profile, and should be considered when planning the way in which you would like to portray yourself. Remember, when it comes to your interests outside of work, you must ensure that you stay professional. For example, it would be unwise to include activities such as ‘partying’ or ‘hanging out with friends’ as your interests. Rather, reveal interests or hobbies that show people what you are passionate about. This could include sporting activities, photography, painting, reading, travelling or doing community work. There are countless interests that disclose an admirable character. Essentially, developing your professional identity helps potential employers to get to know you better, and to understand how you could fit in their companies.

Now that you know how you want people to see you professionally, it is worth taking a critical look at your social media profiles, and deciding whether they convey a professional picture or whether they negate the identity you wish to create.

Three of the most used platforms for enabling and facilitating the creation of online identities are Facebook, LinkedIn and Bizcommunity. Let’s discuss some features of these platforms.

Facebook

Today, when you apply for a job or are offered an interview, the chances are that your potential employer will ‘check you out’ on your social media. Facebook is a commonly used starting point for any would be employer who wishes to discover the ‘real you’. For this reason, you should be very careful about what you share on your Facebook wall. Remember to make sure that you set strict privacy settings to ensure controlled access to your profile. Think twice before you share your thoughts, information and pictures. Be selective and censor yourself before you post comments about situations, people or organisations. Be mindful that other people’s comments on your posts reflect on you because they are your ‘friends’. If you do not like or approve of someone’s comments on your wall, you should delete the comments.

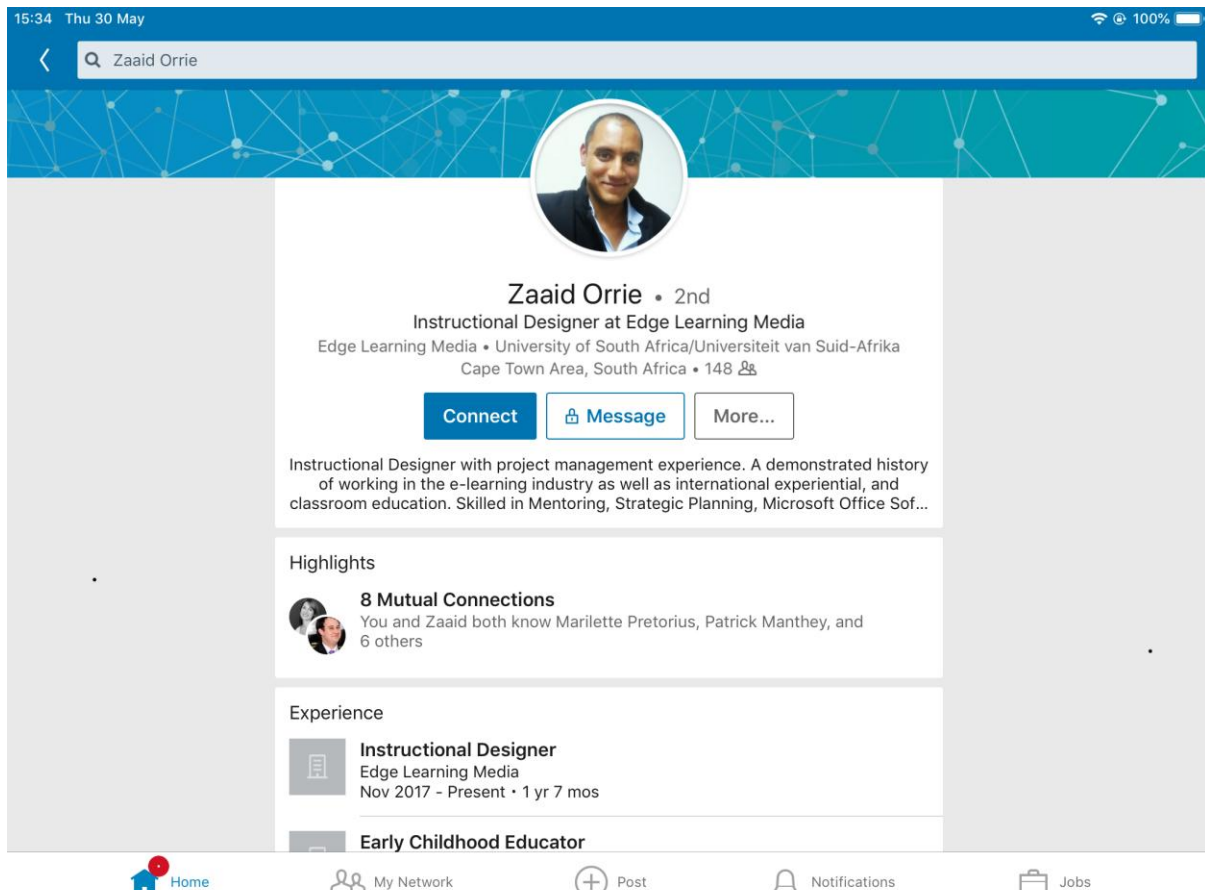
To present a more professional identity on Facebook, you can create a separate professional page on which you share what you do, what you are good at and what your interests are. Make this page more formal than your personal Facebook page.

Professional pages are geared toward people who generate an income from creating items, rather than for people in more traditional lines of work. The professional Facebook page serves as a space in which creatives can showcase their work and talents. Bearing this in mind, do not make a professional or company page simply for the sake of it. It must enhance your professional online identity and endorse your line of specialised or expert capabilities, services and skills.

LinkedIn

As the name suggests, LinkedIn is a network system that links you to other professionals, experts and specialists. These connections cover a wide range of occupations, businesses and sectors of the national and global workforce. LinkedIn is an essential online professional network to join as a young graduate seeking prospects and openings. It offers the opportunity to showcase your profile and your curriculum vitae (CV) to thousands of people. Being on LinkedIn allows you to search for jobs all over the world. You can state in your profile that you are actively seeking employment in a particular field, so that when people view your profile, they will see that you are available for employment. The most beneficial aspect of this platform is that it affords you the opportunity to create a network with other people.

Creating your online identity begins with selecting a picture of yourself. People like to see what you look like. According to Brudner (2016), profiles with pictures are '11 [times] more likely to be viewed' than those without a picture. In posting a profile picture, think critically and purposefully about the picture you choose. Certain types of pictures are bound to ward off potential employers. Pictures of you involved in partying or drinking alcohol are not appropriate for a professional image. Pictures of you pouting or pulling a face do not convey the appearance of a trained or skilled individual. Pictures of you in the mirror (mirror selfies) suggest a sense of self-indulgence and conceit. Think about the picture as your professional identity. Choose a photograph that best exemplifies how you would like to be seen in a work environment. A professional photograph usually has a plain background and depicts someone with a neutral facial expression, or a smile, who is professionally dressed.



(Source: Orrie, LinkedIn, 2019) Figure 4.1: Screenshot of a LinkedIn profile

The summary section of your profile is another important element of your LinkedIn account. This area allows you to stage your personality. A brief, accurate and sincere paragraph of approximately 150 words is enough to convey the essence of who you are. Without this summary of your character, your profile will only contain your CV. Although a CV is an essential component of this online identity, it is not always the most exciting document for a potential employer to read. Try to make your written summary piece conversational. It should be less rigid than a cover letter or a formal response in a job interview. Do not be afraid to reveal unique and quirky idiosyncrasies (habits) that offer the reader more insight into your personality. Avoid writing in the third person; you are having a conversation with the person reading your profile (Brudner, 2016). As with everything that you write, check your spelling and grammar. Nothing makes you seem less professional than numerous spelling and grammatical errors. Poorly expressed and error-filled writing makes it seem as if you do not approach your work with care.

Bizcommunity

This networking platform aims to offer businesses and individuals the opportunity to interface across a wide range of industries. It is a South African platform. This makes it uniquely specialised in facilitating the South African market and the South African economy. Bizcommunity promotes large-scale networking, advertising and events

management. One of the biggest benefits of this platform is the access it provides to a vast spectrum of business news. It does so while supporting both recruiters and jobseekers through branded job advertisements and a comprehensive, professional CV database. It is a site that essentially informs and builds the local industry. As a jobseeker, Bizcommunity provides you with a large portfolio of industry-specific news, market trends and opportunities.

Bizcommunity has an interest in the African economy at large, and has spread its networking into 18 other African countries. This opens doors for South Africans looking for job opportunities beyond South African borders (*Bizcommunity*, n.d.).

4.3.2 Networking professionally

Networking involves connecting with other people to develop professional relationships and exchange information. Networking happens all the time in the business world. Prior to the onset and domination of networking on the Internet, face-to-face networking was an integral part of business development. People left their places of work and interacted with others on a personal level. Your network is built through interactions with friends, colleagues, family connections, business associates, referrals and even strangers. In the physical world, networking occurs when organisations have events such as golf days, business lunches, company social events, conferences and seminars. At these events, people hand out business cards and corporate gifts that have the company logo on them.

Online networking has expanded exponentially in recent decades. It places the world literally 'at your fingertips'. Online networking connects you to contacts all over the globe. However, it is only as successful as the quality of the relationships you build. Online contacts are worth nothing if the relationship is detached and disengaged. You need to harness the potential and the marketing opportunities on offer by establishing good relationships with your online connections.

Your initial networking experiences may include Facebook, Twitter, LinkedIn or blogging. You read what people say and post comments. When someone reads your comment, the network suddenly grows. Searching for potential job opportunities must involve networking if you want to broaden the scope of the jobs on offer. Do not delay in getting your career started – it is something that you should aspire to do sooner, rather than later. Even if you have to start with part-time work, be assured that the more work experience you gain, the more your professional network grows and the easier it will be for you to find more permanent and more desirable employment. Think of it this way: all work is experience and all work exposes you to people who become contacts and could possibly assist you in future.

Networking on LinkedIn is easy and is a good way for you to connect with people in your field. It is a very effective channel for putting your name 'out there'. A great feature of LinkedIn is that when you go onto someone's profile, it lets them know through a 'profile view' notification that you have viewed their profile. If you view the right people's profiles, they are more likely to take an interest in yours. You can discover many people in your field on LinkedIn. Therefore, the first step to promoting yourself is to view people's profiles. Let them know that you are looking at their profiles and that your expertise is in the same field. Do not be afraid to let it be known that you are looking for a job. LinkedIn gives you the option of letting recruiters know that you are seeking employment by adjusting your settings. You can set this feature to limit your search to a particular part of the world – whether it be your home town, a new city or even a whole new country.

If you decide to send someone who you do not know a request to connect, it is best to customise your request. The generic message that is sent by LinkedIn is: 'I'd like to add you to my professional network on LinkedIn'. This may seem to be a reasonable catch line, but if you do not know the person, you need to explain why they should accept you as a connection or why you requested to connect with them in the first place.

Remember that people do not know you, and that they need a genuinely good reason to accept you into their network. In a survey by HubSpot, '64% of buyers and consumers said they'd regard a generic LinkedIn request from a salesperson as "creepy"' (Brudner, 2016).

When someone wants to connect with you, ask yourself whether the person has a valid reason to want to make contact. If the answer is positive, accept the request.

Remember that LinkedIn is not like Facebook. You do not add personal photos or share anything personal on an ongoing basis. It makes sense to build up the number of your connections and grow your network. If someone who is in your field of work or who is a recruiter invites you to connect, there is no harm in accepting the invitation, even if you do not know the person making the request. Use your discretion though. If you do not like the look of someone's profile or you see no valid reason for connecting, reject the request. In summary, what we are emphasising is that you should be more open-minded to strangers trying to connect with you over LinkedIn than you would be on Facebook or Instagram.

4.3.3 Researching the industry: The South African landscape

Once you have established your online profile on various platforms, remembering to include the fact that you are looking for employment opportunities – you cannot simply sit back and wait. You need to be proactive in finding what job prospects there are in the marketplace. The South African job market is highly competitive. Beyond using social

media, you should consider setting up specific job alerts on LinkedIn, Gumtree or Indeed. These sites list a variety of jobs and you can specify exactly which job alerts you would like to receive daily, directly to your inbox. You will be notified every time a new job is posted.

The high rate of unemployment and the challenging economic climate in South Africa make jobs more difficult to find. In addition to this, the South African government has implemented policies to rectify the past inequalities of job opportunities. This places companies and organisations under pressure to comply with regulatory employment procedures. To understand this dynamic better, examine the following terms that are frequently used in job advertisements:

- *A Broad-Based Black Economic Empowerment (B-BBEE) position:* This means that the job is only available to black Africans, coloureds, Indians and some Chinese people, who must be South African citizens.
- *An Employment Equity (EE) position:* This means that the job opportunity is limited by policies that encourage equal representation regarding skills development, company proprietorship, administrative control and special preferential procurement.

There are many positions that need to be filled. The more actively you try to find them, the greater the chance that you will find a job. A good starting point is to let people know that you are actively looking for permanent employment opportunities. You can do this wherever you have chosen to display your profile, whether on a LinkedIn account or a professional Facebook page.

It is important to match your skills and qualifications to job descriptions. You know what your experience, competencies and qualifications are. Filter the job specifications to those that match your profile.

Hundreds of websites offer listings of jobs that are available in South Africa. You can spend hours trawling through numerous jobs looking for something that is viable. Many of the sites allow you to upload your CV and a cover letter. This is a good way of marketing your skills. According to research from the Robert Walters South African Job Index, there was a 36 per cent rise in year-on-year jobs advertised in the second quarter of 2015 (*Robert Walters 1. n.d.*).

If your profile is on a professional social media site, such as LinkedIn, or if your CV is registered on a career-specific website, such as Indeed, the site usually sends you details of jobs that match your skills and profile. This helps to alleviate some of the filtering you must do when researching the job market. Exploring more of these platforms, and in greater depth, will expose you to more job opportunities. Make sure that you examine skill requirements carefully before you begin making further enquiries.

‘[Survey] results revealed that over 79 per cent of job seekers felt that signing up to networks or professional groups devoted to particular business sectors was important to their career progression’ (*Robert Walters 2*, n.d.).

There are also numerous offline platforms that offer job advertisements, opportunities and vacancies. The mainstream newspapers dedicate space to employment offers. Newspapers present features in which they advertise thousands of career and job prospects on allocated days. Local and regional newspapers also allocate space for job advertisements.

Registering with an employment agency could help you to narrow down your research in the marketplace. There are employment agencies that accept your CV either online or at physical office sites. The recruitment specialist will match your CV and skills to vacancies that they have registered on the database. It is advisable to regularly check the websites of recruitment agencies, as they frequently upload new jobs. They often are the best option for finding a placement in a specific profession. However, it is unwise to register with too many agencies in the same area, as your CV may be sent to the same would-be employer by different agencies.

The *Government Gazette* is an official newspaper that lists all available government jobs. You will need to filter your search according to the sector you fit into and your qualification level. The *Government Gazette* is also available online.

Many companies and large organisations have recruitment departments that are available for consultation. Do some online research and networking to find out their contact details. It is a worthwhile exercise to research companies’ employment policies and career opportunities. Explore their company web pages for details of how to contact their human resources departments, and how to make applications for specific jobs or submit your CV for their records.

Many companies (especially larger ones) feature a ‘careers’ link on their website. Some may list all the jobs that are currently available, whereas others may allow you to submit a general application and a CV. When there is a vacancy, the employer could then consult this online database to find a suitable applicant.

4.4 CV creation basics

Your CV is one of the most influential documents you will ever have to produce. The words ‘curriculum vitae’ are Latin and mean ‘the course of life’. When you offer someone your CV, you are presenting an account of your life thus far. A CV is, very often, the first information you offer about yourself to a potential employer. It gives you the opportunity to make a good impression on people who do not know you. Employers request CVs so that they can form an opinion about your possible suitability for the job. Crafting an effective and impressive CV is an exciting exercise, and you should not

squander a job opportunity by sending a poorly constructed CV to a potential employer. You need to create a document that expresses your virtues and your character well. As we have mentioned, the job market is exceedingly competitive. You need to get the attention of the person who first reads your CV, so that they do not simply put it at the bottom of the pile. You cannot rely on making a good impression at the interview, because you may not secure the interview without a good CV. Your CV precedes any physical meeting; it is the document that gets you interviews.

Typically, your CV includes your personal details, your education and academic qualifications, your previous working experience, your special achievements and the names of your referees. Your CV accompanies a job application. It may also be given to prospective employers to keep on record, should any future opportunities arise. It is important to update it regularly.

Your CV must be polished and concise. Make sure that you check your spelling and grammar when you type up your CV. No one wants to read through reams of irrelevant, poorly expressed information. It is worth getting someone reliable to read and edit your CV before you submit it.

The layout of your CV is important for two reasons. Firstly, it demonstrates your logic and your ability to present an appealing and professional document. Secondly, a well-formatted layout will assist the would-be employer to easily find pertinent information. Use headings and bullet points. Avoid writing long paragraphs. It is easier to navigate the profile when key words and specific points are highlighted.

A recent tendency in CV writing is the use of colour. Use colour with caution: it should be appropriate and should not make your CV look gaudy. If you are not sure whether the colour is effective, it probably is not, so take it out. You can achieve visual appeal using effective fonts and spacing. Do not use an elaborate font, and keep the font type and size consistent in the headings and in the body text. There are hundreds, if not thousands, of templates for CVs on the Internet. We recommend that you create your own original version of your CV and avoid templates. When you create your own CV, it indicates that you have given thought to the process and that you are an independent thinker.

4.4.1 Profile summaries that work

Your profile summary is an important outline of who you are, what drives you to achieve and what makes you unique. The information that you include in your CV profile must be authentic and sincere. You should neither exaggerate your accomplishments, nor sell yourself short. You need to ensure that what you say is true and accurate. Do not say something about yourself that sounds impressive but is not true of you. It is also

important to provide a summary that conveys an impression of confidence in your own abilities, but not arrogance. If you have just finished your studies, the employer to whom you are sending your CV knows that this will be an entry-level position. You would be ill-advised to apply for a job that is beyond your skills and qualifications.

Your profile must include your personal details, but include only what is necessary. Your full name is important, and your title should be added. Your age is usually expressed by giving your date of birth. Write out your date of birth using the format 'DD Month YYYY'; for example, 24 April 1993. You should also include your gender. Your marital status is generally required and should be expressed as 'married', 'single' or 'divorced'.

Your physical and your postal addresses should be written out in full. Provide your contact number and email address.

It is important that you read the advertisement for the job you are applying for carefully. You may be requested to submit your South African identity number or details of your driver's licence.

Although it is not common practice to have a photograph in your profile, some prospective employers may call for one. If this is the case, use a high-quality photograph that is current and appropriate. Sending a photograph of yourself on the beach with a drink in your hand would not, for example, be appropriate.

Separate the specific points of information so that what you write is clear and easy to read. The person reading your profile should be able to scan or skim read the document to easily identify important points that are relevant to the application.

In your personal profile summary, include details of your school education, tertiary education, awards, special achievements, and any relevant courses that you may have completed and passed. Present the information on qualifications in reverse chronological order. This means that the most recent qualification will appear first in the list. The following example clarifies how to present your personal profile.

Examine the following fictitious example of what a personal profile might look like in a CV.

Curriculum Vitae: Samuel J. Radebe

Personal details

Full name: Samuel John Radebe
Date of birth: 24 April 1993
Marital status: Single
Residential address: 43 Main Road, Clyde, 2610
Postal address: PO Box 56, Clyde, 2640

Contact details

Cell phone: 089 7879793
Email: smauejrabede@gmail.com

Qualifications

2016 Postgraduate Certificate in Education (PGCE), Unisa
2016 Level 1 First Aid – Medical Support Certificate, Red Cross Society
2015 BA Honours (*cum laude*) English, University of Pretoria
2014 BA (*cum laude*) English and Philosophy, University of Pretoria
2011 Matriculation (university exemption), Pretorius High School
2011 Certificate in Computer Literacy Skills 101 (ITC)

(Source: EDGE Learning Media (Pty) Ltd, 2019) Figure 4.3: Example of a profile summary included in a CV

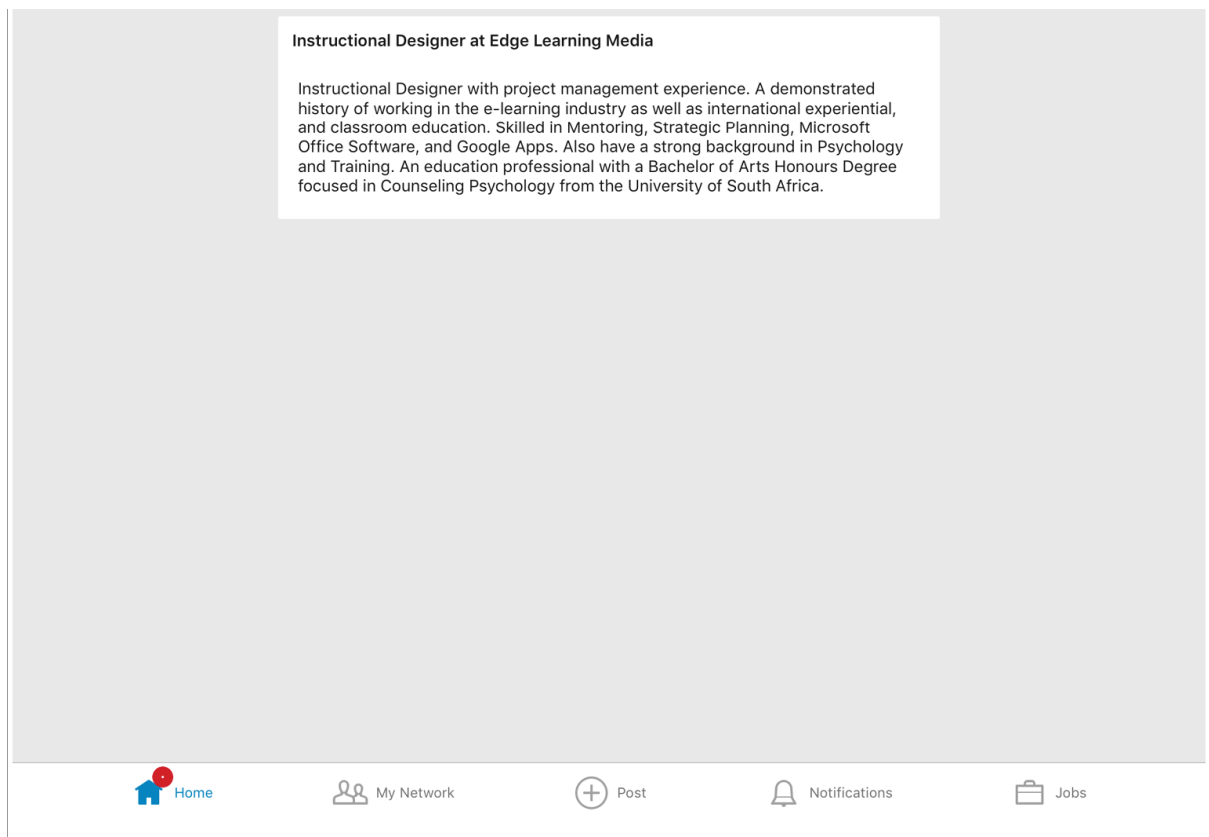
The profile summary on your hard copy (printed) CV, or a CV that you keep as a filed document on your computer, is different to the profile summary commonly included in an online profile such as on LinkedIn. The online profile summary has a more conversational tone and should appeal to the reader in a personal way.

Be authentic when writing your summary. Aim for a unique approach – eliminate the boring, clichéd phrases that fail to make a reader feel inspired or excited. This is a chance to define yourself and your motivations in as few words as possible. Here, you can reveal your goals and ambitions without sounding over-ambitious or unrealistic, provide an insight into your plan of action and offer your services in a field of expertise. You need to write a short narrative about yourself. Consider the following questions when composing your online summary profile (Foote, 2017):

- What motivates you to achieve your objectives?
- What sets you apart from others in your field?
- Are you up to date with current trends?

- What personal traits make you likeable?
- Can you find humour in life and laugh at yourself?
- What special and inspiring talents and experience do you have?
- How can a reader, potential employer, recruiter, potential client or fellow professional contact you?

The following profile summary is a good example of successful interaction between the writer and the reader. The writer engages with prospective interested parties in a personal way. Read the summary in relation to the questions we have just discussed.



(Source: Orrie, LinkedIn, 2019)Figure 4.4: Screenshot of a LinkedIn profile summary

4.4.2 Information that works

The next section of the CV contains other information that you want the would-be employer to know. Remember, you are targeting a specific person on whom you want to make an impression.

Traditionally, the information that comes next is a history of experience relevant to the post for which you are applying. Since you may not have much actual work experience in the specific area (other than the experience of perhaps studying it at university), you

could focus on what you can offer the company or organisation based on other strengths and accomplishments.

Research the requirements of similar positions and investigate the company itself. You need to know about the organisation's mission statement, community upliftment policies, values and ethics.

If you do have previous work experience in the specific field you are applying for, provide the information under an appropriate heading, such as 'Employment history – immediate relevance to the position'. Include the dates of your employment, the name of the employer and the nature of the position you held. If you have had more than one previous job, list them in reverse chronological order. Do not be afraid to highlight specific skills related to the job you held.

It is important to have one basic template of a CV that can be amended according to the requirements of a certain job. As different jobs require different things, you may wish to emphasise some skills over others. Naturally, you should always remain honest and ethical, however, some skills can be emphasised if they are particularly relevant to the job being applied for.

In the section where you include work experience, do not include part-time work such as waiting on tables in a restaurant, bartending, babysitting, housesitting, working as a shop-teller or other jobs that many students do to earn extra money. We are not suggesting that these jobs are not of value. Rather, briefly include details of this kind of work in a cover letter or at the end of your CV under a heading such as 'Other employment'.

If you do not have previous work experience relevant to the position, you may include a heading such as: 'Accomplishments' or 'Achievements'. In this section, indicate, for example, that your research on the company revealed that it advocates an ethos (spirit) of integrity and ethical business practice. You could include details of accomplishments in the university community or details of awards for service to the community. You could also provide details of any academic awards you received as a result of your sound work ethic. Details about memberships of relevant professional associations or affiliations could also be included. Remember to give the full name of the association and the date on which you became a member. If the association is pertinent to the job, your membership could impress a potential employer. This section should also detail any leadership positions you have held. Include the title of the position and the responsibilities it entailed.

Avoid using clichés in any written part of your CV. For example, 'I have a passion for people' and 'I am a hard worker' do not add any credibility to your CV. Remember, concise and sincere expression is more convincing than flowery language. Avoid quotes and slogans – for example: 'Following the inspiration of Nelson Mandela's words,

“Education is the most powerful weapon which you can use to change the world,” my dream is to be an educator.’ If you need to add clarifying commentary on a point in one of the sections, do so subtly but persuasively.

Do not use illustrations or elaborate borders. These only detract from the information that is pertinent in your CV. You do not know the personality type of the person who will read or peruse your CV, so do not risk anything that they may find irritating or unnecessary.

Your CV allows you to disclose interests and hobbies. However, only include these if they have some bearing on the application. Interests and hobbies can reveal much about the type of character you are. Once again, for clarity and ease of navigation, these should be under a specific heading.

The final requirement in a CV is the details of your referees. A referee should be someone who knows you well in either a professional or academic capacity. People such as your teachers, lecturers and managers for whom you may have worked make viable and credible referees. Referees may be called upon to validate what you have said about yourself and to give a character reference. They are required to speak as objectively as possible about you according to their experience and interactions with you.

Avoid giving the names of personal friends or people who only know you socially, as their perspectives may be subjective and emotional. You should always ask permission from people to give their names as referees, and inform them of the reason why you are asking them to be referees. Most prospective employers ask you to include the contact details of two referees.

Your CV is an ever-evolving document. You will have to tailor it to the requirements of the reader you are sending it to. If you are responding to an advertisement for a position, the requirements of your CV will be explicitly conveyed in the advertisement. You must adhere to the required length and nature of the CV. For example, you may be asked to submit a CV of no more than one page. On the other hand, you may be requested to submit a comprehensive CV with no length limitation. Adjust your CV according to specified requirements. If you are delivering your CV to various organisations, or sending it electronically to various recruitment agencies or company personnel departments, keep it concise and impactful.

The following is a fictitious illustration of what these sections of information might look like in a CV. In keeping with the previous example, this CV is tailored to the education sector.

Employment history: Immediate relevant work in the education sector

January 2016 – December 2016: Intern English teacher (High School) at Hillside School, Pretoria
January 2015 – December 2016: Private support English lessons for grade 8 pupils

Responsibilities relevant to teaching

Hillside School – 2016:

Teacher in charge and coach of senior debating team
Assistant coach of 1st team hockey
Library assistant

Hillside School – 2015:

June/July Winter school – English lessons
(These lessons were specifically geared to improve reading and comprehension skills for grade 8 and grade 9 learners.)

Accomplishments relevant to teaching

2012: Dean's commendation list for academic achievement and bursary toward tuition fees
2011: Matric English prize for writing and oral presentations (Pretorius High School)

Interests and hobbies

I am an enthusiastic debater and a member of the Pretoria Alumni debating team that won the National Debating League in 2016.

I volunteer for the Red Cross emergency services at sporting events on the weekends.

I write a current affairs blog: sjsmithabouttown@typed.com. Blogging is a useful ability for students in the 21st century and forms part of their transactional writing skills.

I belong to The Young Writers' Association of Southern Africa. This association connects promising writers from all over Southern Africa to exchange ideas and current trends. I believe I can offer learners some useful insight into the continent's writers.

Referees

Mrs K. Knight – Head English teacher, Hillside School (087 567 8435)
Professor J. Gains – University of Pretoria (093 578 4659)

(Source: EDGE Learning Media (Pty) Ltd, 2019) Figure 4.5: Example of information that should be included in a CV

4.4.3 Attachments that work

Read the advertisement for the position carefully. Some advertisements require the applicant to submit certain documentation with the CV. All academic certifications, such as your matriculation certificate, diplomas, degrees and academic record, should be copied and authenticated by a commissioner of oaths. Your identity document, birth certificate, driver's licence and marriage certificate (if applicable) should also be copied and certified, and must be readily available should they be requested.

In addition, you should have any other relevant certificates of qualifications mentioned in your CV copied and certified. Documents such as testimonials and reference letters should be available if requested.

This does not necessarily mean that you should attach all these documents when sending your CV over email. These documents make up your portfolio and should be kept on file, either as hard copies or electronically. Attach only documents that have specifically been requested. As previously mentioned, any written reports, references or recommendations must be professional, objective and persuasive. Clear and concise expression is very important. Long-winded and exaggerated descriptions of your ability and your character result in a loss of credibility.

Further attachments

Let's consider some of the documents that should not be included in your job application, but that may be requested once your CV and cover letter have won over the recruiter (*Careers24*, n.d.).

Your identity document, driver's licence, certificates and other pre-formatted and printed documents may be requested after your application. They cannot be altered in any way. You will need certified hard copies.

Reference letters and testimonials are usually presented in MS Word document format. Do not illustrate them in any way, and do not put decorative borders around the text. Most reference letters and testimonials will be printed on a company or educational institution letterhead. If you are sending hard copies, you should send the original letter, but keep certified copies for your records. Alternatively, carefully scan the letters and attach them to an email.

When requesting a reference letter, ensure that the letter is appropriately formatted. Include the following information in a logical sequence:

- The address of the referee should be on the top right-hand side of the page
- The address of the addressee (if there is one) should be on the left-hand side of the page, below the writer's (i.e. the referee's) address
- The letter should be addressed 'To whom it may concern', unless a specific addressee is required

- The letter must indicate the name of the person about whom it is written, in an underlined heading (this is usually positioned below ‘To whom it may concern’)
- The following information is also required:
 - How long the referee has known you
 - The capacity in which they have known you
 - Positive qualities about your character
 - Positive attributes about your work ethic
 - Details of the work you performed for the referee
 - Reasons why you are recommended for the position
 - Contact details, including telephone numbers and the email address of the referee
- The referee should sign the letter to prove its authenticity
- The referee should give their full names

In most cases, your testimonial will be an official document from the principal of your high school. Do not attach the testimonial unless asked to do so. A certified copy will suffice.

Photographs are not generally attached when submitting a CV (*Careers24*, n.d.). However, in some cases, the prospective employer may request a recent photograph. Ensure that you send an original high-resolution colour print and that the photograph is suitable. If a photograph is requested, details of the size will be given. Adhere to the specifications. Do not send a larger photograph than requested so as not to impose unnecessarily on the reader. The focus should be on your skills, not on your looks.

The photograph should be recent. A head and shoulders portrait (medium shot) is appropriate. Your dress code in the photograph does not have to be formal, but avoid photographs that are too informal or excessively social in nature (*Careers24*, n.d.).

In some cases, you may need to send a portfolio or samples of work with your application. Design work, architectural work, graphics, fine art and journalism are some fields of employment in which the prospective employer may ask for portfolio samples. In these cases, there are likely to be specifications in the advertisement that you should follow. If you are sending hard copies, do not send your original work. Rather, get the material professionally copied so that it looks exactly like the original. Be selective about what samples you send – it is usually quality rather than quantity that is

important. Therefore, you should choose your most successful pieces for submission (Careers24, n.d.).

If you need a hard copy of your CV, it is worth getting it printed professionally. At the very least, it should be printed on high quality paper using a reliable printer. You should convert your CV to a PDF or Word document if you are submitting it electronically. Do not ever write your CV out as the body text of an email. You should send your CV as an attachment, usually with a cover letter as a separate attachment. If you are sending any other attachments electronically, send them as separate attachments. If you are sending portfolio samples electronically, you need to consider the following:

- Some companies stipulate a maximum file size that can be sent with an application – do not overstep their restrictions.
- Do not send single pieces in separate, individual files. Remember, someone has to open and download these files. It is better and more professional to create a file with several pages that can be saved and browsed through with ease.

The most important attachment is your cover letter. We will discuss this in detail in the next section of this unit.

4.5 Cover letter basics

A cover letter is usually a one-page letter that accompanies (covers) your CV for a job application. A cover letter may also be sent when you are submitting your CV to an organisation or a recruitment agency to keep on record for future positions that may arise. Essentially, the letter endorses your CV and gives a little more insight into your character. It also links the information in your CV to the position for which you are applying.

Avoid writing more than one page for your cover letter. A well-written letter should be concise and to the point, without being curt or aloof. Bear in mind that the prospective employer may have to read hundreds of CVs and cover letters. It is therefore in your interest to make your cover letter stand out and catch the attention of the reader. Begin by making it clear who you are professionally (hint: introduce yourself in a way that aligns with what the recruiter is looking for) (Forbes, 2016). You should also briefly explain why you want the position. This enables the recruiter to know more about you as a possible candidate for the position.

The purpose of the cover letter is to inspire or encourage the reader to take note of what you have presented in your CV, and give the information consideration. A good cover letter gives you the opportunity to sell yourself as a viable employee. The following subsections will guide you in the compilation of an effective cover letter, so that when the opportunity arises, you will be confident in your job application.

4.5.1 Effective cover letter content

As mentioned previously in this unit, content, format and style are all elements of a document that leave an impression. Your objective is to leave the reader of your cover letter feeling positive and enthusiastic about setting up an appointment for an interview with you.

Let's discuss the body of the letter – the 'nitty gritty' content – before we move on to the structure of the letter. Transactional writing is 'writing that is part of a chain of written communication intended to communicate, persuade or inform' (*Reference*, n.d.). Logical paragraphing and sequencing of information are essential to any good transactional writing. In addition, it is crucial that your grammar, language and spelling are faultless. A poorly written letter peppered with errors is very disconcerting for prospective employers to read. There are many different formats for setting out a formal letter. The structure discussed next is the most common and widely accepted of them.

Your letter must include your (the applicant's) address, the name and/or title of the person to whom you are writing, the date, and the address of the person or company to whom you are writing. Figure 4.5 shows the format for setting out this information.

The next section of the letter must be a salutation (greeting). The salutation should preferably address a specific individual, rather than being a generalised salutation such as 'To whom it may concern'. If you do not have a specific name to whom the cover letter must be addressed, write 'Dear Sir' or 'Dear Madam'. You could be proactive and contact the organisation to find out the name of the person dealing with job applications. If you do have a name to whom the letter should be addressed, you should write the name in the salutation; for example, 'Dear Mr Smith' or 'Dear Ms Smith'. In South Africa, we do not insert full stops after the title, and we do not insert a comma after the name or after the word 'Sir' or 'Madam':

- *Incorrect:* Dear Mr. Smith,
- *Correct:* Dear Mr Smith

If you are addressing a female, do not presume that she is 'Miss' or 'Mrs'. The neutral term for addressing an adult female is 'Ms'.

If you are responding to a specific advertisement, you are more likely to have the name of the person to whom the application should be addressed. Ensure that you begin the salutation on the left-hand side of the page, directly below the addressee's address.

Next, the letter needs a reference line or topic heading. This heading should be underlined and should clearly indicate what the letter is about. An advertisement will usually give a reference number for the specific job application. This number should appear in the heading. You may use the job title as a reference for this heading and you

may also include the start date. This reference line is written below the salutation, against the left-hand margin. Consider the following examples:

- Application: Set Designer – Reference 209, The Daily News, 10 January 2017
- Application: Physics Teaching Post, January 2017
- Application: IT Manager, Bloemfontein Branch
- Submission for your records: Curriculum Vitae – Samuel J. Smith (B. Com.)

4.5.2 Effective cover letter structure

We will now discuss the requirements for a correctly structured cover letter.

Paragraphing in a formal cover letter should be in block format. This means that every paragraph begins at the left-hand margin and we leave a line between each paragraph. The opening paragraph of the cover letter is generally a short paragraph referring to the position, and states your intention to apply for it. It is important to remember that your goal is to get an interview. Avoid stating the obvious: ‘I am applying for the above position’. Rather, make an encouraging suggestion or statement such as can be found in the following examples:

- ‘I refer to the above-mentioned position advertised in The Daily News on Sunday, 10 January 2017. As a recent Bachelor of Commerce graduate from the University of South Africa, I would like to be considered for this position.’
- ‘Thank you for the opportunity to apply for the position of IT Manager in your company. My research reveals that you are a forward-thinking and innovative team. I have enclosed my curriculum vitae for your perusal, and I would be happy to meet with you should you wish to know any more about me.’

You may also have been advised by a mutual connection to submit your CV to a company or organisation. If this is the case, you should inform the addressee accordingly. Consider the following example:

Mrs Kaye Maitland has suggested I submit my curriculum vitae for your perusal, as you are looking for enthusiastic commerce graduates. I am keen to start my career as a junior trader and would appreciate your time to consider my credentials.

The second paragraph begins to establish why you are a suitable candidate for the position. ‘It has to clearly show why your skills meet the company’s needs, and make the connections very specific’ (EnglishGrammar, 2014). Avoid repeating all your qualifications. Rather refer the reader to your CV. The following example may help you to get the gist of linking your skills to the job criteria:

As my CV indicates, I worked part-time at XKX Accounting Ltd during my university studies and I have gained valuable work experience as an accounts clerk. My academic record shows that my commitment to excellence is uncompromising. I am confident that my work ethic and my drive for success will meet the expectations you have of motivated young accountants. It would be a privilege to do my articles at an accounting firm whose reputation and prowess in the South African accounting world is exceptional.

What you have done in this paragraph is highlight your work experience and draw attention to what your excellent academic record indicates about you. You have also informed the reader that you are motivated and driven, and you have shown that you have knowledge of the company's status in the business market.

The experience I gained as an intern teacher at Adam's High School while I completed my Postgraduate Certificate in Education (PGCE) during the past year has developed my confidence and my proficiency in the field of teaching Physics. Your advertisement indicates that you are looking for young and dynamic physics teachers, and I am sure my knowledge of integrating teaching and IT will be of interest to you. As specified in my CV, the completion of the integrated course entitled "Teaching the Digital Generation" will bring innovative ideas and skills to the physics classroom.

In this paragraph, you have told the prospective employer that you have experience. You have advised the reader that you have developed personally. You have aligned your character and skills with a specific requirement in the advertisement by reminding the reader that you are young and progressive. You have shown that you are proactive, that you have extended your skills beyond a single sphere and that you are IT literate. You have endorsed your belief that you will bring something special to the classroom.

There is no need to write more than one A4 page's worth of content in your cover letter. Choose a simple font, such as Arial or Trebuchet MS. The font size should be 11 or 12. Highlight what is important in two or three paragraphs, then close off the letter.

The concluding paragraph generally expresses an expectation of an action from the reader. It shows anticipation of a positive outcome such as an interview. You can reiterate some contact details or remind the reader to contact your referees to validate what you have said. Examine the following example to get an idea of these elements:

I hope you have found this information helpful. Should you require any further details, please do not hesitate to contact me on my cell phone number, 089 787 9793 or email me at smaueljsmith@gmail.com. My referees are available at any time should you wish to contact them.

(In this paragraph, you are showing that you expect the reader to be satisfied with the information given. You have indicated that you anticipate some sort of contact, but you

are not demanding that the reader contact you. You have also informed the reader that your referees are available and may be worth contacting.)

You must close your letter off properly. In South Africa, we generally end a formal letter with the words, 'Yours faithfully'. Please note that there is an upper case 'Y' for 'Yours' and a lower case 'f' for 'faithfully'. There is no comma after 'faithfully'. The words should be written against the left-hand margin. The word 'faithfully' suggests that you are writing the letter in good faith and that you have not given any information that is false or inaccurate. Many people make the error of ending a letter such as this with the words, 'Yours sincerely'. The general rule for using 'Yours sincerely' as a sign-off phrase is that you may use it if you know the person to whom you are writing. In most cases, you will not be sending your cover letter to someone you know. The safest option, therefore, is to use the phrase, 'Yours faithfully'. Avoid any other phrases, such as 'Yours truly'.

You must sign the cover letter by hand as an indication that what you have said in the letter is the truth. It seals all that you have stated as accurate and trustworthy. Print your name in full below your signature.

Now that all the details and the suggestions of the cover letter have been offered, examine the following fictitious example of all the components in one cohesive document.

60 Main Road
Belville Park
Greenpark
4140
24 January 2017

The Personnel Manager
Pankhurst Productions
Wellview Centre
Westbrook
4610

Dear Sir

Application: Junior Account Manager (Marketing Division)

Thank you for the opportunity to submit my curriculum vitae in applying for the above position, advertised in *The Herald*, 10 January 2017. I completed my Diploma in Marketing Management through Unisa and Varsity College in November 2016. The prospect of working for your organisation is exciting and I would very much like to be considered for the position.

Since graduating, I have been employed at MM Marketing Enterprises as an assistant to the marketing manager in the research and development division. This experience has afforded me valuable insight into the diverse nature of marketing, and I am looking forward to a career in this stimulating environment. I believe you are looking for independent and forward-thinking employees.

Your organisation has offered young graduates openings to successful careers in marketing, and I am sure my enthusiasm and sound work ethos will fit into the spirit of your company. You will notice from my CV that I have also done a short course in Information Technology that is specifically geared to innovation in marketing strategies. The skills I gained in the field of digital technology as it relates to marketing in the global community has given me a more global view of the marketing industry. I am sure this will stand me in good stead for the position that you wish to fill.

I trust you will find this information helpful. I am available to start work immediately as my current employers are supportive of me getting a permanent position. Please do not hesitate to contact me on my cell phone number 086 479 8766 should you require any further information.

Yours faithfully

S J Radebe

(Source: EDGE Learning Media (Pty) Ltd, 2019) Figure 4.6: Example of a cover letter

Please use this example as a guideline for ideas on content and structure. Remember, original letters are much more impressive than templates filled with clichéd stock phrases. You are a unique individual with distinctive qualities to offer, so your cover letter should be unique and should set you apart.

4.5.3 Tone and register in cover letters

Before we end this section on the cover letter, we need to address tone and register. Tone and register are two important elements in any written document. Tone can be described as the attitude you convey in your writing. When composing a cover letter to accompany your CV, be mindful of tone. Your tone influences the reader's reaction to the information you want to transmit. Kennedy (2012) states that 'the tone of the letter will be dictated on [sic] the position you are seeking'. In other words, the level of the position for which you are applying will determine how formal your letter ought to be. A cover letter for a position that entails authority and status within the company requires a more formal tone than that for a junior post or entry-level position.

Your aim is to generate a positive response. Evoking a negative reaction because of your tone could distract the reader from the valuable information about your credentials. The reader may even set your CV aside without reading it. Your intent must be clear in your letter. The appropriate tone will help to convey the correct intent.

The following examples illustrate different tones:

- *Example A:* If you guys don't give me an interview, your company will be the loser.
- *Example B:* I would appreciate an interview so that I can contribute meaningfully to your organisation.

In Example A, the attitude is arrogant, threatening and demeaning. In Example B, the attitude is respectful, hopeful and optimistic.

We now consider register. This concerns the way you use language for a purpose. Register describes the degree of formality of the language you use. We have already established that the purpose of a cover letter is to persuade the reader that you are the most suitable candidate for the position. Because you are trying to create a professional impression, it is unlikely that you will use informal language in a cover letter, even if you are sending it to an innovative and vibrant organisation.

Informal language includes contractions, colloquial language, slang and idiomatic language. In Example A, not only is the tone unsuitable, the register is also inappropriate. The use of the word 'guys' (colloquial language), 'don't' (contractions) and 'loser' (slang) is detrimental (harmful) to the image you want to convey. In Example B, the register is more fitting and works with the attitude of the message.

This does not mean that the cover letter should be dull and flat. The tone, while respectful, can be enthusiastic and anticipatory. Your register can be formal without being ceremonial or grand. Each time you apply for a job, consider performing the following actions:

- Ensure that you read the advertisement carefully to get an idea of the company's image.
- If there is a job description, look for clues that will guide you as to the style of cover letter you should write to accompany your CV.
- Do some research on the organisation so that you get a feel for its nature and personality. This will influence the tone and register of your cover letter.
- Write a separate cover letter for each application, adapting the language and style as necessary.
- Draft and redraft your cover letter until you are satisfied that it is logical, cohesive and accurate.
- Do not be afraid to ask someone to read your cover letter before sending it. The writer often misses errors or inappropriate elements that are obvious to others.
- Always check spelling and grammar before sending a cover letter. It is a cardinal sin to spell the name of the company or addressee incorrectly.

• 4.6 Interview basics

The job interview can be an extremely stressful experience. Both the interviewer and the interviewee are under pressure to ask the best possible questions and give the most appropriate answers, so that an informed decision resulting in the most positive outcome for both parties can be made. You must be fully prepared for the interview so that you do not flounder either emotionally or in your verbal responses.

The interview is an important exchange that reveals valuable information to both the candidate and the organisation.

4.6.1 Responding to interview invitations

You may be notified in a variety of ways that you have been invited to an interview. This is the first positive step in the process of getting the job. Your first reactions will probably be feelings of excitement, anxiety and cautious optimism.

Let's consider how the invitation to the interview may take place.

The interview invitation via telephone / cell phone call

You are likely to receive a telephone call from the personnel department to set up a date for the interview. You may even get a direct call from the person who is going to make the decision as to whether you will get the job or not. If your CV has been forwarded via a recruiting agent, someone from the agency will contact you to arrange the interview.

It is in your interest to get into the habit of answering your telephone politely. The call may come from an unknown number, so be mindful that it could be the prospective employer or someone from their office.

Once you know who is speaking, respond with an enthusiastic greeting. Listen for the tone of the speaker and the formality of the greeting. You need to be responsive to the attitude of the speaker. Look at the following examples. You need to respond differently to each of these greetings. Read the greeting and consider an appropriate response:

- Greeting:* Hi, Sam. It's Mike here from Pankhurst Productions. I'd like you to come in for a chat. Your CV looks great. (An informal, friendly attitude is clear here. The interview is offered immediately.)

Response: Hello, Mike. Thank you for the call, that sounds good. When does it suit you for me to come in?
- Greeting:* Hello, Samuel. Michael Green here from the marketing department at Pankhurst Productions. Are you available to set up an interview some time? (A slightly more formal but congenial attitude is evident. He is calling to give you an interview, but you need to respond carefully.)

Response: Hello, Mr Green, thank you for your call. I would very much appreciate an interview. What date suits you?
- Greeting:* Good afternoon, this is Michael Green, marketing manager at Pankhurst Productions. Is that Mr Samuel Smith? (Formal tone is evident here, and the caller is identifying himself and the company he represents. There is no offer of an interview in the opening words, but you should anticipate that this is what the call concerns.)

Response: Good afternoon, thank you for calling, Mr Green. Yes, this is Samuel Smith.
- Greeting:* Mr Smith, it's Ms Hill from Great Careers recruiting agency. I have an interview for you with Pankhurst Productions. (This is quite a neutral attitude the agent presumes you know her and that you are anticipating a call from her.)

Response: Thank you for calling, Ms Hill. That is great news, when do they want to see me?

It is important that you ascertain (determine) the speaker's tone and respond accordingly. Remember, this is the first verbal interaction with a representative of the organisation. You must ensure that the impression you leave is positive. First impressions are powerful. Etiquette and confidence are key factors in making a good impression.

If you are contacted on your cell phone and your connection is poor, try to explain your location and offer to call back as soon as you have a good signal. The same applies if you are in a situation where you cannot have a conversation: explain that you are not able to talk and offer to call back as soon as you are able.

Once you have established that the call is for an invitation to an interview, you should, in all cases, thank the caller for the invitation and establish a suitable time for the interview. You are advised to do everything possible to agree to the time scheduled by the caller, unless it is impossible. If you need to schedule a different time to that offered, apologise and explain that you have a timing conflict. This may also be an opportunity to reveal that you know something about the company or organisation, and that you would consider it a privilege to work for them. It is unlikely that you will engage in a long conversation during the call. We will deal with telephonic interviews (in which you will have a slightly longer telephonic conversation) later in this unit. End the conversation by reasserting your gratitude and expressing that you are looking forward to the meeting.

The interview invitation via email

Much correspondence takes place via email in today's busy business world. As you will likely submit your CV and cover letter by email, you may receive an invitation to an interview by return email. The following procedure will help you to acknowledge the invitation appropriately:

- Reply to the person who sent the email as soon as possible. If you delay your reply, it may signify to the sender that you lack enthusiasm or interest.
- The email subject line should include your name and the purpose of the email, such as 'S. J. Smith reference interview request dated 20 January 2017'.
- Address the sender of the email in person: 'Dear Mr Green'. Spell the person's name correctly!
- Thank the sender for the email and for the prospect of an interview – e.g. 'Thank you for your email inviting me for an interview on 23 January 2017'.
- Consider the scheduled time carefully and make sure that you are available. You are, once again, cautioned against making an alternative date. If you do have to change the scheduled or requested time, ensure that you advise the sender timeously and ask for an alternative date. If you are available, accept the time

and date enthusiastically – e.g. ‘I shall be very pleased to meet with you on 23 January at 10h30’.

- If you feel that you want to acknowledge the email immediately, but you cannot check your diary for the date, send the reply and explain that you are currently unable to confirm the date but will do so as soon as possible.
- Remember that less is more. Do not write reams of unnecessary information in your acknowledgement email. Keep your writing concise and to the point. Use block paragraphs that are properly formatted, and ensure that your grammar and spelling are correct.
- If possible, ask a colleague or a friend to check the email.
- Avoid using colour and page backgrounds in your reply emails. Hussain (2017) suggests that you ‘stick to simplicity to keep the focus on your actual message and not on your funky formatting’. Keep the correspondence professional and objective.
- Do not add the option of the automatic receipt request. You will know by a delivery failure notice if the email is not delivered. It is not up to you to demand an acknowledgment of the receipt of your email.

4.6.2 The interview process

The following steps in the job interview process cover the possibilities that usually occur. However, companies and organisations have different policies regarding the hiring of personnel. An overview of the types of interviews follows (Doyle, 2017).

The interview process

Applying for the job

The interview process begins with your application for the job. You need to ensure that the job for which you have applied is within your capability and skill capacity. Once you find an appropriate job vacancy, you apply by sending your CV and cover letter.

Screening interview

This is an initial interview that very often takes place telephonically. It essentially determines your motivation for applying for the job, and whether you have the necessary skills and qualifications for the position. It also allows for the employer to establish your background, salary expectations, and availability. The screening interview is a type of filtering process that gives the employer the opportunity to decide whether further interviews with you are viable. It helps the employer to reduce the pool of applicants to a manageable number before proceeding to the first interview. Most of

the questions in this type of interview will be closed questions. (There is a description of closed questions later in this unit.)

First interview

This is a face-to-face interview and is typically the first time that you meet the recruitment manager or the human resources (HR) manager. This interview gives the potential employer the opportunity to find out more about your experience and skills relating to the position. The interviewer will also check up on your qualifications and previous working experience. The details of the job are likely to be discussed and you may be required to answer some open or leading questions (described later in this unit). After the round of first interviews, the employer will make a shortlist of the most suitable candidates to proceed to the next round of interviews.

Second interview

This is a more in-depth interview. It may entail face-to-face meetings with one or more people involved in the selection process, such as departmental or other managerial staff. The second interview provides the opportunity for both you and the prospective employer to make insightful observations. You may have to deal with reflective, direct or behavioural questions (explained in detail later in the unit). The employer will now have a sound idea as to whether you are the right person for the job. Depending on the nature of the position, the employer may not proceed with a further interview. However, more interviews may be required if the employer is undecided or if the position is a very senior one.

Third interview

If you are invited back for a third interview, you know that the person in charge of hiring is impressed with your previous interviews and that you have a real chance of getting the position. This interview may require you to meet more members of the management team and possibly some potential colleagues. You may be required to participate in some hypothetical situations and show problem-solving skills. Although there may be further interviews, these would most likely only take place in cases of very high-profile positions. A second or third interview is often the final interview in the process of selection.

The interview process involves the interviewer and the interviewee: it is a two-way communication process. Both parties have the same end goal in mind: to decide whether to offer the job to the prospective employee. It can be a stressful process for the optimistic candidate. The key to a successful interview involves three elements: preparation, confidence and engagement. If you are selected and invited for an interview, the following procedures will help you to be prepared for the experience.

Preparation

There are several important things you should do in preparation for the interview. The following are some of the most crucial aspects:

- Make copies of your identity document and your certifications.
- It is crucial that you are familiar with the job description of the position you are seeking. Familiarise yourself with the details of the job description well in advance of the interview. Ensure that you know the responsibilities and duties listed in it. It is common practice for the writer of a job description to place the duties and responsibilities in order of importance. Make sure that you have something to say regarding your experience with similar responsibilities, or about the way your set of skills links directly to the responsibilities that you will bear in the position.
- It is imperative to know the exact location of your interview. When you are preparing for the day of the interview, you must consider the logistics involved in getting to the location. You may need to arrange transport, so timing is crucial. Allow some time for possible delays. Locate the venue and ensure that you know the floor and room number of the office in which the interview will take place. Write down, or save on your cell phone, the name and number of the contact person. If there is any chance that you will be delayed, inform the contact person as soon as possible.
- Know what is on your CV – it shows positivity if you establish a link between your skills and the skills required for the position. Ensure that you can articulate your specific skills by referring to occasions when you have had to draw on your abilities. Connect these experiences to the skills and responsibilities required in the job description.
- You should arrive at the interview location timeously so that you can relax and emotionally prepare yourself for the interview. Take time to freshen up and use the bathroom if necessary before you go in to the interview.
- Once you arrive at the venue, it is likely that someone will meet you and inform you about what to expect.

Interview questions

The interview, as we have already mentioned, is a two-way communication process. You are there to be questioned and ask questions. Identifying types of questions will help you to establish how best to answer them. Examples of different question types are listed in what follows.

Closed questions

With closed questions, there is only one correct answer, which is usually not negotiable. For example, 'Do you have any children?' You either do have children, or you do not.

Open questions or open-ended questions

These questions leave room for the interviewee to offer some sort of opinion. The opinion should be valid and viable. You cannot go off into an outlandish exposition that would lead to doubt. For example, 'What motivated you to follow a career in law?' Answer the question honestly and convincingly.

Leading questions

These questions are tricky to answer because the interviewer directs you with an element of their own opinion in the question. This can sometimes lead you to say something that you do not actually mean. Be careful that you do not fall into the trap of saying something to impress, rather than telling the truth. For example, 'I've always felt that women cannot really achieve the same business results as men, what do you think?'

Reflective questions

These questions require you to look back and contemplate. For example, 'Do you live a well-balanced life?' This is used to help the interviewer to gain insight into your current and past behaviour.

Direct or specific questions

These questions usually refer to something definite to which the interviewer wants a direct answer. For example, 'When can you start training?'

Behavioural questions

These are questions that ask the candidate to describe how they reacted or behaved in previous situations where things did or did not go well. For instance, 'if the role requires creativity and critical thinking, then the employer may want to hear about challenging issues/situations that required some innovation or outside-the-box thinking' (*Skillings*, n.d.). This, too, provides insight into the past behaviour of the candidate.

Hypothetical questions

Sometimes the interviewer provides an imaginary or conjectural situation and asks: 'How would you handle this situation?' This generally involves a made-up scenario that could feasibly occur. The interviewer wants to assess your solution or your behavioural tendency.

Confidence

Nothing gives a better impression than genuine confidence. The significance of confidence cannot be underrated. Confidence must come from within you. Certainly, your preparation for the interview should boost your confidence. However, it is human nature to doubt ourselves when we are nervous or put under stress. The following points should equip you to be more confident in the interview:

- Your body language should be positive and confident. Sit upright and face the person talking to you. Make eye contact with the person asking the questions and speak clearly and plainly.
- If you do not understand a question or do not hear the question properly, ask the interviewer to repeat the question or explain more about what they are asking.
- False smiles and forced laughter can be off-putting. Your aim is to be as genuine as possible. Gestures and nods should be appropriate. Assess what hand and head movements are fitting, and avoid exaggerated gesticulations (gestures). Wood (in Desta, 2014) states that ‘keeping your hands hidden can be misinterpreted as a distrustful behaviour’. Remember, non-verbal communication is as important, and may even be more telling than, verbal communication.
- If you are asked to perform an in situ (on the spot) hypothetical exercise or complete some type of assessment questionnaire, do not hesitate. Agree to participate willingly and confidently. Your reaction will be noted. Being ‘put on the spot’ is sometimes a strategic way of assessing a candidate.
- Be prepared for the unexpected. Phone calls and invitations to interviews may come unanticipated.
- If the policy of the organisation is to record the proceedings, the interviewer must ask for your permission. Do not be alarmed by the thought of having the interview recorded. Remember, it is in your own interest for the interviewers to be able to refer to your responses at a later time.

Engagement

When we talk about someone engaging well in an interview, we mean that the candidate fully participates in meaningful conversation and communicates with depth and insight. Being able to initiate thought-provoking discussion and dialogue shows that you are willing to take initiative and create a positive impression. A sincere two-way conversation in which both parties contribute bodes well for the outcome of the interview.

If you feel discouraged during the interview, it is important to keep your composure (*Robert Walters* 3, n.d.). All your reactions are being observed and your demeanour must remain positive. It may happen, albeit uncommonly, that the interviewer offers you

the job immediately. If you are sure that you want the job and all your questions have been answered, including details of salary packages, you can accept the offer immediately. It may be advisable to request some time to think it over, though. You can ask, respectfully, for a period to consider the offer, but offer a date by which you will give them your decision. You should not need more than 24 hours.

Once the interview ends, and the interviewer and interviewee have completed the questions and discussion, the interviewer will indicate that the interview is over. Importantly, *Robert Walters* 3 (n.d.) advises the following: 'Don't be too discouraged if no definite offer is made nor a specific salary discussed. The interviewer will probably want to consult colleagues or interview other candidates (or both) before making a decision'.

Generally, the interviewer will thank the candidate for coming and offer a handshake. In most cases, the interviewer will stand and usher the candidate out. The interviewer will generally offer some indication that they will make contact within a certain time frame. It is important for the interviewee to thank the interviewer for the opportunity and to make some sort of hopeful comment such as: 'I look forward to hearing from you'. It is a good strategy to show that you are anticipating a positive outcome.

Regardless of how things turn out, leave the interview with a positive demeanour.

4.6.3 Types of interviews

If you research 'types of interviews', you will find labels for nearly every interaction between a prospective employee and their would-be employer. Remember, when you are searching for a job or have put your profile out in the job marketplace, the potential employer (or a person acting on their behalf) may contact you at any time. Always answer your phone politely. Greet the caller and give your full name. If it does turn out to be someone regarding a job application, make sure that you can hear properly and that you will not be disturbed or distracted during the call. If you are unable to hear properly or make yourself heard, ask whether you can call back as soon as you are in a suitable location.

In this section, we consider six main types of interview (*Undercover Recruiter*, n.d.). Within each of these types of interview, the questioning strategies are generally divided into two categories. One trend is a more conventional strategy based on the tradition of broad-based questions (open questions). Employers using this type of questioning want to establish whether the candidate has the required skills for, and a genuine interest in, the job. The notion of being a cooperative member of the team is often determined by the answers to these types of questions.

The second trend in questioning strategy involves the potential employer determining how the candidate will behave in certain situations, based on past experiences. The

interviewer asks a series of questions (follow-on questions) to get an idea of whether the candidate will cope with the job. They are trying to assess what you have gained from your experiences and actions, and whether the actions you took were successful. Referring to these behavioural type questions, Randall Hansen suggests that ‘job-seekers should frame their answers based on a four-part outline: (1) describe the situation, (2) discuss the actions you took, (3) relate the outcomes, and (4) specify what you learned from it’ (Hansen, n.d.).

The dialogue in the following example provides a hypothetical, but feasible, situation that will help you to understand Hansen’s four elements. The interviewer is Mr Khuzwayo, and the interviewee is Ronald Mlambo.

Mr Khuzwayo:	Can you tell me about a time when you had to make a difficult decision regarding a work colleague?
Ronald Mlambo:	Yes, I can. Last year there was a situation in the office that troubled me. The office has an open-plan layout, and four of us have work stations close together. One colleague spent most of the day on personal phone calls and checking personal emails. I found this very distracting and his behaviour impacted negatively on our targets.
Mr Khuzwayo:	That is a difficult position to be in. How did you resolve this situation?
Ronald Mlambo:	I spoke to my colleague about it and told him that I found it difficult to work while he was chatting on his cell phone. I also tried to tell him, in a friendly way, that we were going the extra mile while he was getting away with skiving and shirking his responsibilities. We are quite good friends, but he laughed and didn’t take me seriously. I didn’t want to cause animosity between us, so I went and spoke to our manager.
Mr Khuzwayo:	That sounds like a wise decision, how did the manager help you?
Ronald Mlambo:	The manager was great. He thanked me for being willing to point the problem out to him, and he suggested that he call a meeting with all four of us to discuss new office protocols to be put in place, and to discuss the importance of our individual contributions in relation to the team effort. He did this without letting on that anyone had complained and without making

	any one of us feel embarrassed.
Mr Khuzwayo:	Did this help the situation in the office?
Ronald Mlambo:	Yes, it did help. My colleague told me that he realised that management was aware of what is going on in the office, and that he had better not get caught making all his personal calls during office hours. He became much more conscientious and the personal phone calls were mostly made during his lunch breaks.
Mr Khuzwayo:	Do you think you did the right thing, or do you think you were disloyal to
Ronald Mlambo:	I definitely think I did the right thing. My colleague is a much more independent worker now. I think I have learned that you can't let these kinds of situations get to a point where you get so angry that you make an irrational decision. I also learned that our manager handled the circumstances in a professional manner. He protected me, and he also motivated us to be more conscientious.

We will now consider different interview types in more detail.

One-on-one interviews

This type of interview is also known as a face-to-face interview. Traditionally, the interview process involves two parties: the interviewee and the interviewer(s). Having more than one person in on the interview does not necessarily make it a panel interview (which will be discussed later in this unit). The one-on-one interview is quite intimate. It is a carefully planned process whereby the interviewer probes for insight into you and your character.

This type of interview may sometimes feel a little stressful, as only the perceptions of one or two people are taken into account. It is, however, an opportunity for you to really sell yourself. You can make a good initial impression with your attitude and personal engagement. Even if the person interviewing you is not the final decision-maker, and you know that you must go through a follow-up interview, making a good impression in the one-on-one interview will mean having someone supporting you in the next round.

Telephonic interviews

If you have submitted your CV and letter of application, you should not be shocked if you are called and asked some screening questions telephonically. Most telephone

interviews are a screening test that will hopefully result in a face-to-face interview at a later stage. The interviews are not long, but they are influential. You need to follow the guidelines given under the *Preparation, Confidence and Engagement* sections discussed earlier in the unit.

A would-be employer may phone at any time. If you cannot take the call or if you miss it, return the call as soon as possible. If you take the call and conditions are not conducive (favourable) for talking, respectfully request that a more suitable time for a call be arranged.

The questions in a telephone interview are usually quite direct. Often, they are questions that confirm your qualifications and your interest in the job and the organisation. Sometimes, the interviewer will ask directly about your salary expectations or explain that there is a ceiling offer. If the amount is not within your acceptable range, decline the offer and the interview process will not go any further. However, if there is no mention of remuneration, the telephone interview is not the occasion to discuss salary. It is then better to avoid discussing salary until you are in a one-on-one situation.

Panel interviews

A panel interview usually consists of three or more people making up a panel that will ask the interview questions, make notes and, later, discuss the competency and suitability of the candidate. This may seem daunting, but it is in your best interest to get a range of perceptions as to who you are and what you have to offer. The panel is there to assess your responses and reactions.

When you respond to a specific question from a specific panel member, you should begin by directing your reply to that person. However, make eye contact with the other panel members as you progress so that you are engaging them all.

In rare cases, there may be two or more candidates in a panel interview. If this is the situation, your aim is to be the candidate who is set apart and shines out from the rest. Ways to achieve this include offering to answer questions when there is hesitation from the other candidates. Do not, however, dominate every question. Initiate responses relating to what the other candidates have offered. Avoid being competitive, but develop ideas that one or other candidates have brought to the interview. Show innovative ideas and initiative. Always show respect for the other candidates as this will reflect your ability to work cooperatively with other people.

Meal interviews

You may be invited to a meal interview, perhaps not as a first-time jobseeker after graduation, but later in your life. This type of interview occurs when the potential employer invites you for an interview over a meal. In most cases, it will be a lunch

meeting. However, due to busy work schedules, meeting for breakfast has recently become popular. It is unlikely that the interview will take place over dinner, but remember to be prepared and expect the unexpected.

A meal interview may make you feel nervous, as it involves eating in front of someone you do not know. It is, however, a good opportunity for you to showcase your social skills while maintaining a professional approach to the questions. Keep in mind that it is an interview and, as such, you should remember all the advice given in the preceding paragraphs of this section.

On a practical note, you should carefully consider what you order to eat and drink. Take your lead from your host in terms of drinks and food orders. Order something that is not too complicated to eat and ensure that you remember your manners. Pace your eating, so that the conversation is fluid and you are not caught with mouthfuls of food or greasy hands. You should also take the lead from your host in terms of price. In other words, if your host orders a mid-price item, do not go and order the most expensive dish on the menu. Try to order a meal of equivalent or lower value than your hosts' meal.

If it is a luncheon, be careful with ordering alcohol: there are non-alcoholic drinks that are acceptable substitutes. You do not want anything to impair your judgement. You need to be fully aware of what you are going to say and how you are going to react.

Teleconference or webcam interviews

With the huge advances in technology, some interviews can be conducted over long distances. For these interviews, ensure that the room you are in is quiet and that you have no distractions. With video calls – for example, via Skype or FaceTime – you should ensure that the angle of the camera is positioned at eye level: you should not be looking up or looking down at the camera. The camera should also be positioned for what we call a medium shot, so that your head, shoulders and most of your torso are visible. Ensure that the setting is appropriate and avoid looking at your image on the screen. Rather look at the interviewer. Do not simply read notes that you may have prepared. You need to engage with the interviewer as you would in a face-to-face interview.

Follow-up interviews

Sometimes employers will ask you to come back for a follow-up interview. This is not uncommon. Follow-up interviews occur mainly when a panel has discussed and evaluated you from a first-round interview and invited you back for further evaluation. When this happens, the panel may or may not comprise the same members as in the first round. You cannot afford to take anything for granted. Very often, the questions will be behavioural questions, such as 'What did you do in a similar situation?' or 'How would you deal with this kind of situation?'

There are many overlapping elements in each type of interview. It is essential to be prepared for whatever may be put to you. Take responsibility for the preparation and do not leave anything to chance.

4.6.4 Preparing for various types of interviews

By way of consolidating the key points leading up to the interview, examine the following essential aspects of each.

One-on-one interviews

- Prepare all your documentation. Documents may include copies of your identity document, driver's licence, education certificates, degrees and diplomas, academic records from college or university, reference letters, and any other documents the caller may have asked you to bring.
- Examine the job description and familiarise yourself with the responsibilities and duties of the offered position.
- Anticipate the types of questions you may be asked about the job for which you are applying.
- Research the organisation so that you are familiar with the company culture and ethos. Look up their company policies, the community projects they may support, and the profiles of the senior management members.
- Re-examine your CV and cover letter – you cannot fumble over the profile you have presented to them.
- Ensure that you have made arrangements to arrive on time and that you know how to locate the venue or offices where the interview will take place.
- Make sure that you have a contact person and contact details. Make sure you know the name of the person who will interview you.

Telephonic interviews

- Make sure that the place where you are going to be taking the call is quiet.
- Set up a table and chair so that you can have all your documents in front of you and ensure that you will be comfortable.
- Reconfirm the time ahead of the interview with the interviewer or the interviewer's secretary.
- Ensure that you will not be distracted or interrupted by anyone.

- Some people like to have a friend or colleague listening in on the interview for moral support. If this suits you, make the arrangement.

Panel interviews

- If you know that you are going to be in a panel interview, think about the different personalities that you may encounter. Prepare to respond to these different personalities without being caught off guard.
- If possible, contact the personnel or human resources secretary, or ask the person who invited you to the interview to find out how many people will be on the panel.
- Make sure that the details provided in your CV, cover letter and certifications are firmly in your mind, so that you can recall and discuss them easily.
- As in a one-on-one interview, you should research the nature of the organisation and prepare some appropriate questions of your own.

Meal interviews

- Locate the restaurant or coffee shop and ensure you know how to get there.
- You may even check their menu, either online or by visiting the restaurant. Get a feel for the atmosphere of the chosen venue.
- Have copies of your documentation on hand, just as you would have for other types of interview.
- Try to ascertain the name and face of the person you are meeting – either contact the personnel department or confirm who will be conducting the interview when you are invited.

Teleconference or webcam interviews

- With a teleconference, you should prepare the venue in which you will be sitting beforehand.
- Ensure that any noise is shut out and inform whoever may be around that you cannot be disturbed.
- Have all your documentation easily available for reference.
- For video/camera interviews, ensure that the equipment is set up correctly and is working properly. Do a test run beforehand to ensure that the audio output from your speakers, and input from your microphone, are set to the appropriate volumes. Using earphones during a video interview will eliminate the chance of reverberation.

- Place the camera or cell phone in an appropriate position. It should be at eye level.
- Check the lighting. You do not want to be seen against a severe glare or in shadow.
- Eliminate noise and ensure that there are no distractions or interruptions.
- Seat yourself in a comfortable position at a desk or table. Do not sit on a leisure chair or couch.
- Follow the same procedure as you would for a one-on-one interview.

Follow-up interviews

- After the initial interview, do further research into the position and the organisation.
- Investigate examples of other follow-up interviews of a similar nature.
- If possible, try to establish how many people will be interviewing you and what their names are.

General preparation tips

- Spend time prior to the interview thinking about questions that may come up in a conversation about the position for which you are applying. Does the job entail independent work? Does the job entail teamwork? What leadership skills might the interviewers question you about? What are your skills in conflict resolution? How do you deal with the various relationships in a workplace? What is your communication competency? Are you an extrovert or an introvert? Know your strengths and weaknesses. Often interviewers want to know about the real you. It is important, therefore, to be honest and authentic in all your responses and questions. Do an online search for examples of interview questions.
- Research the organisation. Find out about their initiatives. What sets them apart from other companies or organisations? Formulate a concrete reason as to why you want a job in that particular company. Treat the company as someone that you are interested in interviewing.
- Decide what you are going to wear. Make sure that your dress is appropriate for the nature of the interview. (Further discussion on dress follows this section.) If you need to purchase a new item of clothing for the interview, do it in advance so that it can be washed and ironed in time.
- Schedule your appointments or activities so that you have enough sleep the night before, and so that you are not busy immediately prior to the appointment time on the day of the interview.

4.6.5 Dress to impress

The interview is an opportunity to present yourself in a positive light. There are dress codes for all sorts of occasions, including the occasion of an interview. Appropriate dress depends on several factors. You want to leave a lasting and positive impression. Lorenz (2016) argues that ‘while your skills, experience and capability will ultimately win you the job, how you present yourself can have a big impact on the hiring manager’s impression of you’. If what you wear can contribute to a positive impression, take advantage of the opportunity to dress to impress.

There are no fixed rules. We have encouraged you to be unique and present yourself as an individual. This does not mean that you should be an attention seeker. Appropriate dress depends on many factors. The aim is to present the best image of yourself possible.

When the initial invitation arrives, you could enquire from the person phoning you as to whether there is a dress code to which the company or organisation adheres. You could contact the human resources department and ask for advice on dress. You could also do some online research into the company culture and ethos. What follows are broad guidelines on dress. Use your discretion and good taste to determine what is suitable attire for the interview. Looking good does not always translate into looking professional or dressing appropriately. Personal grooming is very important. Hair should be clean and appropriately styled. Nails should be clean and groomed. Clothes should be clean and fresh.

Differences in dress in various sectors

We will now consider different sectors of the workplace, to establish recommendations for the way to dress for various interviews.

In the professional sector – in areas such as law, finance, corporate business, professional consulting and specialised practice – the dress code is quite conservative. For men, the trend is to wear suits with collared shirts, a tie and formal leather shoes and socks (laced shoes are more formal-looking than slip-on shoes). Limit your choice of socks to black, grey or brown. Do not wear secret socks. For women, the style is also quite formal with a tendency toward a skirt, blouse and jacket or a smart dress or pants suit with smart sandals or court shoes. Sometimes, the company dress policy requires women to wear nylon stockings. Many articulated clerks are required to dress in this way. However, sometimes the policy is more relaxed, and a tie or stockings are not required unless you are consulting with a client. Jewellery and other accessories, such as handbags and scarves, should be tasteful and should not be ostentatious (flashy) or garish.

This does not mean that your dress should be devoid of character. A quirky piece of jewellery or a complementary colour combination may set off your outfit. Wear appropriate and well-fitting clothes. Make sure your clothes are washed and ironed. Do not take a shirt out of the cellophane packaging and put it on without ironing it. If you are going to wear makeup, apply it conservatively. You do not wish to appear pretentious or affected in any way.

In the media industry – in areas such as advertising, marketing, graphic design, artistic or digital design and information technology – the dress code is less formal. Some media industry companies are trendsetters and enjoy the notion of being innovative, non-conformist, creative and inventive. This often means that the dress codes are not restrictive and people within the organisations show off their creativity and individuality through what they wear. The clothes are more trendy, chic and popular. If you have an interview in this industry, ensure that you dress accordingly. Men may wear more casual trousers, a long- or short-sleeved collared shirt and casual, closed shoes. There is the option of wearing a less formal jacket than a suit jacket. Although many of these companies allow jeans, it is better to err on the side of caution and not wear jeans to an interview. ‘Casual’ does not mean that you should wear a T-shirt or slip-slops. Women could consider fashionable dresses, skirts and pants with a suitable blouse or shirt. Shoes should always be well maintained and not scuffed. Accessories should not dominate the attire or distract the interviewer.

The education, training and consulting sector does not generally require a formal dress code. For men, smart pants with a belt, and an open neck shirt (long- or short-sleeved) with closed, casual shoes is acceptable. Women may wear fashionable clothing and shoes, but are advised not to wear jeans, excessively short dresses or skirts, transparent garments or unnecessarily tight-fitting clothing.

In all cases, dress with careful consideration of your height and weight. While there is much commentary on freedom of expression through dress and a call for a policy of non-discrimination in dress tastes, we are visual beings. If your appearance is appealing, engaging and pleasing, the impression left on the interviewer is positive. Avoid being remembered for the wrong reasons. The discussions that take place after you have left the interview should concern your suitability for the position, not an analysis of what you were wearing and why you may have chosen to wear such an outfit.

4.6.6 Interview etiquette

In this section, we will discuss the step-by-step protocol for a successful interview:

1. Do not be late.

2. On arrival at the interview venue, announce yourself to the receptionist or personal assistant. Give a friendly greeting and then give your name and the reason you are there. If asked to take a seat, sit calmly and patiently. Do not display any evidence of agitation or anxiety, such as by shaking your legs or sighing.
3. When you are called for your interview, thank the person you first met and enter the interview room confidently. Look up in anticipation of meeting the interviewer. On meeting the person or persons who will interview you, shake hands and greet them cheerfully and sincerely. You may thank them for the opportunity. Give your name if it has not already been mentioned by the person interviewing you. Wait to be offered a seat before you sit down. Your body language says much about you, so do not slouch or keep brushing your hair back or aside, and do not look down. If you have to cough or sneeze, remember your manners: cover your mouth and nose, and say 'excuse me'. Do not sniff.
4. Address the interviewer by name. If you are not offered a first name, address the person as Mr or Ms and the surname.
5. If you are meeting someone for a meal, ask the maître d (head waiter) if the person has been seated. If so, go directly to the person and introduce yourself with a handshake. Thank them for the opportunity to meet. The same tips mentioned in relation to other interview types apply. In addition, be aware of your eating and drinking habits. When leaving this type of interview, thank the interviewer for the meal.
6. During the interview, be sure that you wait until the question to be asked in full. Do not pre-empt the questioner and interrupt with an answer. If you do not understand the question, it is better to ask for clarification than to respond inappropriately. Responding to difficult questions will be discussed in more detail later in this unit.
7. If you are advised that the policy is to record the interview for review later, remember this is common practice. Interviewers are required to ask your permission and you are well advised to give it.
8. Do not have your cell phone on or in 'vibrate' mode during the interview. Above all, do not take a call on your cell phone during the interview.
9. Do not take a bottle of water or any other drink into the interview.
10. The outcome of the interview is unlikely to be given immediately. Do not ask how the interview went or whether you got the position.

11. Do not initiate any discussion about salaries, leave policies, incentives or bonuses in the interview. If the interviewer brings up these subjects, however, you should engage with them.
12. Never berate or make disparaging comments about any other employer for whom you have worked.
13. When the interview is over, stand and thank the interviewers for their time. Shake hands and leave with a positive comment about the interaction. It is not necessary to send a note or an email to thank the interviewer. This is done when you are invited to the interview or when you meet the interviewer.
14. Do not take a call or make a call until you are away from the venue. You do not want your interviewer to overhear your conversation with a friend or relative concerning the interview.
15. Do not take any kind of gift to the interviewer, as this would appear as manipulative and coercive.

4.6.7 Responding to difficult questions

Throughout this section, we have stated that the purpose of an interview is to determine whether a candidate is suitable for a position. Most interviews are undertaken in a manner aimed at best determining whether the applicant will fulfil all the criteria required for the job. Sometimes, the interviewee will have to face difficult questions. This is part of the evaluation process and is the reason why preparation and groundwork are important. You need to anticipate the possibility of challenging questions and be equipped to deal with them.

It is impossible to give examples of difficult or challenging questions that cover every category of the job market. Instead, we will discuss different forms of thought-provoking questions. You are encouraged to think about the following topics before the interview and consider how you would respond in a sincere and effective manner.

Personal weaknesses

You may be asked what your weaknesses are. You need to be able to identify these and articulate ways in which you manage them. Acknowledging your weaknesses shows humility and self-reflection. Admitting that your weaknesses will never change and that there is nothing you can do about them shows arrogance and stubbornness. Questions may revolve around the way you see your weaknesses impacting your job and how you see them affecting other people with whom you work. Questions may also concern the way you overcome your weaknesses. Questions about the way you deal with other

people's weaknesses may also be asked. The interviewer may be looking at your tolerance, support for others, compassion and empathy.

Theoretical questions

The 'what if' questions may be very tricky to answer as you may not have experienced the situation being presented or anything similar. It is challenging to put yourself in a difficult situation and say how you would cope with it. The questions concerning these hypothetical scenarios usually centre on conflict, disaster and crises. They may also focus on achievement, success and epiphanies (significant realisations). The interviewer may be looking for your initiative, resourcefulness and integrity based on your experience and knowledge of what the position may entail.

Reflective questions

When asked to look back and consider how you handled a difficult situation, the interviewer may be evaluating your logic, practical skills, efficiency and integrity. These are your 'soft skills'. They include character traits that indicate a high level of emotional intelligence and your interpersonal and intrapersonal skills. This type of question also gives the interviewer the opportunity to assess your willingness to change and adapt, your problem-solving competency, your critical thinking ability and your teamwork skills.

The interviewer may ask you to reflect on situations that involved your dealings with sexism, racism, cultural differences, discrimination and sexual orientation. These are pressing issues that face people in the workplace every day. You should have clear and sound opinions on these matters.

Personal questions

Your personal affairs should not be addressed during an interview. Your religion, marital status, sexual orientation, HIV status and personal wealth are personal issues and should not be topics of discussion. If the interviewer enters this territory, you have every right to politely refrain from answering the questions. You may suggest that none of these aspects will affect the way you perform your duties. This is a delicate space and should be handled with respect.

You should also consider how to answer questions about former employers or companies for which you have worked. Do not engage in conversation about individuals. Avoid slandering your previous employers or companies. Be discrete and diplomatic when answering questions about previous employers or colleagues.

Retain your poise and self-control. It is better to ask for a minute or two to contemplate your answer than to reply hastily and inappropriately. Be a responder, not a reactor: taking time to respond shows that you are thoughtful and considerate. Your preparation

prior to the interview is crucial for responding to these types of questions in a mature and refined manner.

4.6.8 Interview outcome

Whether the outcome of the interview is successful and secures you a position or not, it should be a constructive experience for you. If you are successful and are offered the position, you will be notified either by telephone or email. You may be asked to go into the office or workplace to get notified of the outcome. You should respond to the job offer immediately with your acceptance or non-acceptance. If you accept, you may be requested to sign contract documents, finalise salary packages and arrange to start your job. You may be required to go for a medical check-up, get uniforms or company clothing, and meet with your immediate boss and colleagues. Show your appreciation and enthusiasm sincerely. Do not overreact.

If the interview results in you not getting the position, you will also be notified by telephone or email. Acknowledge the outcome by thanking the person and expressing that it was, indeed, a positive experience. You may also add the hope that there will be a time in the future with other opportunities for you. Remember that your CV is on the database. You must leave a lasting positive impression. Never question why you did not get the job. You may, however, ask if there is anything you should be working on to improve your opportunities in the future. Use your discretion as to whether the occasion is appropriate or not. If you retain your confidence despite the outcome, you will be remembered.

4.6.9 Money talks (salary negotiation)

To be in a strong position regarding salary negotiation, you need to have done background research. Firstly, find out about the economic position of the company before you go for the interview or meet to finalise your employment conditions. You should also find out what benefits the company offers, such as pension/provident funds, medical aid/gap cover, employee incentives and bonus policies. Investigate how the company's salaries compare to salaries for similar jobs in the marketplace. This will give you some indication as to whether you can expect a salary offer that is market-related, or whether the company has a reputation for paying either above or below the current market rate.

It is important that you carry out a careful analysis of your own circumstances and financial situation. You can do a financial needs analysis and create a fair and reasonable budget that will give you your bottom-line figure. You should know what similarly qualified people are earning. Remember to take experience into account.

There are many online indicators that provide the results of research into comparative salary packages in specific fields. Remember that your context is the South African environment. Do not be afraid to evaluate your worth. Leave room for compromise, but do not be manipulated and end up in a situation that is not financially viable for you.

At the outset of salary negotiations, it is imperative that you remain composed and congenial. A good-natured discussion with sincere and authentic questions will leave the prospective employer impressed with your integrity. You already have the job offer. The discussion concerning salary and remuneration is the final element of sealing the deal. The salary may have been stipulated in the advertisement or the interview. If this is the case, you will not have queries about the amount. However, you will have questions about the salary package and what it includes. Be aware that you may be so relieved to get the job that your emotional elation clouds your judgment concerning salary. Fortunately, there is usually a time lapse between being offered the job and signing the contract. Usually, the employer will bring up the discussion about salary when you meet to sign your employment documentation. Show that you are well prepared for a dialogue about a fair market-related salary. The information that follows is not absolute; however, it will help you to conduct your discussion in a professional manner:

- You will need clarity on your gross salary (your salary before any deductions are made). This is sometimes referred to as a 'basic salary'. If the salary is not what you expected, be prepared to explain why. For this, you need to have done research. You can then refer to what others in similar jobs are earning in the marketplace. You could ask for an explanation as to why the salary is lower than other market-related salaries.
- You need to know whether, after a probationary period, your salary and/or benefits change. If they do, you could consider the lower starting salary. However, do not undercut your bottom-line salary.
- Establish whether the company has a pension or provident fund. Make sure that you understand what percentage of your salary goes into this fund and what percentage the employer contributes.
- You may be given a breakdown of your salary in terms of your cost to company. The cost to company figure includes your basic salary, a fixed bonus (if there is one), the company's contributions to the pension/provident fund, company or group insurance, Unemployment Insurance Fund (UIF), and Skills Development Levy (SDL) if applicable.
- Ensure that you understand the deduction of pay-as-you-earn (PAYE) tax from your gross salary. This deduction is paid to the South African Revenue Services (SARS).

- Ensure that the employer deducts contributions to the UIF. The total contribution amount is two per cent of your salary, of which your employer pays half, and you pay half. (Some employers pay the full two per cent.)
- Discuss available medical aid funds and whether the company subsidises your contributions.
- Get clarification on any unique deductions such as gift funds or in-house saving options.
- You should clarify the circumstances under which bonuses are paid. For example, consider the following questions:
 - Is your bonus paid annually as a 13th cheque?
 - In what month is your bonus paid? (Some companies pay the bonus in the employee's birthday month.)
 - Is the bonus a discretionary amount? If so, how are you evaluated for the bonus?
 - Is the bonus calculated based on an incentive? If so, how is it calculated and when is it paid?
- If the salary package involves a basic salary plus commission, it is vital that you fully understand the commission structures and how you will be taxed.
- You may ask what the company policy is on annual reviews of salaries and whether appraisals take place annually.

FAST FORWARD: Rethinking CVs – What future employers will expect to see

In previous years, CVs would be two to three pages in length (on average), and would include all the typical pieces of information – i.e. the individual's biographical details, education, qualifications, work experience, and finally, some referees to provide insight into their work ethic and overall performance. Applicants would often hand in hard copies of these CVs when applying for a position.

Over time, and with the expansion of technology, there has been a shift toward submitting CVs via the Internet. This is mainly done via email, or by uploading them onto the website portal of the employee or recruitment agency.

It has also become increasingly popular to have a single-page CV, which summarises the individual's essential skills, qualifications and experience in a more concise format. These CVs are often artistically designed and very 'to the point'. The following figure shows an example of a modern CV.



Belle Einder

Full-Time Teacher

Part-Time Artist

EDUCATION

- 2017** Higher Certificate in ECD Teaching
City Campus University
- 2015** First Aid Certification
- 2014** National Senior Certificate
Goldleaf High School

WORK EXPERIENCE

- 2019** **Teacher** at Mountain Springs
Primary School
- 2018** **Student Teacher** at Learning Tree
Primary school
- 2017** **Assistant** at Learning Tree
Primary school

AWARDS

- 2017** **Top practical student**
City Campus University
- 2016** **Academic merit list**
City Campus University

(Source: EDGE Learning Media (Pty) Ltd, 2019; and Shutterstock.com)Figure 4.7:
Example of a modern CV

However, modern CVs have often been criticised for ‘boxing employees in’, in the sense that they force individuals to summarise their offerings, rather than allowing them to showcase their full range of skills and capabilities. They may also cause complications for job applicants; for example, if there are gaps in their employment history, or too many jobs with short employment periods listed. Additionally, they leave little room for including skills that have been acquired outside of work, and explaining how they are relevant.

In the fast-paced work environment of today, employers simply do not have the time or resources to read through every applicant’s CV in detail. As such, many resort to using online databases and searching for key terms within a number of submitted CVs, in order to find candidates who are suitable for a particular position. They are also increasingly coming to rely on social media – in particular, LinkedIn – to gain insight into applicants and their offerings. Social media has made it substantially easier for employers to ‘headhunt’ candidates for high-profile positions – often, without ever looking at a formal CV.

Another trend in recruitment is an increased focus on who the individual is (and ultimately, whether they can realistically perform well in the job), rather than on what they have done. This focus on competency, rather than achievement, has also led to the increased popularity of competency tests. Here, prospective candidates are asked to complete an assessment that reflects the day-to-day requirements of the position, which will ultimately reveal whether the individual is suitable or not. If the applicant completes the test successfully, they will move on to the next round of recruitment, and will likely be invited for an interview.

These aspects show how modern job application and recruitment are becoming more complex and advanced, and are relying less on CVs, and more on technology and competency tests.

(Source: EDGE Learning Media (Pty) Ltd, 2019; adapted from Staley, 2018)

4.7 Summary

The purpose of this unit was to equip you with important, fundamental career skills. Getting that job is an important step in building a career. Young South Africans need employment if we are to grow the economy and have a favourable future. Engage with the material offered in this unit. Practise and develop the skills covered in the various sections of it. Develop and rework your online profile, CV and your cover letter to secure yourself the greatest chance of landing that dream job!